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TABLE OF CONTENTS

	Pag.
B. Karlgren: Word Families in Chinese	9
A. Waley: The Book of Changes	121
J. G. Andersson: Selected Ordos Bronzes (with 16 Plates)	143
T. J. Arne: Die Funde von Luan P'ing und Hsuan Hua (mit 14 Tafeln)	155

WORD FAMILIES IN CHINESE

BY

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One of the great goals of Chinese historical phonetics is to prepare the ground for comparative Sinitic linguistics — a systematic comparison of Chinese, the T'ai languages and the Tibeto-Burman languages, which are all undoubtedly cognate though widely differentiated idioms. But in my opinion it will not do to pick out isolated Chinese *words* and compare them with isolated Tibetan or Siamese words. It stands to reason that Chinese does not consist of so and so many thousands of independent monosyllables, none of them cognate to any others; in Chinese, as in all other languages, the words form families, groups of cognate words formed from one and the same primary stem. It is not allowable to identify Chinese 目 Arch. *mǐók*¹) 'eye' with Tibetan *mig* 'eye' so long as we have not first established the word family to which *mǐók* belongs. Akin to *mǐók* is undoubtedly the word 睛 Arch. *mǐóg* 'pupil of the eye': and it is just as likely that it is this *mǐóg* which corresponds directly to the Tibetan *mig*. In other words: before Sinitic comparative linguistics can be safely tackled there remains a great task to be solved in each of the language groups concerned. In Chinese the words must be sorted and grouped according to genetic affinity, and the same must be done in T'ai and in Tibeto-Burman. Then, but only then, we can start comparing the *word groups* of these three great branches and hope for reliable results.

That some words in Chinese are cognate to others is of course no new idea. Already August Conrady in his pioneer work: *Eine Indo-Chinesische Causativ-Denominativ-Bildung*, 1896, had this as a fundamental point of departure. In my *Analytic Dictionary* I have pointed out, in many cases, such affinities between words, not only examples in which one and the same word happens to be represented by two different characters, e. g. 集 *dz'ǐəp*: 輯 *dz'ǐəp*, and hence appears to be two different words, but also cases like 夾 Anc. *kap* 'to press': 狹 Anc. *yap* 'narrow' etc., which are clearly cognate words. Indeed, even the Chinese script often indicates two forms as cognate by designing them with one and the same

¹) In the present paper Arch. means Archaic Chinese, the language of the Shi king, and Anc. means Ancient Chinese, the language of the Ts'ie yün, time of the Sui dynasty.

character: 長 *d'iang*¹⁾ 'long': 長 *iāng* 'grow long, grow up'. But it is important to take up this problem of the Chinese word families for a more systematic investigation. The present paper is intended to be a short preliminary notice, as a kind of introduction to a larger work, which I hope to be able to publish soon.

Before entering upon this principal theme I shall have to give some long preparatory chapters. We must not build our study of Chinese word families on the language of the Ts'ie yün (Anc.) which is comparatively late (6th c. A. D.), since we can attain to a fairly detailed knowledge of Archaic Chinese, the language of the Shī king and the (slightly older) *hie sheng* characters (phonetic compound characters) dating from the early part of the Chou epoch. It is about this latter language I first wish to make some fairly extensive remarks.

In my »Shī king Researches» (this Bulletin vol. 4) I have studied certain phonetic categories in Archaic Chinese, and I wish to revert here to some points connected with those categories; and in that work I left the reconstruction of certain other Shī king rime categories for future discussion, and I wish to take them up here for a detailed analysis.

CATEGORIES ENDING IN ARCH. DENTAL CONSONANT.

In Ancient Chinese (Ts'ie yün) there are two rimes *-iən* and *-iĕn* between which the words are distributed in an obviously mechanized fashion (Phonol. Chin. p. 174):

kĕn, kĕuĕn; iĕn, iĕuĕn; lĕn, lĕuĕn; tsĕn, tsĕuĕn; piĕn, 〇 ;

kĭən, kĭuən; 〇 〇 〇 〇 〇 〇 〇 〇 piuən.

The final *iən, iuən* does not exist after palatals and dentals, only after gutturals and labials. It is natural to suspect that this is a result of the general tendency of nivellation, so strongly at work in Chinese, and that Arch. Chinese had both types: *iĕn* and *iĕn*, but that these have been confounded in Anc. *iĕn*.

That this really was so is confirmed by the Shī king rimes. Let us state first that from the word groups here concerned we have to keep entirely separate words with Archaic *ā, a* (long) and *ǎ* (short). They form a Shī king rime category here called A, which is no. 14 in Tuan Yü-ts'ai's famous Liu shu yin kün piao and cat. 9 in Wang Nien-sun's equally important Ku yün pu (in Kao-yu Wang shī i shu).

¹⁾ The typographical device, always used in my earlier works, of indicating palatal explosives thus: *t', d'* etc. is inconvenient, since the apostrophe is easily confused with the aspiration mark, and is particularly clumsy in conjunction with it, e. g. *d''iang*. I therefore replace it, in the present article, by a bow over the consonant: *i, d̂*.

I need not reproduce their tables completely; they can be conveniently summed up in a few type words, for which I insert the Arch. and Anc. sounds:¹⁾

干管 顏關 展轉 見涓 聞患 言原。

Arch. *kân, kwân; ngan, kwan; tian, tiwan; kian, kiwan; kân, g'wân; ngiân, ngi'wân.*

Anc. *kân, kwân; ngan, kwan; i'ân, i'iwân; kien, kiwen; kan, ywan; ngim, ngiwm.*

From this rime category A are well distinguished two other Shī rime categories: B, cat. 12 of Tuan's = cat. 7 of Wang's. Here the principal words are:

1 賢年天田甸顛闌電填千翩。2 淵。

3 姻駟引榛漆臻綦陳臣塵身申神人仁麟親信薪盡燼
 膚濱頻瀕泯。4 均旬詢洵。

1. had Anc. *-ien : yien* etc.;

2. had Anc. *-iwen : i'wen;*

3. had Anc. *-i'ên : i'ên* etc.;

4. had Anc. *-i'üên : k'üên* etc.

C, cat. 13 of Tuan's = cat. 8 of Wang's. Here the principal words are:

1 艮跟。2 昆噲存餽孫遜奔瑞門

3 勤芹欣隱。4 君章訓魚愠云雲員芬焚翦聞問。

5 振辰晨畛忍負絡瘡。6 春濟順鵠掎輪洽。

7 先洒殄。8 艱。9 鰥盼。

10 巾堽殷慙。11 說閑。12 隕園。

1. had Anc. *-ən : kən;*

2. had Anc. *-uan : kuan* etc.;

3. had Anc. *-i'ên : g'i'ên* etc.;

4. had Anc. *-i'üên : k'üên* etc.;

5. had Anc. *-i'ên : ts'ien* etc.;

6. had Anc. *-i'üên : ts'üên* etc.;

7. had Anc. *-ien : sien* etc.;

8. had Anc. *-an : kan;*

9. had Anc. *-wan : kwan* etc.;

10. had Anc. *-i'ên : k'ien* etc.;

11. had Anc. *-i'ên : ts'ien* etc.;

12. had Anc. *-i'wen : j'ien* etc.

Two preliminary remarks:

Firstly, I have entered, in a few cases, characters which do not occur in the Shī rimes but which belong to the category, as revealed by rimes in other Archaic texts or by their 'phonetics'. Secondly, the second character in line 11 according

¹⁾ The 7th character not in the Shī, yet belonging to this category.

to its Anc. sound should stand in the *w* line (12); I have placed it in the *w*-less line (11), since it had no *w* in Archaic Chinese.

This latter phenomenon is an important point, which concerns several common words. I have arrived at the opinion that in Ts'ie yün time there were two kinds of *ho k'ou w*: one is genuine and Archaic, occurring after all kinds of initials, one is secondary and late, occurring only after *p*, *p'*, *b'*, *m* and due to an exaggerated labial articulation of the initial. Whereas genuine *piw*- regularly gives *f(w)*- at the very time of the Ts'ie yün: 方 *piwang* > *fwang*, 分 *piwən* > *fuən*, 非 *pjwɛi* > *fwɛi* etc., a secondary and evidently more volatile and weak *w* causes no such change. That 丙 Anc. *piwəŋ*, 平 *b'iwəŋ*, 明 *miwəŋ* had really a *w* in Ts'ie yün time is certain (丙 is spelled by 永), but since this *piw*- has not given *fw*- I conclude that the Arch. forms were *piǎŋ*, *b'iǎŋ*, *miǎŋ*, and that the *w* is secondary and parasitic. Such cases are to be found, besides in the table «*keng*», also in the tables «*chī*», «*shan*» and «*chen*» of the Sung rime tables (Phonol. Chin. pp. 149, 171, 177, 185).

If we now revert to our Shī king rime categories B and C and scrutinize them, we find that cat. B in the Ts'ie yün language had exclusively *e* vowels: *-en* or *-čn*; cat. C had a rich part with *ə*: *-ən*, *-uən*, *-iən*, *-iuən*, and then both *-čn*, *-en*, *-ən* and *-en*. To my mind there cannot be the slightest doubt that here in C the *-ən* vocalism is primary and principal, and that all the *-čn* and *-en* are secondary. Thus line C 5 was Arch. *i̇iən* etc. and line 6 was Arch. *i̇'iuən* etc.; but whereas *-iən* after the gutturals and labials with *ho k'ou* was preserved down to Ts'ie yün time: line 2 *g'i̇ən*, line 4 *k̇iuən*, *ṗiuən*, after palatals and dentals and after labials with *k'ai k'ou*, *-ən* became > *-čn*: line 5 Arch. *i̇iən* > Anc. *tsičn*, line 6 Arch. *i̇'iuən* > Anc. *ts'iučn*. This explains the gaps in the scheme of p. 2 above! In Ts'ie yün there are only types *k̇iən*, *k̇iuən* and *ṗiuən* but no types *i̇iən*, *i̇'iuən*, *tsi̇ən*, *tsi̇uən* etc. — because the latter, which existed in Arch. Chinese, have passed over to *tsičn*, *ts'iučn*, *tsičn*, *ts'iučn*, thus coinciding with the original (Archaic) *i̇čn*, *i̇wčn* (lines 3 and 4) of cat. B. Line C 7 is easily explained in consistency with this: just as Arch. *ṡiən* (with short *i̇*) in line 5 became Anc. *ṡičn* (after dental), so Arch. 先 *ṡiən* (with long *i̇*) became Anc. *sien* (equally after dental).

There seems to be a great difficulty which vetoes this general theory: the words which I have placed in line C 10: 巾 Anc. *k̇ičn* etc. If an Arch. *k̇iən* kept its *ə* (after guttural): line 2 *g'i̇ən* etc., how could we explain the *k̇ičn* etc. of line 10 with *č* after guttural in this same rime category? The answer to this riddle will be given presently after we have discussed lines C 8, 9, 11 and 12.

The *-an* in lines 8 and 9 cannot be original, for then the words of these lines would have rimed in cat. A above. What their Arch. value was is not difficult to find. In my *Shī king Researches* (pp. 157 and 160) I have shown that 革 Arch. *kek* (with an open, short, slack *ā*- sound: *ε*) and 戒 *keg* (> Anc. *kqi*) rime with *-ək*, *-əg*. Thus *ε* and *ə* regularly go together in the *Shī* rimes. I conclude that the Anc. *-an*, *-wan* (lines 8,9) in our cat. C derive from Arch. *-en*, *-wen*: 8 *ken*, 9 *kwen*.

Next we have lines C 11 and 12. I am happy to be able to improve here my reconstruction system of Anc. Chinese (Ts'ie yün) on this point. For line 11, which is a rime of its own in the Ts'ie yün (no. 19 in the *Nei fu ts'ang T'ang sie pen Ts'ie yün* and in the *Kuang yün*, p'ing sheng) I had not been able to give any independent final at all; for line 12 I had given a very unsatisfactory reconstruction: I had distinguished it from line 6 only by a difference in the *ho k'ou w*: 6 *-iüēn*: 12 *-iüēn*. Now this was very artificial and doubtful, and I have myself stated earlier that a new solution had to be found (*Shī king Researches* p. 126). Our *Shī* rime system helps us to solve the riddle. Lines 11 and 12 are the *ḡ* correspondences to the *-en*, *-wen* of lines 8 and 9: 11 *tsien*, 12 *jiwen* (Arch. *giwen*). Anticipating this discussion I have already entered these values in the table on p. 3 above.

We can now revert to the mysterious line 10 巾 Anc. *kien*. It is explained by the system of lines 11 and 12. If we take the latter two together, we have *tsien* and *mien* and we have *giwen*, but we have no type *kien*, i. e. the *k'ai k'ou* final *-ien* after gutturals and laryngals (which are the most frequent of the Chinese initials) is missing. It is obvious that in line 10 巾, 殷 Anc. *kien*, *ien*, which the *Shī* rimes do not carry to cat. B (Arch. *-ien*, *-ien*) but to cat. C (bringing them together with Arch. *-ən*, *-ien*, *-wən*, *-iwen*, *-ien*, *tsien*, *-iwen*) we have the missing type Arch. *kien*. Thus we get an explanation on the one hand of their placing in this rime category C (cf. rime cat. *ek:ək*, *eg:əg* of the *Shī*), on the other hand of their evolution down to Anc. Chinese:

Arch. *k|ien* > Anc. *kien*;
 » *ts|ien* > » *tsien*;
 » *k|iwen* > » *kiwen*.

It might be expected that in *ho k'ou*, as well as in *k'ai k'ou*, after guttural *-iwen* would become *-iüēn*. And indeed, there is a strong tendency in this direction. For the word 困 *Kuang yün* gives double readings *g'üēn* and *kiwēn* (rime 眞, not rime 諄 諄!), for 闕 it gives *kiwēn* (not *iüēn*!).

There is one more very strong support for our theory that the Anc. *-ien* of

line 10 (巾 etc.) has a different Arch. origin from the *-iĕn* of cat. B (type 因), that indeed it stood closer to Arch. *-iən* (type 斤), C 3, than did type 因 (B 3). In Go-on, the most ancient dialect of which we have a detailed knowledge, type 因 (B 3), Arch. *-iĕn*, is rendered by *in*; type 斤, 隱 (C 3), Arch. *-iən*, is regularly rendered by *-on*; now, the words 巾, 殷, 慇 of line C 10 are, in Goon, not *kin*, *in*, *in*, but *kon*, *on*, *on*. Evidently, in the dialect that was the basis of Go-on, the Arch. *kien*, *'iĕn*, *'iən* in these words had become *kien*, *'iən*, *'iən*, thus joining the C 3 type (斤 *kien*, 隱 *'iən*) and not, as in Ts'ie yün, becoming *kien*, *'iĕn* (joining the B 3 type 因 *'iĕn*).

We are now able to fill in the Arch. values of our tables B and C:

B.	Arch.	Anc.		Arch.	Anc.
	1. <i>ien</i>	> <i>ien</i> ;		2. <i>iwen</i>	> <i>iwen</i> ;
	3. <i>iĕn</i>	> <i>iĕn</i> ;		4. <i>iwĕn</i>	> <i>iwĕn</i> .
C.	Arch.	Anc.		Arch.	Anc.
	1. <i>ən</i>	> <i>ən</i> ;		2. <i>wən</i>	> <i>uən</i> ;
	3. <i>k iən</i>	> <i>k iən</i> ;		4. <i>k iwen</i>	> <i>k iwen</i> ;
	5. <i>î iən</i>	> <i>tš iĕn</i> ;		6. <i>î iwen</i>	> <i>tš iwĕn</i> ;
	7. <i>iən</i>	> <i>ien</i> ;			
	8. <i>en</i>	> <i>ən</i> ;		9. <i>wen</i>	> <i>wən</i> ;
	10. <i>k iĕn</i>	> <i>k iĕn</i> ;			
	11. <i>tš iĕn</i>	> <i>tš iĕn</i> ;		12. <i>iwen</i>	> <i>iwen</i> .

To the three *-n* categories A, B and C studied above there correspond three categories D, E and F ending in *-t* and *-d*. Before taking them up for discussion I wish to make a preliminary remark. In my Analytic Dictionary of Chinese (1923) I pointed out that numerous *hie sheng* cases like 割 Anc. *kāt*: 害 *yái*, 列 *liät*: 例 *liäi* reveal an Arch. final dental in *-ái*, *iäi* lost, or rather vocalized into *-i*, before the time of Anc. Chinese; and since there is regularly a falling tone in such cases, I concluded that the loss of the dental which I interpreted as *-d* (*yäd*, *liäd* in contradistinction to *kāt*, *liät*) had entailed the falling tone. In some later articles I modified my theory and said that the Arch. final dental was a *-t* in *yái* and *liäi* as well, and that the falling tone was primary and decided the evolution: whereas 割 *kāt*, 列 *liät* preserved their *-t*, 害 *yät* and 例 *liät* vocalized it because of the falling tone (similarly 白 *b'ok* > *b'ok*; 怕 *p'ok* > *p'a*). This modified theory had great advantages (see

Shī king Researches p. 119); and yet here I make a sudden *volte-face* and revert to my original theory such as it was sketched in my Analytic Dictionary. My reasons for so doing will be given on p. 23 below.

We now revert to the *-t*, *-d* correspondences to categories A, B and C. Two of them (the *-at* and the *-at* groups) Tuan Yü-ts'ai has erroneously confused into one: his cat. 15 (ju sheng section). This is one of the weakest spots in his otherwise excellent Shī rime treatise. Wang Nien-sun is superior on this point; he has clearly distinguished the three categories.

D. Cat. 14 of Wang's = part of Tuan's cat. 15, ju sheng section. The principal words are:

- 1 葛 葛 渴 渴 恒 達 園 ○ 2 帶 大 害 艾 藹 拔 肺 茂 ○
 3 括 估 活 闊 滅 撥 奪 撮 捋 菱 撥 赫 ○ 4 外 噉 殺 兌 脫 翹 ○
 5 殺 ○ 6 薑 ○ 7 八 ○ 8 噲 敗 蓮 ○
 9 桀 揭 揭 孽 舌 烈 滅 ○ 10 厲 柳 趙 哲 世 泄 總 ○
 11 說 悅 絕 威 ○ 12 悅 說 衛 歲 蹶 ○ 13 截 ○ 14 端 ○
 15 臺 ○ 16 瘞 ○ 17 刮 ○ 18 拜 ○ 19 搗 竭 保 ○ 20 刈 ○
 21 蔚 闕 月 越 鉞 發 伐 發 ○ 22 吹 吹 ○

- | | |
|---|--|
| 1. had Anc. <i>-āt</i> : <i>kāt</i> etc.; | 2. had Anc. <i>-āi</i> : <i>tāi</i> etc.; |
| 3. had Anc. <i>-uāt</i> : <i>kuāt</i> etc.; | 4. had Anc. <i>-uāi</i> : <i>nguāi</i> etc.; |
| 5. had Anc. <i>-at</i> : <i>sat</i> ; | 6. had Anc. <i>-ai</i> : <i>i'ai</i> ; |
| 7. had Anc. <i>-wat</i> : <i>pwat</i> ; | 8. had Anc. <i>-wai</i> : <i>k'wai</i> etc.; |
| 9. had Anc. <i>-iāt</i> : <i>g'iāt</i> etc.; | 10. had Anc. <i>-iāi</i> : <i>l'ai</i> etc.; |
| 11. had Anc. <i>-iwāt</i> : <i>iwāt</i> etc.; | 12. had Anc. <i>-iwāi</i> : <i>siwāi</i> etc.; |
| 13. had Anc. <i>-iet</i> : <i>dz'iet</i> ; | 14. had Anc. <i>-iei</i> : <i>tiei</i> ; |
| 15. had Anc. <i>-at</i> : <i>kāt</i> ; | 16. had Anc. <i>-ai</i> : <i>t'ai</i> ; |
| 17. had Anc. <i>-wat</i> : <i>kwat</i> ; | 18. had Anc. <i>-wai</i> : <i>pwai</i> ; |
| 19. had Anc. <i>-iāt</i> : <i>k'iat</i> etc.; | 20. had Anc. <i>-iāi</i> : <i>ng'iai</i> ; |
| 21. had Anc. <i>-iwāt</i> : <i>k'iwāt</i> etc.; | 22. had Anc. <i>-iwāi</i> : <i>b'iwai</i> . |

There is first a strict parallelism between lines with odd and even numbers. To the final *-t* of the former corresponds *-i* of the latter. This *-i* is the vestige of the lost *-d*, dropped and causing the falling tone (k'ü sheng): whereas 葛 *kāt* has preserved its *-t*, 帶 *tād* has become > *tāi*.

Further the whole of this category corresponds faithfully to the *-n* class in cat. A above. We therefore obtain the following Arch. values:

Category D:

1. *ât*, 2. *ād*; 3. *wât*, 4. *wād*;
 5. *at*, 6. *ad*; 7. *wat*, 8. *wad*;
 9. *iat*, 10. *iad*; 11. *iwat* 12. *iwad*;
 13. *iat*, 14. *iad*; ○ ○
 15. *ât*, 16. *ād*; 17. *wât*, 18. *wād*;
 19. *ĩât*, 20. *ĩād*; 21. *iwât*, 22. *iwād*;

Category A:

- ân*; *wân*;
an; *wan*;
ian; *iwan*;
ian; (*iwan*);
ăn; *wăn*;
ĩăn; *iwăn*.

We can now pass on to cat. E, being the *-t* and *-d* correspondence to cat. B. The principal words are:

- 1 結結願啞莖垣株節。2 噫噫。3 血穴。
 4 吉一逸室實窒秩桎櫛瑟日漆七疾栗慄匹輶駟密。
 5 至。6 恤。

1. had Anc. *-iet* : *kiet* etc; 2. had Anc. *-iei* : *tiei* etc.;
 3. had Anc. *-iwet* : *xiwet* etc;
 4. had Anc. *-ĩet* : *kĩet* etc.; 5. had Anc. *-i* : *tĩi*;
 6. had Anc. *-iuit* : *siuit*.

Here again 1 and 2, and 4 and 5 respectively had the same Arch. final, differentiated only by the contrast *-t*: *-d*, the latter having vocalized into *-i* and causing a falling tone. We thus obtain the following Arch. values:

Category E:

1. *iet*, 2. *ied*;
 3. *iwet*;
 4. *ĩet*, 5. *ĩed*;
 6. *iwet*.

Category B:

- ien*
iwen
ĩen
iwěn.

Somewhat more complicated is cat. F, being the *-t* and *-d* correspondence to cat. C. The principal words are:

- 1 ○ 2 既受俊 3 卒沒。4 對對退潰蒼妹內。5 仡。
 6 氣堅。7 鬱用弟拂。8 貴謂渭蔚味。
 9 麗。10 利泣肄四駟似比紕寐界。
 11 出卒述馱率律。12 類遂椹稔醉萃諄瘁穗。
 13 擊。14 戾棣淥。15 閔。16 惠噫。17 頁。18 屈
 19 滑。20 簣。21 暨。22 棄器 23 橘。24 匱季悽位。

- | | |
|--|---|
| 1. not represented; | 2. had Anc. <i>-di'</i> : <i>kái</i> etc.; |
| 3. had Anc. <i>-uat</i> : <i>tsuat</i> etc.; | 4. had Anc. <i>-uái'</i> : <i>tudi</i> etc.; |
| 5. had Anc. <i>-iat</i> : <i>ngiat</i> ; | 6. had Anc. <i>-jei'</i> : <i>k'jei</i> etc.; |
| 7. had Anc. <i>-iuet</i> : <i>'iuet</i> etc.; | 8. had Anc. <i>jwei'</i> : <i>kjwei</i> etc.; |
| 9. had Anc. <i>-iét</i> : <i>liét</i> ; | 10. had Anc. <i>-i'</i> : <i>li</i> etc.; |
| 11. had Anc. <i>-iuét</i> : <i>ts'iuét</i> etc.; | 12. had Anc. <i>-wi'</i> : <i>ljwi</i> etc.; |
| 13. had Anc. <i>-iet</i> : <i>p'iet</i> ; | 14. had Anc. <i>-iei'</i> : <i>liei</i> etc.; |
| 15. had Anc. <i>-iwet</i> : <i>k'iwet</i> ; | 16. had Anc. <i>-iwei'</i> : <i>yiwei</i> etc.; |
| 17. had Anc. <i>-at</i> : <i>kai</i> ; | 18. had Anc. <i>-ai'</i> : <i>kai</i> ; |
| 19. had Anc. <i>-wat</i> : <i>γwat</i> ; | 20. had Anc. <i>-wai'</i> : <i>k'wai</i> ; |
| 21. had Anc. <i>-iét</i> : <i>k'iet</i> (family name); | 22. had Anc. <i>-i'</i> : <i>k'ji</i> etc.; |
| 23. had Anc. <i>-iuét</i> : <i>k'iuét</i> ; | 24. had Anc. <i>-wi'</i> : <i>kjwi</i> etc. |

Here again, as in categories D and E, the even numbers had the same Arch. finals as the odd numbers, but for the *-d* of the former and the *-t* of the latter. The *-d* has become *-i* and given falling tone. *-ax* is not represented in the *-t* series (1. *-at*) but only in the *-d* series: 2. *kəd* has become > *kái'* (cf. the *-g* category, where 來 Arch. *lag* has become > Anc. *lái*, see Shī king Researches p. 124). In the same way 3. *-uat* (*-wət*) has been preserved, but 4. *-uəd* (*-wəd*) has become *uái'*. This is nicely confirmed by the char. 逮, which has both readings Anc. *t'uat* and *t'uái'* (Arch. *t'wət* and *t'wəd*).

Furthermore, that 6. *-jei'* is the *-d* correspondence to 5. *-iat*, and *-jwei'* to *-iuet* is proved by a large number of double readings and *hie sheng*, e. g. 气 both readings *k'iat* and *k'jei'* (Arch. *k'iat* and *k'iad*); 蔚 both readings *'iuet* and *'jwei'* (Arch. *'iuet* and *'iwəd*); 沸 both readings *p'iat* and *p'jwei'* (Arch. *p'iwət* and *p'iwəd*); 弗 *p'iat* phonetic in 沸 *p'jwei'* (Arch. *p'iwət* phon. in *p'iwəd*) etc.

In this group, just as in cat. C, Arch. *-iat* had a different evolution according as it was preceded by a guttural and labial or by a palatal and dental. Just as Arch. *k|iən* > Anc. *k|iən*, but Arch. *ŋ|iən* > Anc. *ts'ien* (see cat. C above), in the same way line 5. Arch. *k|iət* > Anc. *k|iət*, but 9. Arch. *ŋ|iət* > Anc. *ts'iet*; and in the same way 6. Arch. *k|iəd* > Anc. *k|jei'* but 10. Arch. *ŋ|iəd* > Anc. *ts'i'*. And the case of the *ho k'ou* words is exactly the same: 7. Arch. *k|iwat* > Anc. *k|iuat*, but 11. Arch. *ŋ|iwat* > Anc. *ts'iuét*; and 8. Arch. *k|iwəd* > Anc. *k|jwei'*, but 12. Arch. *ŋ|iwəd* > Anc. *ts'wi'*.

We can continue: just as, with long *i*, Arch. *-iən* > Anc. *-ien*, so here 13. Arch. *-iat* > Anc. *-iet* and 14. Arch. *-iad* > Anc. *-iei'*; 15. Arch. *-iwət* > Anc. *-iwet* and 16. Arch. *-iwəd* > Anc. *-iwei'*.

Finally, just as Arch. *-en* > Anc. *-an*, so 17. Arch. *-et* > Anc. *-at* and 18. Arch. *-ed* > Anc. *-ai*; 19. Arch. *-wet* > Anc. *wat*, and 20. Arch. *-wed* > Anc. *-wai*. And corresponding to Arch. *-ien*, *-iwen* we have here 21. Arch. *k|iət* > Anc. *k|iēt* and 22. Arch. *k|iəd* > Anc. *k|jī*; 23. Arch. *k|iwet* > Anc. *k|iūēt*, and 24. Arch. *k|iwed* > Anc. *k|jwī*.

We can sum up all this in the following table:

Category F.				Category C.	
Arch.		Anc.		Arch.	Anc.
1. (ət)	2. əd	1. (ət)	2. áì	ən	ən
3. wət	4. wəd	3. uət	4. uái	wən	wən
5. k iət	6. k iəd	5. k iət	6. k jēi	k iən	k iən
7. k iwət	8. k iwəd	7. k iūət	8. k jwēi	k iən	k iūən
9. i ət	10. i əd	9. ts iēt	10. ts i	i ien	ts iēn
11. i iwət	12. i iwəd	11. ts iūēt	12. ts wi	i ien	ts iūēn
13. iət	14. iəd	13. iet	14. iei	ien	ien
15. iwət	16. iwəd	15. iwet	16. iwei	o	o
17. et	18. ed	17. at	18. ai	en	an
19. wet	20. wed	19. wat	20. wai	wen	wan
21. iet	22. iəd	21. iēt	22. i	ien	ien, iēn
23. iwet	24. iwəd	23. iūēt	24. wi	iwen	iwen, iwēn

In this last category, F, we have had a large number of words with Anc. final *-ei* and *-i*, and I have shown that these are remnants of an Arch. *-d*. Now, the same finals, *-ei* and *-i*, occur in still another great rime category of the Shī, which it is necessary to take up for examination: cat. G, which is cat. 13 of Wang Nien-sun's = cat. 15 of Tuan Yü-ts'ai's (one half of this latter only). The principal words are:

- 1 哀。2 回崑隕推頰摧罪雷壘枚。
- 3 幾畿豈頤晞衣依。4 歸達煒葦圍韓畏威飛菲腓駢
霏微薇尾。
- 5 脂指旨坻蓄底邸抵鴟砥擇遲師尸尿矢視壽莠梲訛
資沃咨姊泚玼秣私死兕履毗臙匕妣悲美眉泥靡郿。
- 6 追水維唯鴛惟崔綏纍藟。
- 7 氏弟涕體妻妻樓淒齊齊濟濟濟濟濟濟涕犀福福鯉
鯉黎泥迷。8 咍偕階階階。9 懷懷。10 几飢祁伊。
- 11 葵駸遺。12 通瀟。13 證萎。

- | | |
|---|--|
| 1. had Anc. - <i>di</i> : - <i>di</i> ; | 2. had Anc. - <i>u^hdi</i> : <i>y^hu^hdi</i> etc.; |
| 3. had Anc. - <i>xi</i> : <i>k^hxi</i> etc.; | 4. had Anc. - <i>w^hxi</i> : <i>k^hw^hxi</i> etc.; |
| 5. had Anc. - <i>i</i> : <i>tsi</i> etc.; | 6. had Anc. - <i>wi</i> : <i>ts^hwi</i> etc.; |
| 7. had Anc. - <i>iei</i> : <i>ts^hiei</i> etc.; | |
| 8. had Anc. - <i>ai</i> : <i>k^hai</i> etc.; | 9. had Anc. - <i>wai</i> : <i>y^hwai</i> etc.; |
| 10. had Anc. - <i>i</i> : <i>k^hi</i> etc.; | 11. had Anc. - <i>wi</i> : <i>g^hwi</i> etc.; |
| 12. had Anc. - <i>ie</i> : <i>ts^hie</i> etc.; | 13. had Anc. - <i>wie</i> : <i>xi^hwie</i> etc. |

The interpretation of this category might seem to be very simple: nearly all the words end in -*i* and thus rime, and we could, for that matter, suppose it to represent words with original, Archaic final -*i*. But the question is in fact infinitely more complicated and necessitates an extensive investigation.

In the discussion of cat. C above I have purposely left out a few curious rimes, in which Anc. -*ən* rimes with Anc. -*xi*, e. g. 晨 : 輝 : 旂 Anc. *ts^hien* (< Arch. *d^hien*): *xi^hw^hxi*: *g^hxi* (ode T'ing liao); 芹 : 旂 Anc. *g^hien*: *g^hxi* (ode Ts'ai shu). These cases are all the more interesting since 輝 *xi^hw^hxi* has for phonetic 軍 Anc. *k^huən* (with -*n*) and 旂 has for phonetic 斤 *k^hien*. They naturally call to mind cases with other vowels in which similarly words with -*n* have riming or *hie sheng* connections with words ending in vowel, e. g. 儺 Anc. *nā* with phon. 難 *nān*; 幡 *b^huā* with phon. 番 *b^huəm* and riming with 翰 *yān* (Yi king, kua 22); 𪛗 *d'ā* with phon. 單 *tān* and riming with 𪛗 *yān* (Tso chuan, Sūan 2nd year). In all these cases it is very natural to think of nasalization phenomena, so that certain -*ān* have become -*aⁿ* > -*ā*, certain -*ien* have become *iⁿ* > *ieⁿ* > *xi*. I suggested this in this Bulletin, vol. I, p. 182, and the same has been proposed by Prof. Lin Yü-t'ang in his Yü yen lun ts'ung pp. 82 ff.

The theory would purport that 儺 was originally **nān*, 幡 **b^huān* and 𪛗 **d'ān*, and that 旂 was originally **g^hien*, which by nasalization became Anc. *nā*, *b^huā*, *d'ā* and *g^hxi* respectively. But if so, we have to answer the question: how did this nasalization work? If 單 was Arch. *tān* and has always kept its -*n* (Anc. *tān*, Peking *tan*), how could 𪛗, if it was Arch. **d'ān*, get its -*n* eliminated by nasalization and become Anc. *d'ā* (Peking *t'o*)? How could they develop differently? In the same way, if both 旂 and 芹 were Arch. *g^hien*, how could the former become Anc. *g^hxi* (Peking *k'i*) and the latter Anc. *g^hien* (Peking *k'in*)? There is no possibility of this within the same line of the language. If it were so, it must be due to a mixing of dialects. Whereas -*n*, in the main line of the language, that of the Shī king and the *hie sheng* characters, High Chi-

nese, was preserved and lived down to Ts'ie yün time, it has been nasalized in one or several dialects which were on the side of (parallel with) the High Chinese: from this side-track dialect, so to speak, a few forms like 𪛗 *d'á*, 旂 *g'jɛi* have then penetrated into High Chinese and there ousted the regular forms **d'án*, **g'jɛn* for these words and taken their place. Thus, in Ancient Chinese (Ts'ie yün) we have obtained 𪛗 *d'á* (dialectal loan word) but 單 *tán* (regular form), 旂 *g'jɛi* (dialect form) but 芹 *g'jɛn* (regular form).

In principle, there would be nothing against such an explanation. We could find numerous parallels in other languages. In French, for instance, we have the words *cage*, *canevas*, *caillou*. In High French they should properly read *chage*, *chanevas*, *chaillou* according to the regular phonetic laws of that language, and such forms with *ch-* have really existed earlier, but have been ousted, in High French, by the dialect forms (Picardie, Normandie) *cage*, *canevas*, *caillou*. Again, in High Swedish, the words *spär* (*spör*), *lén*, *stråk* (*strök*), *påse* (*pöse*) should regularly have been *spörr*, *lënn*, *strökk*, *pösse* (cf. *börr*, *tënn*, *lökk*, *mösse*), but have got long vowels because they are forms loaned from other dialects than the one which is the regular basis of High Swedish. The same phenomenon can be observed in Pekinese in certain sporadic cases. 孕 and 貞 should regularly give Pek. »ying» and »cheng», but they are pronounced »yün» and »chen» through influence of some dialect in which *-ng* > *-n*. And certain Archaic words have similarly jumped over into Anc. Chin. categories where they should not properly belong: 生 Arch. *səng* should be Anc. *seng* (Kuang yün rime 13) but was really *smg* (Kuang yün rime 12); 川 Arch. *i'iwən* (cat. C above) should give Anc. *ts'i'uən* (Kuang yün rime 18) but has given Anc. *ts'iwän* (Kuang yün rime II, 2). It is here always a question of sporadic irregularities.

For a long time, indeed, I have imagined that this solution was the correct one. But the longer I have studied the question, the more I have become convinced that it must be wrong. I have gradually been brought to an opinion which approaches that expressed by W. Simon (Zur Rekonstruktion der altchinesischen Endkonsonanten II, p. 8), though it is by no means identical.

It is necessary here to make a survey of all the most important cases of interchange of final *-n* with final vowel in Arch. rimes, hie sheng characters and kia tsie. I limit the list to those cases that are pre-Han or Han; later examples out of the Ts'ie yün and Tsi yün, concerning words not attested before Liu ch'ao time of course prove nothing about Archaic Chinese. The readings given in this survey are all in Anc. Chinese.

1 儻 2 左 3 難 4 瘴 5 麗 6 龜 7 疹 8 嘽 9 揣 10 揣 11 團
 12 搏 13 瑞 14 瑞 15 溫 16 戰 17 縵 18 造 19 瑞 20 端 21 擊 22 嬰
 23 嬰 24 嫵 25 嫵 26 裸 27 濯 28 果 29 番 30 嘽 31 翰 32 瞻 33 翰
 34 睥 35 碯 36 碯 37 播 38 盤 39 靦 40 洗 41 殄 42 烜 43 燬 44 火
 45 解 46 販 47 駭 48 敦 49 遺 50 摧 51 鏐 52 韋 53 慙 54 傲 55 灑 56 燿
 57 雷 58 威 59 賁 60 變 61 辰 62 屯 63 巨 64 堆 65 翠 66 揮 67 輝 68 輝
 69 暉 70 晨 71 暉 72 暉 73 暉 74 暉 75 暉 76 暉 77 暉 78 暉 79 暉 80 暉
 81 偕 82 通 83 近 84 微 85 遠 86 祈 87 祈 88 祈 89 祈 90 祈 91 祈 92 祈
 93 衣 94 妻 95 私 96 怨 97 至 98 圻 99 畿 100 垠 101 沂 102 衣 103 冠 104 殷
 105 匪 106 分 107 洗 108 西 109 栖 110 棲 111 茜 112 晒 113 西 114 施 115 先 116 施
 117 駸 118 殷 119 辰 120 巡 121 戰 122 嶺 123 民 124 門 125 安 126 眠 127 閒 128 洒 129 殄 130 汎 131 犀
 132 寅 133 北 134 死 135 伊 136 璽 137 祇 138 振 139 震 140 櫓 141 禮 142 選
 143 兕 144 先 145 還 146 泚 147 滌 148 鮮 149 水 150 弟 151 準 152 悲 153 門。

- I. This is the already mentioned 1. *ná* which in Shī (Chu kan) rimes with 2. *tsá* and yet has *nán* as phonetic.
- II. The word *ná* 'ample' (Shī, Si sang) is written (kia tsie) 3. *nán*.
- III. The word 4. has two Kuang yün readings *tán* and *tá*; phonetic *tán*. Rimes as *tán* in Shī, Pan.
- IV. Is the already mentioned 5. *d'á* (Ts'ie yün but not Kuang yün has an alternative reading *d'án*!) with phon. *tán* and riming with 6. *ngiwm* (Li ki, Yüé ling).
- V. 7. Ts'ie yün *t'á* and *t'án* (T'ang yün also *tá*, not in the Ts'ie yün); has phon. *tá*. Shuo wen quotes Shī, Sī mu, as 7.; the Mao version has 8. *t'án*.
- VI. 9. *tuá*, Shuo wen 'a horse whip', phon. *tuán*. No pre-Han text example.
- VII. 10. Anc. *ts'wiq* 'to measure', phon. *tuán*. Used kia tsie for 11. *d'uan* in a *fu* written by Kia Yi († 168 B. C.) in Ts'ien Han shu (k. 48, p. 3 a). Kuo P'o, comm. to Fang yen, reads it *xiwän*.
- VIII. 13. *xiwq*, phon. *tuán*.
- IX. 14. *ts'wiq*, phon. *tuán*, rimes (Shī, Siao yüan) with 15. *uwm* and 16. *ts'ian*, and (Chuang-tsī, Ts'i wu lun) with 17. *muán*. For 14. in Meng-tsī II (Legge p. 187) the Ting version has 18. read *ts'iwän*.
- X. 19. Yü p'ien *tuán* and *tuá*, phon. *tuán*; Shuo wen says »read like 20. (*tuán*)».
- XI. 21. *b'uan* and *b'ud*, phon. *puán*.

- XII. 22. *b'uân* and *b'ud*; it occurs in the bisyllabic expression 23. *b'uân-sân* (Sī-ma Siang-ju, Tsī-hū fu, Wen süan 7, p. 14 a; Li Shan indicates the reading *b'uân*), but this is evidently the same as Shī (Tung men chī fen) 24. *b'ud-şa* (so also Er ya).
- XIII. 25. *kuân* 'libation'. All ancient commentators (Mao Heng, Hū Shen, Cheng Chung, Cheng Hūan) define it as 26. *kuân* 'libation', and it must be etymologically cognate to this; yet it has 27. *kuá* as phonetic.
- XIV. 28. *b'iwon* 'a track', *p'iwon* 'a turn', *b'uân*, *p'uân*, *b'ud* various place-names, *puá* 'courageous'. In Shī (Sung kao) it rimes with 29 *tán*, *yán*, *xiwón*.
- XV. 30. *b'ud*, *puá*. Rimes in Yi king (kua 22) with 31. *yán*, in Tso chuan (Süan 2) with 32. *yán*.
- XVI. 33. *b'uân* a place-name, *puá* 'stone used for arrow point', cf. 34. *puá*.
- XVII. 35. *puá*; rimes in Kuan-tsī (Tī tsī chī, chapter 59) with 36. *b'uân* (the present text is corrupted by adding a word which spoils the rhythm and should be eliminated).
- XVIII. 37. *luá*, phon. *luân*.
- XIX. 38. *muá*, has phon. *miän* and rimes in Shī (Sin t'ai) with 39. Arch. *d'im*; there was, however, also a reading *muân* given by the T'ang commentator Ting Kung-chu (*ap.* Sun Shī) to Meng-tsī II (Legge p. 207), and already by Kuo P'o († 324 A. D.) in his comm. to Fang yen (k. 3, p. 4 a).
- XX. 40. *xiwón* 'to dry', Yi king (Shuo kua); *xiwíq* 'fire', Chou li (Sī kuei shī), so read in Kuang yün and King tien shī wen. Both build on Cheng Hūan, who says it is read like 41. (the Ts'i dial. word for 42).
- XXI. 43. *tsiq* (Kuang yün, Yü p'ien, King tien shī wen to Li ki T'an kung, and Tsī lin *ap.* King tien shī wen). Phonetic *tan*. Shuo wen gives a variant 44. with phon. *xién*.
- XXII. 45. *niei*, phon. *nán*.
- XXIII. 46. frequent in various readings and meanings (many of them *kia tsie*): *tuán*, *tsiüén*, *t'uán*, *d'uán*, *tuá*. In Shī (Pei men) it rimes with 47. *jwi*, *ts'udi*; it is then read *tuán* by Mao Heng, *tuá* by Cheng Hūan.
- XXIV. 48. *xiüén* (Arch. *diwón*), *tuán* 'a kind of bell'; *d'udi* 'butt of a spear'. As *d'udi* it rimes in Shī (Siao jung) with 49. *g'iuán*.

- XXV. 50. *d'uđi* with phon. 46. *tuən*. In Sung Yü, Feng fu (Wen süan 13, p. 2 b), it occurs in a bi-nom 51. *d'uđi- yuən*, which I suspect should be read *d'uən-yuən* (after the pattern of innumerable such bi-noms) in spite of the *d'uđi* gloss of the commentary.
- XXVI. 52. *t'uən* and *t'uđi*. Rimes in Shī (Ts'ai k'i) with 53. *luđi*, *'jwgi*.
- XXVII. 54. has a series of Anc. readings: *pji*, *b'jwgi*, *b'iuən*, *p'iuən*, *b'uən*. To the *pji* of Yi king (kua 22) Cheng Hüan (*ap.* King tien shī wen) says: »it means 55. *p'ian*», which is evidently a phonetic gloss (the sense explained by an approximate homophone). In the reading *puən* it rimes with 56. *éiən* (Arch. *điən*) in Tso chuan (Hi 5).
- XXVIII. 57. *i'iuən* serves as kia tsie for 58. *tuđi* in Chuang-tsī (Chī lo, last section).
- XXIX. 59. *xiwgi*, phon. *k'iuən*.
- XXX. 60. *yuən* 'curved handle of a plough'; *xiwgi* 'a clothes-peg'.
- XXXI. 61. *xiwgi* 'light, brightness' rimes in Shī (T'ing liao) with 62. *éiən* (Legge translates it 'smoke' and reads it »*huān*» i. e. Anc. *xiuən* and Couvreur translates 'fumée' and reads *hiun* i. e. Anc. *xiuən*; they both have followed Chu Hi, who has fabricated a »poetical» reading; Mao Heng says it means *kuang* 'brightness', indicating the ordinary sense and reading of the word; Lu Tê-ming underlines this by saying: »read 63. *xiwgi*). Other readings given by Kuang yün and Tsi yün are *xiuən*, *xiwən*, *yuən*, for which, however, there are no pre-Han examples. But it is used as kia tsie for 64. *jiuən* in Chou li (Shī tsin) and for 65. *jiuən* in Li ki (Tsi t'ung).
- XXXII. 66. *ngudi* and 67. *'jwgi* rime in Shī (Ku feng) with *'iwm*.
- XXXIII. About 68. *kiwen* Shuo wen says: »it is read like 69. *jwgi*.
- XXXIV. 70. *kjwgi* rimes in Shī (Tung shan) with 71. *san* (not recognized by Tuan Yü-ts'ai, but by both Wang Nien-sun, Kiang Yu-kao and Chu Tsün-sheng).
- XXXV. 72. *kai* and 73. *ńzi* both rime in Shī (Ti tu) with 74. *g'ien*.
- XXXVI. 75. *mjwgi* rimes in Ta Tai li (Wu ti tê) with 76. *jiwm*.
- XXXVII. 77. *g'jgi* 'to pray' has phon. *kien*.
- XXXVIII. 78. 'name of a herb' read *g'jgi* and *g'ien*, has phon. *kien* and is used as kia tsie on the one hand for 77. *g'jgi* 'to pray' (so often in bronze inscriptions), on the other hand for 79. *kien* in Chang Heng, Si king fu (Wen süan k. 2, p. 7 a).

- XXXIX. 80. *g'jɛi* has phon. *kjən* and rimes in Shī (T'ing liao) and in Tso chuan (Hi 5) with 62. *zjēn*, and in Shī (Ts'ai shu) with 81. *g'jən*.
- XL. 82. *g'jɛi*, which in Shī (Shī jen) rimes with 83. *'jɛi*, *ts'iei*, *si*, has *kjən* as phonetic and is used as *kia tsie* in Li (T'an kung) either for 84. *k'en* (so acc. to Lu Tê-ning) or rather for 74. *g'jən* (so Chu Tsün-sheng, based on Cheng Hūan's gloss: «equal to 85.»), and also as *kia tsie* in Chou li (K'ao kung ki, Chou jen) for a word 'strong' which Cheng Chung (1 st c. A. D.) reads *k'en*.
- XLI. 86. *g'jɛi*, id. with 87. *g'jɛi*, is also used for 88. *ngjən* and has phon. *kjən*.
- XLII. 89. *ngjɛi* has phon. *kjən* and is used as *kia tsie* for 88. *ngjən* (Ts'ien Han shu, Sū chuan).
- XLIII. 90. *'jɛi* rimes in Ch'u ts'i (Pu kü) with 91. *kuən*.
- XLIV. 92. Arch. *'jən* is said by Cheng Hūan (comm. to Li Ki, Chung yung) to be read like 90. *'jɛi* by the people of Ts'i (cf. Lin Yü-t'ang, Yü yen lun ts'ung).
- XLV. 93. *pjwɛi* is used as *kia tsie* for 94. *pjwən* in Chou li (Chung tsai) — so already acc. to Cheng Chung (1st c. A. D.).
- XLVI. 74. *g'jən* is used as *kia tsie* for 80 (77) *g'jɛi* in Li ki (Tsi fa, Couvreur p. 259) — so already acc. to Cheng Hūan.
- XLVII. 95. *sien* and *siei*; phon. *sien*.
- XLVIII. 96. *siei* 'bird's nest' = 97. *siei*. Applied as *kia tsie* to *siei* 'west'. But it is phon. in 98. *ts'ien* and 99. *sjēn*. For the name 100. «Si shī» Mei Sheng in his Ts'i fa (Wen süan 34, p. 5 a) writes 101. «Sien shī», and the T'ang commentator Li Shan in his note to this passage quotes Chan kuo ts'ê (Ts'i ts'ê 4) so as to show that his Kuo ts'ê version had «Sien shī». 96. *siei* rimes, on the one hand in Shī (Liu yüe) with 102. *siei*, *g'jwi*, on the other hand with various *-n* words: in Shī (Sang jou) with 103. *'jən*, *zjēn*; in Li ki (Tsi i) with 104. *zjwēn*; in the Yi lin (3: 1, 11: 54, 12: 37, 12: 51, 13: 7, 13: 34, 27: 14, 32: 54, 37: 54, 43: 12, 51: 8) with a long series of *-n* words (105. *tsjān* etc.).
- XLIX. 106. *šai* 'to sprinkle', *siei* 'to wash'. In the latter reading it is identical with 95. above (*siei*, *sien*). And the character is used as *kia tsie* for various other words in *-n*: *sien* 'respectful' (Li ki, Yü tsao), *sien* and *siei* 'scared' Chuang-tsī, Keng-sang-ch'u). In Shī (Sin t'ai) Lu Tê-ming reads it *ts'udi*, but it rimes with 107. *d'ien*. *šai* 'to sprinkle' should be compared with 108. *sjēn*, *sien* 'to sprinkle'.

- L. 109. *siei*, phon. *sien*.
- LI. 110. is read both *ien* and *i* by Ts'ie yün, by Yü p'ien and by Lu Tê-ming (Shang shu yin i 1, 3 a). 110. does not occur as a rime, but it is phonetic in various words ending in *-n*.
- LII. 111. *b'ien* and *b'ji* (Yü p'ien, Kuang yün, King tien shī wen). It rimes in Lao-tsi with 112. *si*.
- LIII. 113. *i* is used as *kia tsie* for 114. *ien* in the chapter Hung fan of Shang shu on the stone classics of the Tung Han Hi-p'ing period.
- LIV. For 115. *tsi* in Li ki (Nei tsê, Couvreur p. 666) Cheng Hsün gives the variant 116. *tsien*, and the same character in Shang shu (Wu yi, Couvreur p. 291) is rendered by 117. *tsien* by Sī-ma Ts'ien (Chou kung shī kia).
- LV. Instead of the char. 116. (var. 117) *tsien* in Yi king (kua 32) Shuo wen cites 118. *tsi*.
- LVI. 119. *liei* rimes in Li ki (Li yün) with 120. *siwän*.
- LVII. 121. *siei* rimes in Ch'u ts'ī (Chao hun) with 122. *sien*, *ywan*.
- LVIII. 123. *ts'ī* and 124. *mjiq* rime in Shī (Sin t'ai) with 125. *siän*.
- LIX. 126. *swi* and 127. *d'iei* rime in Shī (Mien shuei) with 128. *tsiüen*.
- LX. 129. *pjiwi* rimes in the Yi lin (64: 5) with 130. *muän*.

We see that the contacts of *-n* words with words ending in vowel are quite numerous, and the dialectal nasalization explanation becomes *eo ipso* somewhat dubious. But it becomes all the more so if we examine some of these cases more closely; the theory is indeed quite hopeless.

In the first place we should have to have recourse to *two* different nasalization phenomena:

a). In cases like XXXIX 旂 *g'iei* with phon. 斤 *kien* and riming with 晨 *ien*, Arch. *ien*, we should have to say that the Anc. *-i* word 旂 *g'iei* had *-n* originally: **g'ien* and therefore got its phonetic 斤 and rimed with the *-n* word *ien*. But dialectally it developed **g'ien* > *g'ien* > *g'iei*. In other words, a nasalization which did *not* exist in the Shī king language, nor in the *hie sheng* language nor in the main line of the later High Chinese, occurred dialectally, and from this unknown dialect penetrated (through a certain number of loan words) in the Ts'ie yün language.

b). On the other hand, in cases like LVIII: 泚 *ts'ī* riming with 鮮 *siän*, we should have to suppose that it was the Anc. *-n* word *siän* which in some Archaic

dialect had been nasalized: *sĭän* > *sĭäⁿ* > *sĭäi* and therefore could — in a dialectally coloured Shī ode — rime with the -i word 泚 *ts'i*. In other words: a Shī ode would have revealed to us a nasalization that has left no trace whatever in later times and has not been mirrored in the Ts'ie yün language.

The necessity for two different nasalization theories is already very disturbing; and we should furthermore have to operate with the whole transitions *ien* > *iäⁿ* > *iĕi* and *än* > *äⁿ* > *ä* dialectally already in Shī king time. Whereas 旂 still had its -n: **g'ien* in a dialect which is the base of a rime like 旂: 芹 *g'ien* (Shī, Ts'ai shu), the originally homophonous 頤 **g'ien* would already have become *g'ĕi* in another dialect which is the base of a rime like 頤: 衣 *ĕi*: 妻 *ts'iei* (Shī, Shī jen). Similarly 西 (XLVIII) would have been **sien* in the dialect of the ode Sang jou, but *siei* in the dialect of the ode Liu yüe. Whereas 瞿 was **d'än* in a dialect which is the base of the hie sheng character (phonetic 單 *tän*) and of the rime 瞿: 龜 *ngiwm* in Li ki, Yüe ling, 儺 **nän* would have been **nä* already in Shī time in the dialect of the ode Chu kan (riming with 左 *tsä*). All this is extremely unlikely.

In the third place — and worst of all — it is very difficult to imagine the nature of a dialectal nasalization which could explain the -i: -n contacts listed above. It is all very well to say that 鮮 *sĭän* had become dialectally *sĭäⁿ* > *sĭäi* in order to rime with 泚 *ts'i*. But what about 山 *šan* (Arch. *sän*) riming with 歸 *kjwĕi*, or 冠 *kuän* riming with 衣 *ĕi*? Here we could not very well postulate dialectal transitions *sän* > **sai*, > *kuän* > **kuäi*. And even if we were so bold, it would lead to impossible consequences; for if *kuän* > dial. **kuäi*, in order to rime with *kjwĕi* (with original -i), how could 儺 **nän* > dial. **nä* in order to rime with 左 *tsä*, and not *näi*? All this is plainly impossible.

We have, then, to abandon the nasalization theory as a means of explaining the totality of these -i: -n contacts and search for other ways.

We could, in the next place, imagine the possibility that we have not to do with a dialectal phenomenon but that 旂 was a true Arch. -n word (since it had *kien* as phonetic and rimed with -n words) and yet in some way different from 近 *g'ien*; this, then, would explain why the former has become Anc. *g'ĕi* and the latter Anc. *g'ien*. A glance at the cases listed above, in which there is contact between -n words and vowel-ending words, convinces us that it cannot have been a question of the vocalism — there are all types of vowels, all of which simultaneously occur in words with -n preserved to this day. Nor can it have been a question of tone.

For certain words it might be tempting to assume a palatalized $-n$: 旅 $*g'iən$: 近 $g'iən$. This would explain very nicely why $g'iən$ has become $g'jɛi$ but $g'iən$ kept its $-n$: $g'iən$. On the other hand it would furnish a passable explanation why a supposed 願 $*g'iən$ could rime with a 衣 $jɛi$ — because of its yodized (*i*-tasting) final $-n$. But we realize immediately the impossibility of this explanation. It would explain only a few cases. It could never be applied to cases like 山 Arch. $sān$ riming with 歸 $kjwɛi$, or 匪 $pjwɛi$ kia tsie for 分 $pjwən$, for we cannot suppose a palatal $-n$ in $sān$ and $pjwən$, which have their $-n$ preserved in Ts'ie yün and down to our time; nor would it be applicable to cases like 儺 $nā$ (< $*nān$?) riming with 左 $tsá$, 幡 $b'uá$ riming with 畔 $yán$. And it is obvious that no explanation is plausible which does not solve *all* these contact problems, which are certainly connected and must have a common explanation. It would, moreover, be very bold to construct an Archaic antithesis 旅 $g'iən$: 近 $g'iən$, for then we should have to find a reason why $-n$ in the one case was »mouillé»: $-n$, in the other not: $-n$; simply to say that this is due to unknown earlier phenomena (in Proto-Chinese) would be very unsatisfactory.

We could, finally, imagine that in all these cases (I—LX) the member ending in Anc. vowel has had an Arch. $-n$, but an $-n$ that was weaker than $-n$ that was preserved: 旅 $g'iən$ (short $-n$): 近 $g'iən$ (long $-n$), 儺 $nān$ (short $-n$): 難 $nān$ (long $-n$), etc. But this would be, again, to construe a difference imputable to Proto-Chinese phonology, of which we know nothing, which is in itself risky, as just stated. Moreover, since the various $-i$ words of I—LX above freely rime with the words of the entire Shī rime cat. G above, we should have to construe a weak final $-n$ in the whole of this category; and this is absolutely impossible, for then we could never explain why this category in normal cases is well distinguished from our cat. C ($-ən$ group) above.

We see that all these tentative solutions fail. We cannot arrive at a satisfactory explanation so long as we insist upon all these words of types 儺, 幡 $b'uá$, 旅 $g'jɛi$ etc. having really an Arch. $-n$, which has been lost in one way or another. We shall have to start at another end and look more closely into the big group of words ending in $-i$ (cat. G) and examine whether their $-i$ cannot have represented something else than $-n$ or $-i$ in Arch. Chinese.

Experience from the guttural groups has taught us that Arch. final $-g$ has to a large extent dropped and given rise to $-i$, e. g. 來 $læg$ > $lái$, 戒 keg > kai , 子 $tsiæg$ > tsi , 賄 $xwæg$ > $xuəi$ > $xuái$. Similarly $-d$ has become $-i$, as described above (categories D, E, F): 釐 $liad$ > $liái$, 釐 $l̥iət$ > tsi , 釐 $g'ád$ > $yái$ etc. When we now, in our

present *-i* group (cat. G), find words of the types *-äi*, *-ai*, *-xi*, *-iei* etc., it is very natural to suspect that all these *-i* are vocalizations of some final consonant; that would only be in accordance with the well-attested general evolution of the Chinese language. The frequent interchange with words in *-n* shows that in such a case it must be a question of some kind of *dental* final.

This is underlined by the fact that besides the numerous contacts between *-i* and *-n* words in rimes, hie sheng and kia tsie adduced above, also an etymological connection between *-i* and *-n* words can be traced with a great amount of certainty in many cases:

- a. Between 衣 *'jxi* 'clothes' and 隱 *'iæn* 'to cover, conceal';
- b. Between 屛 *'jxi* 'a screen', 翳 *'iei* 'a screen' and 隱 *'iæn* 'to cover, conceal';
- c. Between 依 *'jxi* 'to lean on' and 隱 *'iæn* 'to lean on' (common expression: *nyin ki* 'to lean on a stool');
- d. Between 幾 *kjxi* 'near to' (common expression *ki hu* 'near to') and 近 *g'iæn* 'near to';
- e. Between 畿 *g'jxi* ('close quarters':) 'Royal domain proper' and 近 *g'iæn* 'near to';
- f. Between 饑 *kjxi* 'famine', 飢 *kji* 'famine' and 饑 *g'iæn* 'famine';
- g. Between 水 *swi* 'water' and 準 *tsiücn* (Arch. *iüwän*) 'a water level';
- h. Between 圍 *jwxi* 'to encircle, surround': and 運 *jüwän* 'to turn round';
- i. Between 緯 *jwxi* 'woof': and 緯 *jüwän* 'woof'.
- k. Between 飛 *pjwxi* 'to fly' and 奮 *pjüwän* 'to start flying'.

Here, still more than in the rimes, the hie sheng and the kia tsie, it comes out clearly that *-i* is the vestige of a lost dental.

When it now comes to determining the nature of this dental final, it will not do at all to pose, as W. Simon (*op. cit.*) does, the same dental here as in categories D, E, F: cases like 例 *liäi* which has 列 *liät* as phonetic (Simon writes 例 *liäd*, 旂 *g'jed*). They are absolutely different. In the *liäi* type (cat. D, E, F) — Arch. *-d* — there is an interchange with *-t* in rimes and hie sheng; here, in the 旂 *g'jxi* type (cat. G) there is an interchange with *-n* (cases I—LX above). The two types practically never mix.

In the *-i* words of our cat. G, the dental final cannot have been a *-t*, for as a rule they do not rime with ju sheng *-t*; we cannot suppose 旂 Arch. *g'jet* etc.

It cannot have been a *-d* for the same reason. The *-d* words are in cat. F, and rime frequently with the *-t* words, just as experience from other groups shows us that e. g. *-ak* and *-ag* rime quite freely. But with cat. F our cat. G here has

very few rime connections (see p. 24 below); on the other hand, the *-t*, *-d* words practically never rime with *-n* words, but we have just seen how our cat. G here has quite considerable *-n* connections. A *-d* is therefore just as much excluded as a *-t*; we cannot construe 旃 Arch. *g'jed* etc.

It cannot have been an *-n*. We have already discussed extensively why 旃 cannot have had an *-n*. To suppose 旃 **g'ien* (>*g'jɛi*) 禮 **lien* (>*liei*), 回 **g'uán* (>*yuái*) is impossible, for then we could never explain why we have Anc. 近 *g'ien*, 憐 *lien*, 丸 *yuán* with preserved *-n*.

In other words: the final dental cannot have been *-t* or *-d* or *-n*. What is then left? Evidently *-r*, *-l* and *-s*.

It is a remarkable fact that Tibetan, to which Chinese is undoubtedly cognate, has a great number of words of the types *-r*, *-l*, *-s*, e. g. *dur*, *dul*, *dus* etc. It is but reasonable to expect some corresponding word types in Chinese, and the only phonological group in which these can very well be suspected of lurking is our very category G of the Shī rimes, i. e. words ending in Anc. *-i*. I have stated elsewhere and on p. 1. above that I consider it premature to try to compare isolated Tibetan and Chinese words; the following cases are therefore not meant as positive identifications but only as examples of how it might turn out that Chinese words correspond to Tibetan *-s*, *-r* and *-l* words:

Tib. *g-nis* 'two' = Chin. 二 *ní* (W. Simon, Tibetisch-Chinesische Wortgleichungen 1930, p. 29);

Tib. *lus* 'body' = Chin. 體 *t'iei* (Simon p. 30; the phonetic of the char. is *liei*);

Tib. *bras* 'rice' = Chin. 米 *miei* 'rice' (Simon, p. 30);

Tib. *'p'ur* 'to fly' = Chin. 飛 *pjwɛi*;

Tib. *k'or* 'to return' = Chin. 歸 *kjwɛi*;

Tib. *ser* 'finger' = Chin. 指 *tɕi*;

Tib. *ts'il* 'fat, grease' = Chin. 脂 *tɕi* (Simon, p. 27).

A possible indication of a final *-r* is to be found in the word 師 Anc. *ɕi*, belonging to our cat. G, which was applied in early Han time to denote the 'lion', an animal earlier unknown in China. The question is interestingly linked with another loan word, designated by various words of the category. Pelliot has cleverly seen that all these forms point to an *-r*. He writes (T. P. XXVI, 1929, p. 141): »La boucle de ceinture en métal fut désignée d'un nom Hiong-nou, qui apparaît dans les textes chinois sous les transcriptions 鮮卑 *ɕiän-pjiɛ*, 師比 *ɕi pji* . . . 犀毗 *ɕiei-b'ji*, 犀比 *ɕiei-pji* . . . 私銀 *si-b'ji*; la forme originale du nom est inconnue . . . naturellement le nom . . . évoque spontanément celui des tribus *Sien-pi*

(écrit de même manière) . . . il n'est pas impossible que le vieux nom des Sien-pi survive dans les tribus 室韋¹⁾ *che-wei* des T'ang, et ceci supposerait un original ancien du type **Sārbi*, **Serbi* (par une coïncidence curieuse, le 師 de 師比 a servi de transcrire sous les Han un nom étranger du lion qui pourrait bien être aussi à -*r* finale et s'apparenter au persan *šēr*)*. Pelliot is very probably right about the final -*r* in these words. *ši* 'lion' would then transcribe an Iranian *šary*, according to what my friend Prof. G. Morgenstierne tells me.

If thus the words of our cat. G. ended primarily in -*r*, -*l*, -*s*, it is clear that in Shī king time they did not end, some of them in -*r*, others in -*l* and others again in -*s*. They all rime regularly and freely with each other, and there is no possibility of subdividing the category into smaller groups according to -*r*: -*l*: -*s*. Either all had -*r* or all had -*l* or all had -*s*.

The choice is not difficult. The final in question was -*r*. An -*s* in them all is easily excluded. In rimes like 旂 *g'jɛi*: 芹 *g'jən* we cannot pose a 旂 *g'jəs*, and 斤 *kjən* could not serve as phonetic in a 旂 *g'jəs*; nor could 匪 *piwɛi*, if it were an Arch. *piwəs*, serve as kia tsie for 分 *piwən*. -*r* and -*l* are equally possible from the point of view of rimes, hie sheng and kia tsie. A rime like 旂 *g'jər*: 芹 *g'jən*, a hie sheng like 旂 *g'jər*: 斤 *kjən* and a kia tsie like 匪 *piwər* for 分 *piwən* are passable; they are not good, and therefore only occur as exceptions (cases I—LX above), being indeed makeshift rimes and somewhat poor hie sheng and kia tsie, but still they might occur occasionally; -*l* would be equally good and equally bad as -*r*: *g'jəl*: *g'jən*, *piwəl*: *piwən*. But my decision for -*r* and against -*l* depends upon the fact that it is easier to imagine an evolution Proto-Chinese -*s* > Arch. -*r* (e. g. 二 *ńjəs* > *ńjər*) than P. C. -*s* > Arch. -*l* (*ńjəs* > *ńjəl*). The latter would go against all linguistic experience. The former is a common and well-known transformation. I need only recall the Germanic final -*s*, which regularly gives Old Icelandic -*r*: Got. *sunus*: Icel. *sunr*; and of the »rhotacism» in Latin (*genes-* > *gener-* in *generis*, *es* > *er* in the verb *esse*). Particularly suggestive, moreover, is the cognate language Tibetan, where there is sometimes an interchange of -*s*: -*r* as final consonant: *mdzes-pa* 'beautiful': *mts'ar-ba* 'beautiful'; *byus* 'misfortune': *byur* 'misfortune', etc.

I conclude, therefore, that the three Proto-Chinese types *a* | *s*, *a* | *l*, *a* | *r* have all become Arch. *a* | *r* and that the whole of our cat. G ended in -*r*.

That I am here on the right track seems to me to be confirmed, once we go back to the table of cat. G on p. 11 above and fill in the Arch. values. It turns

¹⁾ In T'ang time *šičō-jwɛi*.

out that this category with dental final, *-r*, forms an exact parallel to two other categories with dental finals, *-n* and *-t* (*-d*) studied earlier, categories C and F, and this I consider to be a strong corroboration:

Cat. G. (p. 11)		Cat. F. (p. 10)		Cat. C. (p. 10)	
Arch.	Anc.	Arch.	Anc.	Arch.	Anc.
1. <i>ər</i>	> <i>ái</i>	(<i>ət</i>), <i>əd</i>	> —, <i>ái'</i>	<i>ən</i>	> <i>ən</i>
2. <i>wər</i>	> <i>uái</i>	<i>wət</i> , <i>wəd</i>	> <i>uət</i> , <i>uái'</i>	<i>wən</i>	> <i>uən</i>
3. <i>k ǝər</i>	> <i>k ǝǝi</i>	<i>k ǝət</i> , <i>k ǝəd</i>	> <i>k ǝət</i> , <i>k ǝǝi'</i>	<i>k ǝən</i>	> <i>k ǝən</i>
4. <i>k ǝwər</i>	> <i>k ǝwǝi</i>	<i>k ǝwət</i> , <i>k ǝwəd</i>	> <i>k ǝwət</i> , <i>k ǝwǝi'</i>	<i>k ǝwən</i>	> <i>k ǝwən</i>
5. <i>ǝǝər</i>	> <i>tǝǝi</i>	<i>ǝǝət</i> , <i>ǝǝəd</i>	> <i>tǝǝǝt</i> , <i>tǝǝi'</i>	<i>ǝǝən</i>	> <i>tǝǝǝn</i>
6. <i>ǝǝwər</i>	> <i>tǝǝwi</i>	<i>ǝǝwət</i> , <i>ǝǝwəd</i>	> <i>tǝǝǝwət</i> , <i>tǝǝwi'</i>	<i>ǝǝwən</i>	> <i>tǝǝǝwən</i>
7. <i>ǝər</i>	> <i>iei</i>	<i>ǝət</i> , <i>ǝəd</i>	> <i>iet</i> , <i>iei'</i>	<i>ǝən</i>	> <i>ien</i>
8. <i>er</i>	> <i>qi</i>	<i>et</i> , <i>ed</i>	> <i>qt</i> , <i>qi'</i>	<i>en</i>	> <i>qn</i>
9. <i>wer</i>	> <i>wqi</i>	<i>wet</i> , <i>wed</i>	> <i>wqt</i> , <i>wqi'</i>	<i>wen</i>	> <i>wqn</i>
10. <i>ǝer</i>	> <i>i</i>	<i>ǝet</i> , <i>ǝed</i>	> <i>ǝǝt</i> , <i>i'</i>	<i>ǝen</i>	> <i>ǝen</i> , <i>ǝǝn</i>
11. <i>ǝwer</i>	> <i>wi</i>	<i>ǝwet</i> , <i>ǝwed</i>	> <i>ǝǝwət</i> , <i>wi'</i>	<i>ǝwen</i>	> <i>ǝwen</i> , <i>ǝǝwən</i>

This tallies beautifully as far as lines 1—11 of the G table on p. 11 are concerned. There are then but two lines left to be explained: 12. Anc. *ńǝǝǝ* and 13. *ǝǝwǝǝ*. The rime examples of these two lines are very few and have to be considered as exceptional, just as occasional confusions can occur between other regularly distinguished categories. Just as there are irregular rime contacts *a : ə* (categories A : C) e. g. in Shī, Ch'u ts'í: 嫫 *hán* (A): 愆 *k'ian* (A): 孫 *suən* (C); in Shī, Siao jung: 羣 *g'iwən* (C): 苑 *ǝwǝn* (A); so we have here some occasional *a : ə* contacts: line 12. Arch. *ńǝǝr*, *mǝǝr*, 13. Arch. *ǝǝwǝr*, *ǝǝwǝr* riming with *-ər* words.¹⁾

In connection with our table above it is the proper place here for reverting to the question of the Arch. final *-d* (see p. 7 above). That I have had to abandon my idea of *-t* in falling tone > *-i* and go back to my construction *-d* > *-i* of my Analytic Dictionary is just because the *-t* theory does not satisfy the general system of the Archaic language. I have shown earlier (following up

¹⁾ The phonetic series 爾 is very enigmatic. Itself Anc. *ńǝǝǝ* < Arch. *ńǝǝ*, the word had no final consonant. But on the one hand it is used as *kia tsie* for 耳 Arch. *ńǝǝg* and its derivate 彌 Anc. *mǝǝǝ* is used for 弭 *mǝǝǝ* < *mǝǝǝg*, which all points to final guttural; on the other hand it is phonetic in 滿 Anc. *ńǝǝi* < *ńǝǝr* and 通 Anc. *ńǝǝǝ* < *ńǝǝr* with dental final. To say, with Tuan Yü-ts'ai, that in Chou time it belonged to cat. 15 (*-r*) but in Han time to cat. 16 (*-g*) is a poor expedient. Very likely we have here several distinct series confused at a very early epoch.

ideas first advanced by W. Simon) that Arch. Chinese had both *-k* words and very large groups of *-g* words. It would be strange indeed if it possessed *-t* words and *-r* words but no *-d* words. All probability, then, speaks in favour of a *-d*, not *-t*, in the *-i* sections of categories D, E, F above. But probability is not the same as proof. I have obtained the proof in another way. I have stated above that cat. F. does *not* rime, as a rule, with cat. G, i. e. *-r* words. But there are some exceptions, and these are highly significant. I shall give some examples:

1 閔濟 2 至禮 3 利濟 4 利禮 5 致示死 6 隧哀階 7 至
濟死 8 比累水 9 棄飢 10 至視 11 至利視 12 利威指 13 位
氣威 14 痔磴 15 慄猷。

1. Shī, Tsai ch'ī; 2. Shī, Pin chī ch'u yen; 3. Yi, Hi ts'ī, hia; 4. Li ki, Fang ki; 5. Li ki, Ju hing; 6. Li ki, K'ü li; 7. Ch'u ts'ī, Kiu pien 6; 8. Sung Yü, Kao t'ang fu; 9. Shu, Shun tien; 10. Chuang-tsi, Chī pei yu; 11. Sün-tsi, Ch'eng siang; 12. Han Fei-tsi, Chu tao; 13. Han Fei-tsi, Kie Lao; 14. Kuan-tsi, Sin shu; 15. Sung Yü, Feng fu.

The words to the left of the colon belong to cat. F, those to the right to cat. G (*-r*). Now, the striking fact is that in one case only (15) have I been able to find a real *ju sheng -t* riming with *-r*. In all the other cases it is a question of the final dental, which was lost before Anc. Chinese, causing a falling tone, the dental which I had first interpreted as *-d* and later as *-t*. It is quite evident here that the former interpretation must be right. For if it had been a *-t*, there is no reason whatever why *-r* should rime more with *-t* than with *-t*. If, on the other hand, it was a *-d*, it is but reasonable that *-r* rimes more easily with *-d* than with *-t*. We can then well understand the cases above: as a rule neither *-t* nor *-d* rimed with *-r*; yet exceptional rimes *-d: -r* could sometimes occur, *-d* and *-r* being sufficiently similar phonetically, but hardly ever *-t: -r*.

These are the considerations that have forced me back to my original construction of *-d* (and of *-g* in 怕 etc.) in my Analytic Dictionary. It is true that it will then be necessary to find an explanation of the phenomena discussed in my Shī king Researches p. 120. I shall revert to that question on another occasion.

If we now, having arrived at a final *-r* as the solution of cat. G, go back to the cases I—LX on p. 13 above — it must be remembered that after all they are exceptional cases, makeshift rimes, hie sheng and kia tsie — we obtain the following results:

- I. 1. *nār* has phon. *nān* and exceptionally rimes with 2. *tsā* (an occasional contact *-ār: -d* is phonetically not very shocking; cf. the High

- English pronunciation of *far* with the *-r* brought out practically only when followed by a vowel).
- II. *nār* 'ample' is written *kia tsie* by 3. *nān*.
 - III. 4. double readings *tār* and *tān*.
 - IV. 5. *d'ār* and *d'ān* has phon. *tān* and rimes with 6. *ngiŵǎn*;
 - V. 7. *t'ār* and *t'ān* with phon. *tā* (cf. I above).
 - VI. 9. *twār* with phon. *twān*.
 - VII. 10. *t'īwar*, phon. *twān*, used *kia tsie* for 11., 12. *d'wān*, read *ḡiŵān* < *ḡiwan* by Kuo P'ō.
 - VIII. 13. *ḡiwar*, with phon. *twān*.
 - IX. 14. *t'īwar*, with phon. *twān*, riming with 15. *·wān*, 16. *t'ian*, 17. *mwān*. For *t'īwar* in Meng-tsi Ting reads 18. *t'īwan*.
 - X. 19. *twān* and *twār*, phon. *twān*, Shuo wen: »read like 20. *twān*.»
 - XI, XII, 21, 22. *b'wān* and *b'wār*, phon. *b'wān*.
 - XIII. 25. *kwān*, phon. 27 *kwār* (that this »*kuo*» series had all *-r* is confirmed by the fact that it does not rime in the *-ā* category (Tuan cat. 17)).
 - XIV. 28. *b'īwǎn*, *p'īwǎn*, *b'wān*, *p'wān*, *b'wār*, *pwār*.
 - XV. 30. *b'wār*, *pwār* riming with 31, 32 *g'ān*.
 - XVI. 33. *b'wān*, *pwār*. How this stands to 34. *pwā* is an intricate question; it must reasonably show that forms with lost *-r* must already have existed very early.
 - XVII. 35. *pwār*, riming with 36. *b'wān*.
 - XVII. 37. *lwār*, phon. *lwān*.
 - XIX. 38. *mwār*, *mwān*.
 - XX. 40. *ḡiŵǎn*, *ḡiwar*, 41. *ḡiwar*; observe that 42. must have had two Arch. readings: *ḡwər* in the Shī king (rimes regularly in cat. G) and *ḡwār* > T'sie yün *ḡuā* > Mand. »*huo*».
 - XXI. 43. *t'iar*, phon. *tān* (or 44. *ḡiēn*).
 - XXII. 45. *niər*, phon. *nān*.
 - XXIII. 46. *twān*, *t'iwān*, *t'wān*, *d'wān*, *d'wān*, *d'wār* (riming with 47. *g'īwer*, *ts'wər*).
 - XXIV. 48. *ḡiŵān*, *twān*, *d'wār* (*d'wār* riming with 49. *g'īwān*).
 - XXV. 50. *d'wār*.
 - XXVI. 52. *t'wān*, *t'wār*, riming with 53. *lwār*, *·īwār*.
 - XXVII. 54. *p'iar*, *b'īwār*, *b'īwān*, *p'iwān*, *b'wān*; *p'iar* explained (Cheng Hūan) by 55. *p'ian* (< *plian*).
 - XXVIII. 57. *t'iwān* *kia tsie* for 58. *twār*.

- XXIX. 59. *xiwər*, phon. *kjwən*.
 XXX. 60. *g'wən* and *xiwər*.
 XXXI. 61. *xiwər*, riming with 62. *ɖjən*, *kia tsie* for 64, 65. *gɿwən*.
 XXXII. 66. *ngwər* and 67. *'iwar* riming with *'iŋwən*.
 XXXIII. 68. *kjwən*. Shuo wen says: «read like *'iwar*» (a gloss based on an approximate phonetic resemblance).
 XXXIV. 70. *kjwər* riming exceptionally with 71. *sən*.
 XXXV. 72. *ker* and 73. *ɳjər* riming with 74. *g'jən*.
 XXXVI. 75. *mjwər* riming with 76. *gɿwən*.
 XXXVII. 77. *g'jər*, phon. *kjən*.
 XXXVIII. 78. *g'jər* and *g'jən*, *kia tsie* for 77. *g'jər* and for 79. *g'jən*.
 XXXIX. 80. *g'jər* with phon. *kjən* rimes with 62. *ɖjən* and 81. *g'jən*.
 XL. 82. *g'jər* rimes with 83. *'jər*, *ts'jər*, *sɿjər*, has phon. *kjən* and serves as *kia tsie* for 84. *k'ən* or more probably for 74. *g'jən*.
 XLI. 86. *g'jər* serves as *kia tsie* for 88. *ngjən*.
 XLII. 89. *ngjər* has phon. *kjən* and is used as *kia tsie* for 88. *ngjən*.
 XLIII. 90. *'jər* rimes exceptionally with 91. *kwən*.
 XLIV. 92. *'jən* is said by Cheng Hüan to be read, by the Ts'i people, like 90. *'jər*.
 XLV. 93. *pjwər* used as *kia tsie* for 94. *pjwən*.
 XLVI. 74. *g'jən* used as *kia tsie* for 80. (77.) *g'jər*.
 XLVII. 95. *sien* and *sjər* with phon. *sien*.
 XLVIII. 96. *sjər* = 97. *sjər*, *kia tsie* for *sjər* 'West', phon. in 98. *ts'jən* and 99. *sɿjən*. The *sjər* 'West' rimes with 102. *sjər*, *g'jwer*, and with 103. *'jən*, *ɖjən*, 104. *dz'jwən* etc.
 XLIX. 106. *ser* 'sprinkle', *sjər* 'to wash', *kia tsie* for *sien* 'respectful', *sjər* 'scared'. In Shī, Sin t'ai, read *ts'wər*, riming with 107. *d'jən*; cognate to 108. *sɿjən* 'to sprinkle'.
 L. 109. *sjər*, with phon. *sɿjən*.
 LI. 110. *ɖjən* and *ɖjər*.
 LII. 111. *b'jən* and *b'jər*, riming with 112. *sjər*.
 LIII. 113. *'jər*, used as *kia tsie* for 114. *'jən*.
 LIV. 115. *ɭjər*, has the variant 116. *ɭjən*.
 LV. 116. (117.) *ɭjən* has the variant 118. *ɭjər*.
 LVI. 119. *liər* rimes exceptionally with 120. *sɿwən*.
 LVII. 121. *sjər* rimes with 122. *sien*, *g'wən*.

LVIII. 123. *ts'iar* and 124. *mjar* rime with 125. *sian*.

LIX. 126. *šiwər* and 127. *d'iar* rime with 128. *liwən*.

LX. 129. *pjar* rimes with 130. *mwən*.

Though these rimes, *hie sheng* and *kia tsie* are exceptional, yet they are sufficiently numerous to show that the Chinese in Archaic times had a very strong feeling for the close affinity between *-n* and *-r* words. This was *not only* due to the phonetic similarity (a rime like *'iar : kwān* must be said to be phonetically very poor) but also and above all because they had numerous word pairs in *-n*: *-r* which they knew and felt to be cognate, two aspects of the same stem. Here we revert to the examples on p. 20 above:

- a) *'iar* 'clothes': *'ien*, 'to cover, conceal';
- b) *'iar* 'a screen', *'iar* 'a screen': *'ien* 'to cover, conceal';
- c) *'iar* 'to lean on': *'ien* 'to lean on';
- d) *kjar* 'near to': *g'ien* 'near to';
- e) *g'iar* 'close quarters, Royal domain proper': *g'ien* 'near to';
- f) *kjar* 'famine', *kier* 'famine': *g'ien* 'famine';
- g) *šiwər* 'water': *liwən* 'a water-level';
- h) *giwər* 'to encircle': *giwən* 'to turn round';
- i) *giwər* 'a woof': *giwən* 'a woof';
- k) *piwər* 'to fly': *piwən* 'to start flying'.

To these cases we can now add the following, out of our cases I—LX above, which are clearly such double aspects of the same stem:

- III, 4. *tār* and *tān* 'distressed';
- IV, 5. *d'ār* and *d'ān* 'iguana';
- V, 7. *t'ār* and *t'ān* 'exhausted';
- X, 19. *twār* and *twān* 'hanging ears of grain';
- XI, 21. *b'wār* and *b'wān* 'to eliminate';
- XIX, 38. *mwār* and *mwān* 'to defile';
- XX, 42, 41, 40. *xwər*, *xwār*, *xiwar* and *xiwān* 'fire';
- XXVI, 52. *t'wər* and *t'wən* 'in complete array';
- XXXVIII, 78. *g'iar* and *g'ien* 'name of a herb';
- XLVII, 95. *sir* and *sien* 'to wash';
- XLIX, 106. *ser* and *sičn* 'to sprinkle';
- LI, 110. *djar* and *dien* 'respectful';
- LII, 111. *b'iar* and *b'ien* 'female'.

The cases are sufficiently numerous to make a seemingly bold theory of an alternation $-r \sim -n$ within the same word stem plausible. Moreover, we find a very suggestive parallel to this in Tibetan, where there is a frequent alternation both of $-r \sim -n$, of $-l \sim -n$ and of $-r \sim -l$:

$-r \sim -n$:

sbur-ma 'chaff': *spun-pa*, *sbun-pa*, 'chaff';
gčer-ba 'bare, naked': *rjen-pa* 'bare, naked';
'byor-ba 'to arrive': *'byon-ba* 'to arrive';
dkor 'rare, precious': *dkon* 'precious thing';
gnyer-ba 'take pains with': *nyen-pa* 'to be pained, labour hard';
nyer-ba 'to tan, make soft': *mnyen-pa* 'flexible, soft';
star-ba 'to tie fast': *brtan-pa* 'firm', *gtan* 'to bar (a door)';

$-l \sim -n$:

p'ul 'complete, perfect': *p'un* 'complete, perfect';
rtsol-ba 'to be diligent, to endeavour': *brtson-pa* 'to be diligent, to endeavour';
'dral-ba 'to pull down, tear to pieces': *'dren-ba* 'to pull, tear out';

$-r \sim -l$:

dgar-ba 'to separate': *'gol-ba* 'to separate';
'byer-ba to give way, be removed': *'byol-ba* 'to give way, step aside';
gžor-ba, *gžer-ba* 'to weigh': *gžal-ba* 'to weigh';
k'al 'a burden, load': *k'ur* 'a burden, load';
'jur-ba, *'dzur-ba*, *'č'or-ba* 'to evade': *'jol-ba* 'to evade';
sbyor-ba 'to join, mix': *spel-ba* 'to join, mix';
'k'yer-ba 'to carry away', *'k'ur-ba* 'to carry': *skyel-ba* 'to carry away', *'k'yol-ba* 'to be carried'.

'k'or-ba, *k'yir-ba* 'to turn round': *'k'al-ba*, *'kel-ba* 'to twist, to spin', *'k'yil-ba* 'to twist';

sgor-ba 'to boil': *skol-ba* 'to boil'.

The reconstruction system of Archaic Chinese sketched here means that I have come to the conclusion that an $-i$ as final and principal vowel of an Arch. syllable did not exist at all; i (strong, vocalic) or $\dot{i}$ (short, consonantic) occurred exclusively as a »medial i «, a subordinate element inside the syllable, combined with other vowels. This means that on an important point I have to waive my objections (»Tibetan and Chinese«, TP 1931, p. 24 ff.) to W. Simon's reconstructions, in so far as he has concluded a final dental in our cat. G (though not the $-r$ at

which I have arrived). It might seem that I should then also have to give up my there expressed criticism of and objections to his general theories of the Sinitic final consonants. I have (*loc. cit.* p. 31, 32) adduced a great number of forms from all kinds of Sinitic languages for the word stems 'four' (四 Anc. *si*), 'to die' (死 Anc. *si*) and 'water' (水 Anc. *swi*) and shown that everything points to Sinitic roots ending in vowel, not in dental consonant; I have therefore objected to Simon's construing a Proto-Tibetan 四 *bzið* and 死 *sið* on the strength of Chinese forms in dental — a dental which I contested. Now, when because of rimes like 四 : 紕 : 界 (*si:*) *biəd* : *piəd* (the *-d* in the last two is certain for hie sheng reasons) I have to acknowledge the *-d* in 四 *siəd*; and when, for all the various reasons given above, I have to admit the *-r* in 死, it would seem to be a corollary that I accept Simon's Proto-Tibetan *bzið*, *sið* (or such-like, at least some dental final) which would mean a dental final in these words in Sinitic.

But in spite of appearances it does not. In my article just quoted I have given ample examples showing that we have to distinguish Chinese *-k* and *-t* words with primary (general Sinitic) *-k* and *-t*, e. g. 六 Arch. *liók* (p. 18), 八 *pwat* (p. 15) which have their *-k* and *-t* in the great majority of the Sinitic languages, and words with Chinese *-k* and *-t*, in which these *-k* and *-t* must be an innovation, some kind of suffix in one or several Sinitic languages but not primary and common to them all. As such examples I have given 百 Arch. *pāk* 'hundred', (p. 17), 日 *niēt* 'sun' (p. 19), 月 *ngiwāt* 'moon' (p. 21) — they all have typically vowel-ending Sinitic roots. It is just the same with the final *-g* in 九 *kiüg* 'nine', which must be a special Chinese feature, without correspondence in other Sinitic languages (Tib. *dgu* etc., *op. cit.* p. 36). Now, the words 'four' (Tibetan *bzi* etc.), 'to die' (Tib. *si* etc.) and 'water' (Tibetan *č'u* etc.) are typical vowel-ending Sinitic roots (*op. cit.* pp. 31, 32) and from the Arch. Chin. 四 *siəd*, 死 *siər*, 水 *siwər* I dare not, by any means, conclude any Sinitic dental finals. Their final consonants may be just as particularly Chinese as the *-t* in 日 *niēt* and 月 *ngiwāt*.

It should be emphasized that it is only because of the contrast with the well-known cases with real Sinitic *-k*, *-t* (六, 八), in which *-k* and *-t* do appear in a great number of Sinitic languages, showing that Sinitic *-k*, *-t* should not disappear in all languages except Chinese, that I refuse to see a primary Sinitic *-t* in cases like 日 *niēt*, 月 *ngiwāt*, and *-d* in 四 *siəd*. If it were not for that contrast, I would not deny the theoretical possibility of a primary Sinitic *-d* in the latter, having been dropped in all languages except Chinese. This would in itself be theoretically admissible. We must remember that — apart

from Chinese — for one language only, Tibetan, we know a stage as ancient as the 7th c. A. D.; for Siamese only the 13th c. A. D. Most of the Sinitic languages we know only in their modern and certainly very strongly evolved forms. We could never, from all the modern Germanic languages, reconstruct an Ancient Germanic language in the very least similar to the Anc. Germanic we know thanks to Gothic texts and to comparative Indo-European linguistics. From Icelandic *steinn*, German *Stein*, Swedish *sten*, English *stone* we could never suspect the Anc. Germanic *stainaz*. The reason for this is obvious. There were inherent in the Germanic peoples certain common psychological tendencies which have caused their languages to evolve — even after the cohabitation of the peoples was broken — along parallel lines. Hence *stainaz* has lost its final consonant and its vowel of the ending in these languages independently of each other and by a parallel evolution. Just the same may have taken place in all the Sinitic languages, so that by a parallel evolution they all lost their *-d* in 'four' except Arch. Chinese (and later on Chinese as well). That is why I say that I conclude against such a wholesale dropping of a final *-d* in 卅 *sɿəd* only because of the contrast with words with preserved Sinitic final consonants.

It is quite necessary to keep in mind this possibility of independent and yet parallel evolutions, once we think of a comparison between e. g. Siamese and Chinese. In his work «Le dialecte de Tch'ang-ngan sous les T'ang» (BEFEO 1920) H. Maspero has given a series of Siamese-Chinese word comparisons, some of which seem quite convincing.¹⁾ If we dress a table with the Siamese forms, the Anc. Chinese (6th c. A. D.) and the Arch. Chinese as reconstructed by me, the Siamese forms seem to afford crushing evidence against my Archaic reconstructions:

	S.	Anc.	Arch.
九	<i>kəo</i>	<i>kɿəu</i>	<i>kɿŭg</i>
舊	<i>kəo</i>	<i>g'ɿəu</i>	<i>g'ɿŭg</i>
丘	<i>k'əo</i>	<i>k'ɿəu</i>	<i>k'ɿŭg</i>
牛	<i>nguə</i>	<i>ngɿəu</i>	<i>ngɿŭg</i>
告	<i>klau</i>	<i>káu</i>	<i>kóg</i>
袋	<i>tai</i>	<i>d'ái</i>	<i>d'əg</i>
害	<i>hai</i>	<i>ɣái</i>	<i>g'ád</i>
雞	<i>kai</i>	<i>kiei</i>	<i>kiər</i>

¹⁾ They are of course on the whole very uncertain; on p. 84 Siam *k'ao* is given as equivalent to Chin. 丘, on p. 86 Siam. *k'uə* as equivalent to the same word!

It would seem that Siamese forbids the reconstruction of the Arch. final consonants. And yet, in several of these very words there are absolute proofs of their existence:

舊 rimes in Shī with **時**, which frequently rimes with *-k* (e. g. Chuang-tsī: Ta tsung shī, Huai-nan-tsī: Lan ming, Yi Chou shu: Tu hün etc.);

丘 rimes in Ch'u tsī with **之**, which frequently rimes with *-k* (e. g. Shu: Lü hing, Sün-tsī: Kün tao); and it rimes in Sün-tsī: Ta lue with **背**, the final *-g* of which is certain from the element **北** *pək*;

牛 rimes in Shī with **哉**, which frequently rimes with *-k* (e. g. Chuang-tsī: Ta tsung shī and Shan mu etc. and which has the same phonetic as **載**, which latter again regularly rimes with *-k* and is used for **則** *tsək*) and in Chuang-tsī with **來** which regularly rimes with *-k* (dozens of examples);

告 has two Anc. readings *káu* and *kuok* and regularly rimes with *-k* (*passim*); there cannot be the slightest doubt about its Arch. final guttural;

袋 has the same phonetic as **駭** *d'ək*, and as **貸** *t'ai* and *t'ək*; its phonetic **代** has phon. **弋** (*d)ək*;

害 *γai* is phonetic in **割** *kāt* and serves as *kia tsie* for **曷** *γāt*.

Thus, in spite of Siamese, we cannot but acknowledge the Arch. final consonants in these words, and I can see no reason why Proto-T'ai could not have had final consonants as well, lost or changed into *-u*, *-i* in the same fashion as in Chinese, and thanks to parallel evolutions.

It is, indeed, interesting, in this context, to observe the parallelism of Tibetan and Chinese sound evolutions during the last millennium on several striking points. Just as the ju sheng *-t* has been lost in the whole of Northern China: **八** *pwat* > *pa*, **七** *ts'jēt* > *ts'i*, **骨** *kuat* > *ku* etc., so Tib. final *-d* has been lost in the Central provinces: *nad* > *nä*, *bod* > *b'ö*, *dpyid* > *či* etc. And just as certain voiced initials, i. e. explosives, affricates and fricatives, have become surd in the whole of Chinese, except the Wu dialects, so certain voiced initials, namely fricatives, have become surd in Tibetan: Chin. **是** *zí*, > *ši*, **祥** *zjang* > *siang*, Tib. *ža* > *ša*, *za* > *sa*.

CATEGORIES ENDING IN ARCH. GUTTURAL CONSONANT.

Having finished the investigation of the words ending in Arch. dental, I wish to take up once more the question of the word groups ending in *-k*, *-g*, *-ng*, extensively treated in my Shī king Researches. Professor Li Fang-kuei has recently

published an article: «Ancient Chinese -ung, -uk, -uong, uok etc. in Archaic Chinese» (Bull. Nat. Research Inst. Hist. Phil. vol. III, pt. 3, 1933), which is largely a polemic against my conclusions and a system of reconstruction of his own. This article is full of interesting observations and ideas, and on some points I can revise my system thanks to his proposals; on the whole, however, I cannot accept his conclusions, and his reconstruction scheme is in my opinion quite impossible.

Among the points which seem to me to be acceptable, I first mention his opinion that the hie sheng characters must be somewhat older than the Shī king odes, a fact which I had doubted earlier. In fact, in the -at category above (F) there are certain phenomena which confirm Li's opinion. We find there, quite regularly, 內 and 對 riming with -t words, which clearly indicates Shī *nwəd*, *twəd*. But that 內 had originally a -b is quite certain. The labial final is brought out by 納 Arch. *nəp* (Anc. *nāp*, Mand. *na*), originally written simply 內, and it is obvious that this Arch. *nəp* 'to bring in': 內 *nwəb* 'interior': 入 *n̥iəp* 'to enter' are but three aspects of the same stem. And 對 *twəb* 'to answer, vis-à-vis, etc.' stands to 答 *təp* 'to answer' just as *nwəb* 'interior' stands to *nəp* 'to bring in'. In Shī time *nwəb* had become *nwəd* by dissimilation.¹⁾

Another valuable point in Li's treatise concerns the words treated on pp. 136—140 in my Shī king Researches, e. g. 路. This was an original *glág*, since it had 各 *kldk* for phonetic. And yet it rimes in Shī king with words of type 故 *ko* and never with ju sheng *tsdk* etc. Li had assumed earlier that it had lost its final -g already between hie sheng time and Shī king time. But since type 故 *ko* regularly rimes with type 家 *kā*, and type 路 rimes with type 故 *ko* but not with type 家 *kā*, I had concluded that 路 could not be simply *glo* in Shī, and so I had supposed an implosive final: 家 *kā*: 故 *ko*: 路 *glo_k*. Now Li proposes, instead, a final laryngal: 路 *glo*· («glottal stop»), which is an extremely common substitute in modern dialects for an earlier ju sheng -k, and he thus obtains a nice system of rimes: *kā*: *ko*; *ko*: *glo*·; but never *kā*: *glo*·, the latter two being too dissimilar phonetically. This I think is much better than my own explanation. We have therefore to state that final -g was still living, in Shī times, after *e*, *ə*, *o* and *u* (e. g. 來 *ləg* riming with -k) but that after the vowel *a* it very early became· (glottal stop): 路 *glág*, 怕 *p'äg*, 夜 *ziag* became *glá*·, *p'a*·, *ziä*·, and these again *glo*·,

¹⁾ When I speak of the age of the hie sheng characters, I should express myself more precisely. Many of the hie sheng characters of later ages were written in early Chou time without radicals, i. e. they were properly speaking only *kia tsie* characters to which later on specializing radicals were added. From the linguistic point of view it is of course immaterial whether the «phonetic» was used alone or whether it was written with an elucidating signfic («radical»).

p'o', *zjo'* in the Shī language, which explains the rimes in Tuan Yü-ts'ai's cat. 5, which are otherwise inexplicable.

So far, so good. But for the rest Li's constructions are very disappointing. He seems to start from an assumption that every Arch. vowel must exist in combination with every Arch. final consonant — if there are gaps, the construction must be wrong. The chess-board of $8 \times 8 = 64$ squares must have every one of the 64 squares filled; if not, we are on the wrong track. This is a funny axiom, to say the least of it. I know of no language with such a structure, and I fail to see why Chinese should be one. He finds in Anc. Chinese, in the *-əng* group:

1 登 2 肱 3 〇
4 兢 5 〇 6 弓

1. *əng* 2. *wəng* 3. 〇
4. *ɿəng* 5. 〇 6. *ɿung*

Since there is no Anc. *ɿwəng* and no Anc. *-ung* (in this Shī rime group), he concludes for Arch. Chinese:

1. *əng* 2. *wəng* 3. 〇
4. *ɿəng* 5. 〇 6. *ɿwəng*.

This looks very nice indeed, but it is extremely embarrassing, once it has to be applied to the corresponding words with *-k* and *-g*:

1 得 2 國 3 〇
4 五 5 域 6 國
7 來 8 灰 9 母
10 子 11 邇 12 久

In Anc. Chinese they were:

1. *tək* 2. *kwək* 3. 〇
4. *kɿək* 5. *ɿwək* 6. *ɿkuk*
7. *lɿi* 8. *χuɿi* 9. *mɿu*
10. *tsi* 11. *kjwi* 12. *kɿɿu*

On the analogy of his interpretation of the *-ng* words, Li has to assume *the same* Arch. final for 5. and 6.; for 8. and 9.; and for 11. and 12. Thus:

1. *tək* 2. *kwək* 3. 〇
4. *kɿək* 5. *ɿwək* 6. *ɿwək*
7. *lɿg* 8. *χwəg* 9. *mɿəg*
10. *tsiəg* 11. *kɿwəg* 12. *kɿwəg*

But since it is impossible that an Arch. *-iwək* could give sometimes Anc. *-iwək* and sometimes *-iuk*; that an Arch. *-wəg* could give sometimes Anc. *-uđi* and sometimes *-zu*; and that an Arch. *-iwəg* could give sometimes *-wi* and sometimes *-izu*, he has to find explanations for these divergent treatments.

In the first place (5: 6) he has to deal partly with labial-initialled words, e. g. 副 *p'iwək*: 服 *b'iuk*. These cause no real difficulty, for in *p'iwək* the *w* is a »false ho k'ou« (see p. 4 above) and the Arch. form was *k'ai k'ou p'iek*. Partly he has to deal with guttural-initialled words: 域 (*g)iwək*: 囿 (*g)iuk*, and here Li has no better way out of the difficulty than to refer to »analogy»: 囿 **giwək* has become Anc. (*g)iuk* by analogy, through influence of other *-iuk* words in another Shī rime group — but 域 *giwək* (Anc. *jiwək*) has *not* undergone this analogical influence! And he passes this somewhat severe judgment (p. 391): »We have so far in discussing Chinese phonology made little use of analogy, but such a forceful principle so well attested in many languages cannot leave no trace in Chinese Karlgren's reconstructions, I believe, fail because . . . he fails to recognize certain analogical processes which are of paramount importance». I am afraid I know sufficiently well the part played by analogy in various languages to be aware that it cannot be drawn upon in Li's haphazard and hazy way: if we explain an evolution by analogy, we have to show which particular word or words have been influenced by which other particular word or words, and why they have done so; and we are certainly *not* allowed to explain a 囿 **giwək* > *jiuk* by »analogy» unless we show at the same time why 域 *giwək* has *not* equally become *jiuk* but remains Anc. *jiwək*. Li has here left the field of linguistic science.¹⁾

In the second place Li has to explain why certain *-wəg* have become *-uđi* and others *-zu* (8: 9). Here he has found a very clever explanation. He thinks there is a tone difference: *-wəg* in shang sheng (rising tone) became *-zu*, *-wəg* in p'ing sheng (even tone) became *-uđi*. This would be a brilliant expedient — if it were true. In order to prove it to be so Li gives statistics drawn from the Kuang yün. He serves us a series of characters many of which are of Liu ch'ao make and did not exist in Chou, Ts'in or Han time — and consequently prove nothing at all (this is a methodical fault which recurs throughout Li's paper). If we keep to the really pertinent words, above all those existing in the Shī king, we find e. g. 培 *b'zu* in p'ing sheng which according to Li should be *b'uđi*, and 每 *mudī* and 悔

¹⁾ It must be remembered that it is not a question here of a fluctuation between *iwək* and *iuk*, such as in Pekinese, where 'to learn' can be read both *hūe* and *hiao* and *hūo*, 'horn' both *kūe* and *kiao* and *kūo* through a mixing of dialects. Here there are certain words which have exclusively Anc. *iwək* and certain others which have exclusively *iuk*.

χuđi in shang sheng, which according to him should be *mzu*, *χzu*. The latter two are particularly important, since they are two of the most common words in the language. Li has to consider these *mudí*, *χuđi* in shang sheng as «exceptional»!

If Li's tone theory is thus an obvious failure, I think none the less that I was wrong, in my *Shi king Researches*, in supposing 9. Arch. *mug*. It must be observed that 9. -*zu* occurs exclusively after labials: *pzu*, *b'zu*, *mzu*. And on the other hand 7. *đi* occurs after all kinds of initials: *kđi*, *tđi*, *lđi*, *tsđi*, e x c e p t l a b i a l s; there are no *pđi*, *b'đi*, *mđi*. I conclude that 9. *mzu* is the labial-initialled class answering to those 7. *đi*: 7. *kəg* > *kđi*, *ləg* > *lđi*, *tsəg* > *tsđi*: 9. *məg* > *mzu*.

There is one objection to this theory which may seem fatal: a general rule in the hie sheng characters says that k'ai k'ou and ho k'ou words do not serve for each other. A *kán* is very rarely phonetic in a *kuán* or *vice versa*. But here we have 母 *mzu* phonetic in 每 *mudí*. Is it then possible to reconstruct 母 Arch. *məg* phonetic in 每 *mwəg*?

Yes, it is. For the words with labial initials are exceptions from the general rule. A few examples will suffice to show this:

1 非排: 分貧: 麻摩: 曼慢: 門閤: 元晚: 皮波。

1. Anc. *pjwəi*: *b'ai*; 2. *pjwən*: *b'ičn*; 3. *ma*: *mud*; 4. *muán*: *man*; 5. *muən*: *mīčn*; 6. *mjān*: *mjwəm*; 7. *b'jig* (< *b'ia*): *puđ*. Thus a *məg* can very well serve as phonetic in a *mwəg*. Moreover, this same 每 *mwəg* is undeniably phonetic, again, in a plainly k'ai k'ou word: 海 *χái* (< *χməg*).

In the third place Li has to explain why certain -*iwəg* become -*wi* and others -*iəu* (11:12). It is true that half of the enigmatical cases in question can be eliminated. There are both guttural-initialled words (*kjwi*: *kjzu*) and labial-initialled words (*pjwi*: *pjzu*) in our category. The *pjwi* etc. have not become T'ang (and later) *fi*, but have preserved their *p*-, which shows the ho k'ou *w* here to be secondary, a parasitic addition to the initial *p*- (see p. 4 above). Thus they were not Arch. *pjwəg* but *pjəg* and need cause no trouble. But there always remains the contrast *kjwi*: *kjzu* for Li to explain; there he cannot refer to the tones, so he thinks that the contrast is due to «a dialectal difference or maybe variations in one dialect». This, then, would be cases of the kind which I have exemplified on p. 12 above: a mixing of dialects, the Ts'ie yün language having obtained, from sister dialects, certain words -*wi* which have ousted the regular -*iəu* or *vice versa*.

This last idea is of course not impossible in itself, though we shall see presently that it is not at all necessary. But when we find that Li, in order to surmount

the *three* serious difficulties which obstruct his reconstruction scheme, has to resort to *three* different explanations: one theory of analogy which is not scientifically founded; one tone theory which is disavowed by the most common of the words in question; and one theory of »dialectal variations» inside the Ts'ie yün language — then it is impossible to follow him.

The simple truth is that the five Anc. endings *-ək*, *-wək*, *-iək*, *-iwək*, *-iuk* cannot successfully be reduced to *one* Arch. final *(i)(w)ək*; and the six Anc. endings *-ái*, *-uí*, *-əu*, *-i*, *-wi*, *-iəu* cannot successfully be reduced to *one* Arch. final *-(i)(w)əg*. I have every possible reason to remain by my own earlier reconstruction (except for 9.):

1. <i>ək</i>	2. <i>wək</i>	3. <i>o</i>
4. <i>iək</i>	5. <i>iwək</i>	6. <i>iuk</i>
7. <i>k əg</i>	8. <i>wəg</i>	9. <i>m əg</i>
10. <i>iəg</i>	11. <i>iwəg</i>	12. <i>iug</i> .

If I do so, however, I have to give an acceptable answer to two questions: why did *-iung*, *-iuk*, *-iug*, exist in this Shī rime group but no *-ung*, *-uk*, *-ug*? And why does *-iung* rime with *-əng*, *-iəng*, why does *-iuk* rime with *-ək*, *-iək*, nay even with *-ək* — an apparently very unsatisfactory rime from the acoustic point of view? I think it is possible to answer both these questions (see p. 43 below).

If Li has been so keen on eliminating my *iung*, *iuk*, *iug* in this rime category, it is because he thinks he has found these Arch. finals in quite another Shī rime group cat. 9. of T'uan's, cat. 1. of Wang's. We find in this category words of the five Anc. types:

1. 江 Anc. *kāng*: 2. 工 *kung*: 3. 宮 *kīung*: 4. 冬 *tuong*: 5. 恭 *kīwong*.

And, correspondingly in the ju sheng:

1. 角 *kāk*: 2. 谷 *kuk*: 3. 菊 *kīuk*: 4. 酷 *k'uok*: 5. 曲 *k'iwok*.

It has been a much debated theme among Chinese philologists, whether Anc. *-āng*: *-ung*: *-iung*: *-uog*: *iwong* form one rime category in Shī king or two (and whether *-āk*: *-uk*: *-iuk*: *-uok*: *iwok* form one or two). The two greatest authorities, Tuan Yü-ts'ai and Wang Nien-sun both voted for *one* category (Wang, however, only as far as the *-ng* words were concerned), and they were followed by the brilliant linguist Chu Tsün-sheng. But two other great experts, K'ung Kuang-sen and Kiang Yu-kao, thought it possible to distinguish two *-ng* classes (and two *-k* classes). One is formed by Anc. *-āng*, *-ung* and *iwong*; one is formed by Anc. *-uog*

and *-iung*. Li Fang-kuei follows the latter. By adducing extensive materials he shows that not only in the Shī rimes but also in the hie sheng characters there are frequent and close connections between Anc. *-āng: -ung: -i Wong* (and between *-āk: -uk: -i wok*) on the one hand, between *-uong: -iung* (and between *-uok: -iuk*) on the other; but that a mixing of an *-uong* or an *-iung* into the *-āng: -ung: -i Wong* series, or, *vice versa*, of an *-āng*, an *-ung* or an *i Wong* into the *-uong: -iung* series (and similarly in the *-k* groups) is a comparatively rare phenomenon (though by no means unknown). There cannot be the slightest doubt, to my mind, that he is right. A seeming obstacle, the character 降 Anc. *kāng*, which regularly goes together with *-uong*, *-iung* and not with other *-āng* nor with *-ung*, *-i Wong* he shrewdly guesses to be a word different from other Anc. *kāng*. He draws the perfectly legitimate conclusion that my earlier reconstruction system:

	1.	2.	3.	4.	5.
Arch.	<i>āng</i>	<i>ong</i>	<i>iong</i>	<i>uong</i>	<i>i Wong</i>
Anc.	<i>āng</i>	<i>ung</i>	<i>iung</i>	<i>uong</i>	<i>i Wong</i>

is defective, since it does not explain these curious rime and hie sheng interrelations 1: 2: 5 versus 3: 4. He therefore thinks that Anc. *-āng: -ung: -i Wong* had one kind of principal vowel in Arch. Chinese, *-uong: iung* another. And since the former series contains *-āng*, he decides that those were Arch. 1. *-āng: 2. -ong: 5. -iong* (*-ong* becoming Anc. *-ung*, and *iong* breaking into *-i Wong* just as *-io > -iwo*, proved by me), whereas the latter were 4. *-ung: 3. -iung* (*-ung* breaking into Anc. *-uong*). Similarly Anc. 1. *-āk: 2. -uk: 5. -i wok* were Arch. *-āk: -ok: iok*, forming one rime group, and Anc. 4. *-uok: 3. -iuk* were Arch. *-uk: -iuk* forming another rime group.

This looks all very nice, and I admit that at first sight I was strongly tempted to accept it. And yet it is inadmissible. There are various obstacles. So far Li is right that the two sub-categories had a difference in principal vowel; but for the rest he is quite off the right track.

In the first place it must be remembered that a *strict* distinction between the two groups is not maintained. They are confused often enough to cause Tuan and Wang to join them in one great category (the latter only the *-ng* words). When Shī (Lie wen) rimes 邦 *pāng: 崇 tɕ'iung* and Yi king frequently rimes 邦 *pāng: 中 tɕiung*, then Li's Arch. values: *pāng: tɕ'iung*, *pāng: tɕiung* are not very convincing.

In the second place it might be argued that it goes against the testimony of the ancient dialects. When I proposed, in my Shī king Researches, that Anc. 工

-ung, 谷 -uk and 宮 -iung, 菊 -iuk derived from Arch. -ong, -ok, -iung, -iok, (ung < ong accepted by Li), it was because I could show that even in Ts'ie yün time there were considerable dialects which had an *o* vocalism. Now, that is true not only of -uk (谷, Li Arch. -ok) but also of -iuk (菊, Li Arch. -iuk), which has *o* regularly in Go-on (see Shī king Researches p. 127): Ts'ie yün *kjuk*, *tš'juk*, *ljuk*, *sjuk*, *pjuk*, *mjuk* = Go-on *koku*, *soku*, *roku*, *soku*, *poku*, *moku*, and after labials also in Kan-on and Sino-Corean: Ts'ie yün *mjuk* = Kan-on *boku*, Cor. *mok*. And even more serious: the Ts'ie yün rime 冬, which would have been Arch. -ung according to Li, has perfectly regularly -ong in Sino-Corean and -ou in both Kan-on and Go-on! If we conclude Arch. -ong, -ok for 公, 谷, Ts'ie yün -ung, -uk, because sister dialects of the Ts'ie yün language had *o*, we seem forced to pose an Arch. -o also for 冬 (Ts'ie yün -uong), which shows exactly the same phenomenon. This objection, however, is not very fatal, for, as we shall see, I was wrong in surmising Arch. 工 -ong, 谷 -ok for Ts'ie yün -ung, -uk (I shall revert to that presently); indeed, the various Sui time dialects had a most variable pattern answering to these finals in Arch. Chinese, and it is dangerous to draw far-reaching conclusions from them.

In the third place there is an objection which is much more serious. There are certain Arch. rimes which are irreconcilable with Li's interpretation. There crop up, here and there, some freer rimes due to *licentia poetica*, contacts between different rime categories, and these are often very telling. When we find in Shī (Lie wen) 崇 Anc. *dž'iung*: 皇 *γwáng*; in Yi king (ken) 躬 *kjüng*: 正 *tšjäng*: 終 *tšjüng*; in Ch'u ts'i (Kiu chang) 中 *tjüng*: 窮 *g'iung*: 行 *γwng* (< *g'äng*), and so on, then we can safely say that Arch. -iung for Anc. *iung* is absolutely excluded. A make-shift rime -ong: *äng*, an -ong: *äng*, an -ong: *äng* might pass, but not an -ung: *äng*, an -ung: *äng*, an -ung: *äng*. No, Anc. -iung was undoubtedly Arch. -iung of some kind; and that Anc. -uong had some kind of *o* is indisputable.

How, then, are we to explain the two sub-categories, since my earlier reconstruction obviously fails (see p. 37 above) to account for them? I think we had better start our investigation not from the -ng words but from the -k and -g words; for here, as in the -ang category, it turns out that the -ng words are less fully represented in all the possible combinations than the -k and -g words. Let us draw a table of type words, representing the various Anc. finals which fall in the three (for the -ng words only two) categories of the Shī king, which tally very well (it is highly important to observe this) with the distinctions in the hie sheng characters. I arrange them in a peculiar way with a view to the following discussion.

I		II		III	
1 學	2 包	11 樂	12 〇	23 角	24 穀
3 酷	4 老	13 較	14 郊	25 谷	26 瞽
5 旬	6 休	15 沃	16 高	27 曲	28 仆
7 〇	8 陶	17 藿	18 鈇		
9 威	10 蕭	19 虐	20 廟		
		21 的	22 苕		
29 降				32 江	
30 冬				33 工	
31 宮				34 恭	

(24, 26 and 28 are erroneously placed by Tuan in other categories; their phonetics prove them to belong to our cat. III here).

The Anc. values of these type-words were:

I		II		III	
		11. <i>ák</i>	12. 〇		
1. <i>ák</i>	2. <i>au</i>	13. <i>ák</i>	14. <i>au</i>	23. <i>ák</i>	24. <i>zu</i>
3. <i>uok</i>	4. <i>áu</i>	15. <i>uok</i>	16. <i>áu</i>	25. <i>uk</i>	26. <i>zu</i>
5. <i>ɿuk</i>	6. <i>ɿəu</i>	17. <i>ɿuk</i>	18. <i>ɿwo</i>	27. <i>ɿwok</i>	28. <i>ɿu</i>
7. 〇	8. <i>ɿäü</i>	19. <i>ɿak</i>	20. <i>ɿäü</i>		
9. <i>iek</i>	10. <i>ieu</i>	21. <i>iek</i>	22. <i>ieu</i>		
29. <i>äng</i>				32. <i>äng</i>	
30. <i>uong</i>				33. <i>ung</i>	
31. <i>ɿung</i>				34. <i>ɿwong</i>	

I have placed the *-k* and *-u* words abreast in the way hie sheng characters with double readings show them to belong together. There is e. g. 覺 read both *kák* (1) and *kau* (2); 告 read both *kuok* (3) and *käu* (4); 祝 read *tɿɿuk* (5) and *tɿɿəu* (6) — and so on.

Cat. I is Wang Nien-sun's cat. 21, II his cat. 20 and III his cat. 19 (ju sheng). There are a considerable number of rime contacts between the three categories; and in my Shī king Researches I even called in question the correctness of distinguishing I and II. Prolonged deliberation has convinced me that after all it is necessary to accept such a distinction. It is necessary to keep apart I and II, insignificant though the difference must have been, not only because of the rimes in the Shī and other classics, which in spite of numerous contacts on the whole must be

said to keep these classes apart from each other; it is also and above all because the very same three-class distinctions are clearly discernible in the hie sheng characters, an extremely interesting and important fact. Li Fang-kuei has brought out this in a very meritorious way.¹⁾

So far, therefore, he has convinced me. But his reconstruction scheme suffers from such great faults that it is entirely unacceptable. In order to get a different principal vowel in the three categories he introduces, *besides* the ordinary *o* and the open *â*, which is »half-way between» *o* and *â* (as in Engl. *all*), still a third open *o* sound, written *ω* »half-way between» *â* and *â*! This is of course phonetically extremely unnatural and impossible. Still more unnatural is the way in which Li has worked out this idea:

I		II		III	
		11. <i>ωk</i> (> <i>âk</i>)	12. <i>○</i>		
1. ?	2. <i>og</i> (> <i>au</i>)	13. <i>âk</i> (> <i>âk</i>)	14. <i>âg</i> (> <i>au</i>)	23. <i>âk</i> (> <i>âk</i>)	24. ?
3. <i>uk</i> (> <i>uok</i>)	4. <i>ug</i> (> <i>âu</i>)	15. <i>wok</i> (> <i>uok</i>)	16. <i>ωg</i> (> <i>âu</i>)	25. <i>ok</i> (> <i>uk</i>)	26. ?
5. <i>ïuk</i> (> <i>ïuk</i>)	6. <i>ïug</i> (> <i>ïæu</i>)	17. ?	18. ?	27. <i>ïok</i> (> <i>ïwok</i>)	28. ?
7. <i>○</i>	8. ?	19. <i>ïwk</i> (> <i>ïak</i>)	20. <i>ïωg</i> (> <i>ïäu</i>)		
9. <i>iuk</i> (> <i>iek</i>)	10. <i>iug</i> (> <i>ieu</i>)	21. <i>iwk</i> (> <i>iek</i>)	22. <i>iωg</i> (> <i>ieu</i>)		
29. ?				32. <i>âng</i> (> <i>âng</i>)	
30. <i>ung</i> (> <i>uong</i>)				33. <i>ong</i> (> <i>ung</i>)	
31. <i>ïung</i> (> <i>ïung</i>)				34. <i>ïong</i> (> <i>ïwong</i>)	

We can see at a glance how unsatisfactory all this is:

1) Can anyone seriously believe that such different Arch. values as those in Li's cat. I (nearly all with a *u* vocalism) could give in nearly every detail the same Anc. results as those in cat. II (nearly all with a vowel more open than *â*, practically an *â*)?

2) What about all the empty spaces, in particular 1, 24, 26, 28, 29 (8, 17, 18 are of little consequence, as we shall see presently)? How will it be possible, under Li's system, to fill in logically any Arch. values in them? Li has wisely abs-

¹⁾ There would seem to be an important exception to this rule. The Anc. *âu*, *au*, *ïäu*, *ieu*, *ïæu* of categories I and II interchange, in the hie sheng characters, *not only* with Anc. *uok*, *ïuk*, which they should do, in accordance with the Shī rime system of the table above, but also in some cases with Anc. *-uk* (III), which they should not, see my Shī king Researches p. 152 (table). But this is not so bad as it would seem. Against an overwhelming number of contacts with Anc. *uok*, *ïuk*, there are, in the table, only 6 cases of contact with *uk*; 5 of these are *puk*, *p'uk*, *muk* — evidently the labial initial has here confused an earlier *puok* or such-like. Moreover, of the 6 cases only 2 are characters existing in Han time or earlier. So the general rule is very safe.

tained from saying anything about them at all. And yet these words (and many others for which they serve as type words) clearly belong to these categories. If, with Tuan Yü-ts'ai, Li would place 26 and 28 not in III but in I, he would jump from the frying-pan into the fire; it would make it still more impossible for him to find separate Arch. values for these word types. His system here breaks down entirely.

3) If II had a vowel intermediate between *ā* and *á*, how is it possible that II does not interchange frequently with words of type 作 Arch. *tsák* etc. (Tuan's cat. 5) and with cat. III (*ák* according to Li) but does so, in numerous cases of contact between I and II, with *-u-* words? That is perfectly inexplicable.

4) Why do not 13. »ák», 14 »ág» rime with 23 »ák»?

All this is plainly impossible. We shall have to find ways of approach quite different from those of Li.

Let us start with rime cat. III, and acknowledge at once a fundamental fault committed in Shī king Researches: the endeavour to elucidate it by aid of such late dialects as Sino-Japanese and Sino-Corean. My conclusion that Anc. 25. *kuk* was Arch. *kok* was devoid of value. Leaving apart, for the moment, 24, 26, 28 and concentrating upon the principal types: 23, 25, 27, Anc. *ák*, *uk*, *iwok*, which, as vindicated by K'ung Kuang-sen and Kiang Yu-kao, form a rime category distinguished from I and II, we shall have to judge them in the light of Arch. Chinese itself. There are two salient facts to be taken into account:

1) Whereas the *ák*, *uok*, *iwok* of I and II have frequent connections, in rimes and hie sheng, with words of types Anc. *ák*, *jak*, *iek*, *äu*, *ieu*, words which obviously all had k'ai k'ou, the *ák*, *uk*, *iwok* of III have no such connections (yet see note on p. 40).

2) In irregular rimes, III very often (but I and II never) mixes with Tuan's cat. 4, which quite certainly was Arch. *-u*, *-iu* (see Shī king Researches p. 145). Examples:

1 驅績穀 2 奏保 3 木附屬 4 谷穀垢 5 屬貝 6 谷附漏

1. (Shī, Siao jung) Anc. *k'iu*: *ziwok*: *kuk*; 2. (Shī, Ch'u ts'i) Anc. *tsu* (< *tsu*): *luk*; 3. (Shī, Küe kung) *muk*: *b'iu*: *ziwok*; 4. (Shī, Sang jou) *kuk*: *kuk*: *k₂u* (< *ku*); 5. (Li sao) *ziwok*: *g'iu*; 6. (Yi king, *tsing*) *kuk*: *b'iu*: *l₂u* (< *lu*).

Indeed, this phenomenon is so marked that Wang Nien-sun has placed our cat. III (Anc. *ák*, *uk*, *iwok*) as the ju sheng correspondence to Anc. *zu*, *iu* (Arch. *u*, *iu*).

These two facts remove all doubt about the Arch. values of types 25. 谷 Anc. *kuk*. 27. 曲 *k'iwok*. They were not, as I supposed in Shi king researches, *ok*, *iwok*; nor were they, as Li supposes, *ok*, *ïok*. They were clearly ho k'ou words: 25 *uk*, 27. *ïuk*. Therefore makeshift rimes were possible such as those in the table p. 41 above: Arch. 1. *k'ïu*: *dziuk*: *kuk*; 2. *tsu*: *luk*; 3. *muk*: *b'ïu*: *dïuk*; 4. *kuk*: *kuk*: *ku*; 5. *dïuk*: *g'ïu*; 6. *kuk*: *b'ïu*: *lu*. (With Li's system: 1. *k'ïu*: *dziok*: *kok*; 2. *tsu*: *lok* etc. would be quite inexplicable).

The Arch. *-uk* was preserved in the Ts'ie yün dialect, but in other ancient dialects it was broken into *uok* (S.-Jap. spells 屋 *woku*, 翁 *wou*); *ïuk* was broken into *iwok*. Yet in the corresponding *-ng* words the Arch. 34. *-jung* peeps through in the oldest Anc. dialect we know of, the Wu dialect, which was the base of Go-on. We find (Karlgren, Phonol. Chin. p. 853) Ts'ie yün *kïwong*, *g'ïwong*, *'ïwong*, *îwong*, *lïwong*, *sïwong* etc. = Go-on *ku*, *gu*, *iu*, *çu*, *riu*, *šu* etc.

So far all is plain sailing. But type 23. Anc. *âk* seems to form a serious obstacle. If 25. was Arch. *uk* and 27. *ïuk*, what was this 23. *âk*?

It is obvious that *âk* III (23) had a different Arch. origin from *âk* I (1) and *âk* II (13), since none of the three types rimes with any of the others. *âk* III (23), which corresponds to the Arch. *-g* word 24. Anc. *-zu*, whereas *âk* I (1) and *âk* II (13) correspond to 2, 14 Anc. *au*, evidently had a darker vowel than the others. Since it rimes with 25. Arch. *uk* and 27. Arch. *ïuk* and constantly interchanges with them in the hie sheng, it must have been some kind of *u*. But what was the difference? I think the secret lies in the *quantity*.

H. Maspero was the first to emphasize the great and fundamental difference between long-vowelled (tense-vowelled) and short-vowelled (slack-vowelled) syllables in Anc. (and of course in Arch.) Chinese. This was a fecund idea, and it has enabled us to solve a long series of riddles. There was this contrast in nearly every Arch. category. There were *ang*: *ǎng*; *iang*: *ǐǎng*; *an*: *ǎn*; *ian*: *ǐǎn*; *ien*: *ǐĕn*; *ieng*: *ǐĕng*; *am*: *ǎm*; *iam*: *ǐǎm*. If we examine the short-vowelled rimes in the *a*-groups, we find that when having medial *ï* they existed only after gutturals (with laryngals) and labials, but not after palatals and dentals: Anc. types *kïm*, *kïwɔn*, *pïwɔn*; *kïwng*, *kïwɔng*, *pïwɔng* etc.; furthermore that when they had no medial *ï*, they existed after the said gutturals and labials, and besides that, in some cases, after *î*, *ǎ*, *tɕ*, *tɕ'*, *dɕ*, but not after *l*, *t*, *t'*, *d*, *ts*, *ts'*, *dz*, *s*: types *kɔm*, *pɔwɔn*, *ɕɔn*. Here, in our present categories I, II and III, we find that the three Anc. *âk* types (1, 13, 23) had exactly this peculiarity: they existed principally after gutturals and labials (*kâk*, *pâk*) and in a few cases like 提 *tɕâk* etc. Similarly the two

Anc. *ang* types 29, 32, were *kang*, *pang* and an occasional 雙 *šang* etc. I conclude that the types 1, 13, 23, 29, 32 belonged to this class of short-vowelled syllables, and this gives us the key to various distinctions in our tables I, II and III which otherwise would be inexplicable. If we revert, first, to cat. III, we can now fill in the Arch. values:

23. Arch. *ũk* > *øk* > Anc. *ǎk*; 25. Arch. *uk* > Anc. *uk*; 27. Arch. *ɿuk* > Anc. *ɿwok*; And, in the corresponding -*g* series:

24. Arch. *ũg* > *u* > Anc. *ɣu* (just as 句 Arch. *ku* > Anc. *kɣu*); 26. Arch. *ug* > *u* > Anc. *ɣu*; 28. Arch. *ɿug* > Anc. *ɿu*.

In the -*ng* series:

32. Arch. *ũng* > *ōng* > Anc. *ang*; 33. Arch. *ung* > Anc. *ung*; 34. Arch. *ɿung* > Anc. *ɿwong*.

And here we obtain quite unexpectedly the solution of the riddle that puzzled us in the *ə* category above (p. 36). In that Arch. category:

<i>əng</i>	<i>wəng</i>	○
<i>ɿəng</i>	—	<i>ɿung</i>
<i>ək</i>	<i>wək</i>	○
<i>ɿək</i>	<i>ɿwək</i>	<i>ɿuk</i>
<i>əg</i>	<i>wəg</i>	○
<i>ɿəg</i>	<i>ɿwəg</i>	<i>ɿug</i>

we had *ɿung*, *ɿuk*, *ɿug*, but, strange to say, no *ung*, *uk*, *ug*.

In the present category:

<i>ũng</i> , <i>ũk</i> , <i>ũg</i>	○ ○ ○
<i>ung</i> , <i>uk</i> , <i>ug</i>	<i>ɿung</i> , <i>ɿuk</i> , <i>ɿug</i>

we have *ũng*, *ũk*, *ũg* but no *ɿũg*, *ɿũk*, *ɿũg*! When we find that in the *ə* category the 弓 *ɿung* 囿 *ɿuk*, 久 *ɿug* existed only after gutturals (laryngals) and labials, types *kɿung*, *kɿuk*, *pɿuk*, *kɿug*, *pɿug*, but after no other initials, and thus clearly agreed with the short-vowelled types in the *a*-groups (*kǎn*, *kɿwǎn*, *pɿwǎn* but no other initials), we can see at a glance that the 弓, 囿, 久 in the *ə* category are the very types (short-vowelled): *ɿũng*, *ɿũk*, *ɿũg* which are missing in our cat. III here.

Why, then, should they rime, not with *ũng*, *ũk*, *ũg* in our cat. III here but with the *əng*, *ək*, *əg* in the *ə* category? It stands to reason that the short *ũ* must have been modified, when preceded by an *ɿ*, so as to make the sound less labial and less

velar than in *üng*, *ük*, *üg*. It must have been something similar to the Swedish *u* in *kung* or the English *u* in *value*. This, indeed, stands genetically and acoustically fairly close to *ə*, and we easily understand that in rimes and hie sheng it goes together rather with *ə* than with a narrowly labialized and strongly velar *ü* and *u*. This once said, and this peculiarity in script and rimes nicely explained, we can be satisfied to write it, etymologically correct, 弓 *kjüng*, 囿 *giük*, 久 *kjüg*, keeping in mind that the *ü* in these syllable types had a different and more open *timbre* than the *ü* in types *küng*, *kük*, *küg*.

Let us now take up for examination the intricate categories I and II on p. 39 above. If we look at their Anc. values, there is an almost shocking similarity. It would seem to be absolutely futile to endeavour to find a difference in quality of the principal vowel for these two Arch. categories. And yet such a difference must have existed, since they are distinguished fairly clearly as rime categories — with numerous confusions, it is true. And we have necessarily to solve the riddle.

In the first place we shall somewhat reduce the apparent similarity of the two categories. The table on p. 39 above is correct, yet it is somewhat misleading. For all the types there given are not equally normal and frequent. The normal types in cat. I are 1, 2, 3, 4, 5, 6, 9, 10; the type 8. *jäu* is hardly existent. It is represented by the word 陶 Anc. *jäu* in some rimes. Moreover, Tuan brings in 𪛗 *g'jäu* and 椒 *tsjäu* in our cat. I, because of their phonetics, but in Shī they only rime with each other once and with no other words, so we cannot know exactly whether they belong in I or in II. Among the words with Anc. *jäu* in the Ts'ie yün the great majority are obviously words of cat. II, having phonetics belonging to that category; a few words have phonetics belonging to cat. I, but that does not necessarily place those words in the Arch. cat. I, for these characters may be due to contact between the (undoubtedly very similar) categories I and II in the hie sheng. Altogether it can be said that the Anc. *jäu*, regular and frequent in cat. II, as a rule does not exist in cat. I, just as *jeu* (I, 6), regular and frequent in cat. I, does not exist at all in cat. II. Here, then, is a strong and real difference between I and II.

In cat. II the normal types are 11, 13, 14, 15, 16, 19, 20, 21, 22. Types 17 and 18 are represented each only by an isolated word, and these we must disregard, since they cannot be built upon. They may be due to some special conditions in individual cases. We may therefore reduce somewhat our scheme of the normal type words of our three categories, and rewrite it thus:

I				II				III			
				9 樂	10 〇	19 月	20 穀				
1 學	2 包			11 較	12 郊	21 谷	23 替				
3 酷	4 老			13 沃	14 高	23 曲	24 仆				
5 菊	6 休			15 虐	16 廟						
7 威	8 蕭			17 的	18 惹						
25 降						28 江					
26 冬						29 工					
27 宮						30 恭					

In Anc. Chinese:

I				II				III			
				9. <i>ək</i>	10. 〇						
1. <i>ək</i>	2. <i>au</i>			11. <i>ək</i>	12. <i>au</i>	19. <i>ək</i>	20. <i>ɹu</i>				
3. <i>uok</i>	4. <i>əu</i>			13. <i>uok</i>	14. <i>əu</i>	21. <i>uk</i>	22. <i>ɹu</i>				
5. <i>ɿuk</i>	6. <i>ɿəu</i>			15. <i>ɿək</i>	16. <i>ɿəu</i>	23. <i>ɿwok</i>	24. <i>ɿu</i>				
7. <i>iek</i>	8. <i>ieu</i>			17. <i>iek</i>	18. <i>ieu</i>						
25. <i>əŋg</i>						28. <i>əŋg</i>					
26. <i>uŋg</i>						29. <i>uŋg</i>					
27. <i>ɿuŋg</i>						30. <i>ɿwŋg</i>					

Since cat. III had *u*, I and II cannot possibly have had *u*. And since *ək*, *ek*, *ək*, *ak*, *ek* are to be found in other Shī king rime categories, neatly distinguished from our categories I and II here, I maintain what I said in my Shī king Researches, against Li Fang-kuei's proposals: these two categories had some kind of *o* for principal vowels. But Li is surely right in saying that II must have had a more open principal vowel than I. This gives us closed *o*: *ô* for I, open *o*: *o* and *ā* for II. And then we have to apply the distinction we have already determined between long-vowelled and short-vowelled syllables: *ō* as against *o*, *ô* as against *ô* (for typographical reasons, in order to avoid an ugly *ô*, I write the short *ô* thus: *ô*). And we are finally able to fill in our scheme with Archaic values which will nicely explain both the distinction between the three categories in rimes and hie sheng, and the considerable number of exceptional contacts between them, equally in rimes as well as in hie sheng. Observe that the *-ng* group is much poorer than the *-k* and *-g* groups. It has fewer forms with medial *ɿ*, and it has forms corresponding only to I and III, not to II. Whether the latter is due to confusion of

two primarily different groups it is impossible to tell; we can merely state that neither Shī rimes nor hie sheng indicate a distinction here similar to that in the *-k* and *-g* words.

I		II		III	
		9. <i>dk</i> (▷ <i>ák</i>)	10. <i>o</i>		
1. <i>ók</i> (▷ <i>ák</i>)	2. <i>óg</i> (▷ <i>au</i>)	11. <i>ók</i> (▷ <i>ák</i>)	12. <i>óg</i> (▷ <i>au</i>)	19. <i>úk</i> (▷ <i>ák</i>)	20. <i>üg</i> (▷ <i>zu</i>)
3. <i>ók</i> (▷ <i>uok</i>)	4. <i>óg</i> (▷ <i>áu</i>)	13. <i>ók</i> (▷ <i>uok</i>)	14. <i>og</i> (▷ <i>áu</i>)	21. <i>uk</i> (▷ <i>uk</i>)	22. <i>ug</i> (▷ <i>zu</i>)
5. <i>íók</i> (▷ <i>íuk</i>)	6. <i>íóg</i> (▷ <i>ízu</i>)	15. <i>íok</i> (▷ <i>íak</i>)	16. <i>íog</i> (▷ <i>íäu</i>)	23. <i>íuk</i> (▷ <i>íwok</i>)	24. <i>íug</i> (▷ <i>íu</i>)
7. <i>íók</i> (▷ <i>iek</i>)	8. <i>íóg</i> (▷ <i>ieu</i>)	17. <i>iok</i> (▷ <i>iek</i>)	18. <i>iog</i> (▷ <i>ieu</i>)		
25. <i>óng</i> (▷ <i>áng</i>)				28. <i>ung</i> (▷ <i>áng</i>)	
26. <i>óng</i> (▷ <i>uong</i>)				29. <i>ung</i> (▷ <i>ung</i>)	
27. <i>íóng</i> (▷ <i>íung</i>)				30. <i>íung</i> (▷ <i>íwong</i>)	

There is one more group with guttural finals which needs a few words of elucidation. Tuan's cat. 11, Wang's cat. 6 contains words with Anc. *eng*, *jäng*, *ieng*,¹⁾ and since it never rimes with the *ang*, *iang*, *äng*, *jäng* etc. of Tuan's cat. 10, we can see that the palatal vowel was Archaic. To this *-ng* category correspond as *-k* and *-g* words Tuan's cat. 16, Wang's cat. 11, with Anc. *-iɛ*, *iei*, *ai*, *iwei*. The *iɛ* of this category should be well distinguished from the *iɛ* (< *ia*) of Tuan's cat. 17, e. g. 何 Anc. *ɣá*, 遇 *kuá* (< *kwá*), 皮 *b'jiɛ* (< *b'ia*), 爲 *juiɛ* (> *guia*), 加 *ka*, which had *-a* (open syllable) in Arch. Chinese. The *-iɛ* in cat. 16. is the *-g* correspondence to Anc. *-jäng*.

In Shī king Researches (p. 157) I stated that the Anc. *-eng*, *-ek* rimes: 耕 *keng*, 革 *kek* etc., rime, in the Shī, the former in the *e* group, i. e. with *-jäng*, *-ieng*, the latter in the *ə* group, i. e. with *-ək*, *-əg*. This is true, but not the whole truth. Anc. *eng* (*ek*) contains characters of two quite different Arch. origins. One of them, with an open, slack *ä* sound: Arch. *-eng*, *-ek*, rimes with the neutral slack *ə*: *əng*, *ək*; the other, which in Anc. Chinese coincided with the open *ε* (since 耕 *eng*: 清 *jäng*: 青 *ieng* are different rimes in the Ts'ie yün) must have been another kind of *ä* or *e* in Arch. Chinese. On the analogy of cat. B above (pp. 3, 6), where we have Arch. *iēn*: *ien* as rimes, I conclude in our present category that the three principal endings were *əng*: *ǐəng*: *ieng*, and I obtain the following scheme:

¹⁾ The series 生 Anc. *šəng* should be expected to be Arch. *säng* and rime in the *-ang* category. But it rimes quite regularly in the *äng-eng* group, which shows that an Arch. *əng* here has irregularly passed over to the Anc. *vng* rime.

Archaic ε class, riming in the $-ang$, $-ak$ category:

1 橙 2 革 3 戒
4 宏 5 妄 6 徑

- | | | |
|--|--|--|
| 1. εng ($\rightarrow \varepsilon ng$) | 2. εk ($\rightarrow \varepsilon k$) | 3. εg ($\rightarrow ai$) |
| 4. $w\acute{\varepsilon}ng$ ($\rightarrow w\acute{\varepsilon}ng$) | 5. $w\acute{\varepsilon}k$ ($\rightarrow w\acute{\varepsilon}k$) | 6. $w\acute{\varepsilon}g$ ($\rightarrow wai$) |

Archaic e and $\check{e}$ class, forming Tuan's categories 11 and 16:

1 耕 2 危 3 解
4 嶺 5 割 6 挂
7 清 8 易 9 知
10 慈 11 〇 12 〇
13 青 14 錫 15 提
16 局 17 昊 18 圭

- | | | |
|---|--|--|
| 1. $\check{e}ng$ ($\rightarrow \varepsilon ng$) | 2. $\check{e}k$ ($\rightarrow \varepsilon k$) | 3. $\check{e}g$ ($\rightarrow ai$) |
| 4. $w\check{e}ng$ ($\rightarrow w\acute{\varepsilon}ng$) | 5. $w\check{e}k$ ($\rightarrow w\acute{\varepsilon}k$) | 6. $w\check{e}g$ ($\rightarrow wai$) |
| 7. $\check{i}\check{e}ng$ ($\rightarrow \check{i}\check{a}ng$) | 8. $\check{i}\check{e}k$ ($\rightarrow \check{i}\check{a}k$) | 9. $\check{i}\check{e}g$ ($\rightarrow ia > i\check{e}$) |
| 10. $\check{i}w\check{e}ng$ ($\rightarrow \check{i}w\check{a}ng$) | 11. 〇 | 12. 〇 |
| 13. $i\acute{e}ng$ ($\rightarrow i\acute{e}ng$) | 14. $i\acute{e}k$ ($\rightarrow i\acute{e}k$) | 15. $i\acute{e}g$ ($\rightarrow iei$) |
| 16. $iw\acute{e}ng$ ($\rightarrow iw\acute{e}ng$) | 17. $iw\acute{e}k$ ($\rightarrow iw\acute{e}k$) | 18. $iw\acute{e}g$ ($\rightarrow iwei$) |

For 3. $\check{e}g > \check{e}i > ai$ cf. German $ei > ai$. The transition $\check{i}\check{e}g > ia$ must have taken place quite early, for the Anc. $i\check{e} < Arch. \check{i}\check{e}g$ (our present cat., Tuan 16) and the Anc. $i\check{e} < Arch. ia$ (Tuan cat. 17), which are neatly distinguished in the Shī king, rime quite freely already in Lao-tsī and Chuang-tsī.

* * *

It might seem bold to reconstruct in its petty details Archaic Chinese, a language of some 2500 years ago, by aid exclusively of internal evidence, without comparative Sinitic materials, nay, on some points even seemingly against the evidence of e. g. Siamese (cf. p. 30 above). It must be observed, however, that in certain important respects we are much better situated for such a reconstruction than the scholar who has to reconstruct an earlier stage of a language exclusively by aid of divergent but *later* materials. The example adduced on p. 30 above: *steinn, Stein, sten, stone: stainaz* is very instructive as to the dangers the latter runs.

There may be important features in the early language which the late materials never reveal. Our position is very much more favourable in as far as we have sources for Arch. Chinese dating back to the very period of the language (say roughly 1000—600 B. C.), sources which give no concrete sound values, it is true, but which give so to speak the *frame* to be filled out, the *phonological categories* which need only be interpreted. And the value of these early sources is enhanced enormously by the fact that they are of two kinds absolutely independent of each other: the Shī rimes and the hie sheng characters. By a very lucky chance these two sources throw light upon a practically identical language. On a few points, it is true, the hie sheng reveal a slightly older stage of the language (see p. 32 above). But in most categories the accord is astoundingly good: the same distinctions, the same division of words into phonological groups can be observed in both sources. It is evident that the Shī king odes were given their final form and the standard set of hie sheng (originally *kia tsie*) were invented in one and the same centre, presumably the Chou court, and that dialectal aberrations were allowed to appear in the rimes and in the script only in sporadic cases.

There is, however, one great deficiency to be pointed out. If we are favourably placed, thanks to the double sources, for the reconstruction of the Arch. vowels and the final consonants, we are greatly handicapped when it comes to the initials, by our having recourse here only to *one* set of materials, the hie sheng; here, of course, the poetry fails us entirely. It is true that the hie sheng have made it possible on many important points to discern Arch. initials very different from the Anc. ones (*g*' > *γ*, *ā* > *á*, *í* > *íá*, *t* > *tá*, *s* | *a* > *ša*, *gi* > *ji*, *dí* > *í* etc., see my Analytic Dictionary); but many differences between the Arch. and the Anc. initial system, which do not happen to be revealed by this single source, the hie sheng, are sure to have escaped us. In particular I am afraid that many *consonant groups* may have existed where we can only discern single consonants. The possibility of such *x*'s in our equations, which can only be filled out in future by Sinitic comparisons, must never be forgotten. To a certain extent they will make the following investigation less reliable than it would appear at first sight. In spite of this we have to attempt it, confident that though a revision may be necessary on isolated points, the system as a whole must be fairly reliable.

There is one point regarding initial consonant groups on which I wish to say a few words. When we have the well-known alternation *k*-: *l*- and *p*-: *l*- in the hie sheng, e. g. 各 Anc. *kák*: 洛 *lák*, 變 *pian*: 緝 *liwän*, it might seem dubious

whether the consonant group existed in the *k* (*p*) member or in the *l* member or in both members. Thus three interpretations seem *a priori* possible:

A. 各 *klák*: 洛 *lák*; B. 各 *kák*: 洛 *klák* (*glák*); C. 各 *klák*: 洛 *glák*.

There is, of course, no fixed rule to be expected for this, for not all hie sheng characters may have been built on exactly the same principles. But in many cases there is one of these three alternatives which is decidedly the most plausible: the alternative C.

Alt. A. is excluded in several examples where we can build on reliable testimonies. There is, first, the case 藍 Anc. *lám* 'indigo', often discussed earlier. Here we have, fortunately, double *points d'appui*, which fact admits of a definite conclusion. When on the one hand we have 監 Anc. *kam* as phonetic, on the other hand Siamese *k'ram* 'indigo' < older *gram*, then the Arch. guttural before *lám* is certain: Arch. *glám* 'indigo'. Again, there is an interesting case in which the guttural before *l* can be shown to have lived down to early Han time, which Prof. G. Morgenstierne has pointed out to me. The city Lou-lan at Lop-nor, first found and excavated by Sven Hedin, was called 樓蘭 already in Chang K'ien's travel report (2d c. B. C.), and this transcription of the foreign word must be approximately of that age. The *lou* has the same phonetic as 婁 *g'iu*, revealing a guttural, and the city is called *Kroraimna* in Kharoshti documents (Stein, *Serindia* p. 41 a). So *l₂u* was Han *glu*. In both these examples the alt. A. is excluded.

H. Maspero (Le dialecte de Tch'ang-ngan sous les T'ang) has identified 變 *piän* (phon. *l₂wän*) with Siamese *pien*. If this is right, which seems probable, then this is a case where alt. B. is excluded. And here again we can find a trace of a *kl*- in the Anc. *k*- member (各 *kák*) even down in Han time. 各 *kák* has as homophone 閣 *kák*. 各 serves as phonetic in 洛 *glák* (with the *g*- restituted according to what was said of alt. A. above). Was 閣 Arch. *kák* or *klák*? It occurs in Shī king, Sī kan, in a phrase 約之閣閣 where *kák* 'chamber' gives no sense. Mao Heng (middle of 2nd c. B. C.) explains it by a method often resorted to by early commentators. He considers it as a *kia tsie* for another word with a similar sound, and says «*kák kák* is equal to 歷歷 *liek liek*.» This phonetic gloss, impossible and meaningless if 閣 were an Arch. *kák*, is comprehensible if it was an Arch. *klák*: «*klák-klák* is equal to *liek-liek* (such a measure of phonetic dissimilarity occurs sometimes in the *kia tsie*). This decidedly speaks in favour of alt. C, and therefore we obtain 各 *klák*: 洛 *glák* etc.

Even if we can obtain a definite answer in this case, and if it seems probable that there are many analogous cases, we must not, on the other hand, generalize too

rigidly and conclude that it was always so; there is of course no guarantee that the hie sheng creators did not apply sometimes the A type and sometimes the B type just as well as the C type.

* * *

We are now, finally, going to dress a series of tables of words which can be suspected of being cognate, i. e. of forming word families. In order to be cautious at the start, I shall keep within certain fairly narrow limits in the present paper. In the first place I leave out entirely words which consist of only *two* elements, an initial and a vowel (or diphthong). A comparison of words like *ku: ko*, *pá: pia* etc. is very risky, because the word bodies are too short. There is infinitely greater chance of hitting the truth in words with *three* elements: initial, vowel (diphthong) and final: *kán, gían: k'íwan; tung: tók: d'óg* and such-like. In the second place, it is quite possible that words with extremely different initials are really cognate — especially in the light of other Sinitic languages which show us that a simple Chinese initial is often a violent reduction of a long consonant group (Tib. *brgyad* = Chin. *pwal* 'eight') — and that e. g. 時 Arch. *ḍiæg* may be in affinity with 期 *g'ig*; 壽 *ḍióg* with 考 *k'óg* and 老 *lóg*. But at present I leave all such questions open as a *cura posterior*. Here I keep within the limits of phonetically cognate groups, and quite arbitrarily I decide not to go outside the following principal categories.

First I divide the words into three great groups according to finals:

1. *-ng, -k, -g*;
2. *-m, -p, -b*;
3. *-n, -t, -d, -r*.

Then I subdivide these principal groups into categories according to Archaic initials:

- A. *k-, k'-, g-, g'-, ng- ḡ-, '-*;
- B. *t-, t'-, d-, d'-, ṭ-, ṭ'-, ḍ-, ḍ'-; ts-, ts'-, dz-, dz'-, tṣ-, tṣ'-, dṣ'-; ś-, s-, z-, -ṣ;*
- C. *n-, ṇ-, l-*;
- D. *p-, p'-, b'-, m-*.

(Words with initial consonant groups (*kl-, gl-* etc.) I consider to be so risky materials that I have only adduced them in a few cases).

For the vowels, on the contrary, I make no group limitations. Experience from Tibetan teaches us that this language has a richly developed *Ablaut* which allows

of the most varied vocalism inside the same word stem. I have come to the conviction that the same phenomenon obtains in Chinese.

Of the tones I take no notice at all. It would not do simply to apply the Anc. tones to the Arch. words, and it is doubtful if we can ever arrive at a detailed knowledge of the Arch. tone system. Hence the phonetic difference between the words of my tables is often greater than it appears from the forms given: there is often a tonal difference as well, which is not marked in my transcriptions. I hope to revert to this question in a future paper.

The purport of the tables should not be misunderstood. I am very far from affirming that all the words in each group *are* cognate; I only mean to say that they may be suspected of being cognate. In a few cases the affinity is absolutely obvious and certain. In many more it is strongly probable. In the rest it is only possible and at least worth discussion. So each small »family group» has to be considered merely as a kind of *frame*, containing materials from which a choice will have to be made in future. Definite results can only be gained by comparative Sinitic researches, for the phonetic similarity can sometimes very well be deceptive. There is, for instance, such a large number of Chinese words which all end in *-ng* that we may well suspect that many of them derive from Sinitic words ending in quite other consonants. Again, the *č-* and *š-* series may be a result of a simplification of the most varied Sinitic consonant groups. Therefore, at best, only a part of the combinations can be true word families; many similarities must reasonably be due to chance. And yet I do not hesitate to put up these frames, for a start has to be made and I see no other way of tackling the problem.

The present collection of possibly cognate words is not meant to be exhaustive; very many more could be adduced, but at present I only wish to give a few examples.

For an investigation like the present one it is necessary to be critical as to the words adduced. They must be words well known to have been real, living words. If we should draw upon the Kuang yün and Tsi yün, with their tens of thousands of »dictionary words», or even if we took all words for granted which are given in the earliest dictionaries, Er ya, Ts'ang kie p'ien, Fang yen, Shuo wen kie tsī, Kuang ya, we could easily obtain very large groups of »cognate words». But such materials are not acceptable. I adduce only words which either belong to the most common and current words of the language — these are the majority in my tables — or, if less common, are well attested in early *texts*.

A. WORDS OF TYPE K-NG.

1 景 2 鏡 3 光 4 晃 5 煌 6 旺 7 瑩 8 耿 9 頌 10 炯 11 熒
 12 螢 13 杲 14 赫 15 旭 16 熙 17 喜 18 曉 19 映 20 行 21 徨 22
 往 23 迂 24 街 25 巷 26 邀 27 諧 28 告 29 更 30 改 31 麴
 32 酵 33 迎 34 逆 35 糠 36 穀 37 穀 38 癭 39 癰 40 鴻
 41 鵠 42 浴 43 沃 44 渥 45 形 46 營 47 影 48 亢 49 狂 50
 競 51 衡 52 橫 53 扁 54 杠 55 至 56 涇 57 漿 58 澄 59 江 60
 潢 61 津 62 泳 63 澤 64 洪 65 浩 66 瀕 67 洶 68 決 69 注 70 滄
 71 擴 72 恆 73 驚 74 警 75 敬 76 惶 77 匪 78 競 79 恭 80 恐 81
 嬰 82 懼 83 駭 84 忌 85 噩 86 愕 87 恍 88 恂 89 嚇 90 競 91 岡
 92 擎 93 陞 94 嶸 95 扛 96 企 97 起 98 高 99 踞 100 喬 101 翹 102 丘
 103 印 104 昂 105 仰 106 嶢 107 顛 108 嶽 109 崖 110 危 111 傲 112 堯 113 峽
 114 興 115 香 116 馨 117 馨 118 炕 119 曠 120 洞 121 罌 122 稿 123 稿
 124 菱 125 糗 126 熬 127 烘 128 煨 129 頸 130 頸 131 項 132 脛 133 骸
 134 康 135 慶 136 幸 137 祺 138 喜 139 好 140 誑 141 誑 142 惑 143 乖
 144 誑 145 詭 146 怪 147 欺 148 亢 149 狡 150 矯 151 疑 152 誑

1. *kjǎng* bright, light, scenery etc.: 2. *kjǎng* (light-reflector:) mirror: 3. *kwáng* light, brightness: 4. *g'wáng* bright: 5. *g'wáng* bright, to blaze: 6. *giwang* bright: 7. *giwǎng* glittering, as a gem: 8. *kěng* brilliant: 9, 10. *kiweng* light, bright: 11. *g'iweng* bright, lights: 12. *g'iweng* glow-worm, firefly: 13. *kog* bright; 14. *ǎk* burning, brilliant: 15. *xiuk* brightness: 16. *xiǎg* bright: 17. *xiǎg* bright: 18. *xiog* dawn, light; 19. *ǎng* bright.

20. *g'ǎng* to walk, go, a street: 21. *g'wáng* to go to and fro: 22. *giwang* to walk, go: 23. *giwang* to walk, go: 24. *kǎg* street: 25. *g'ǎng* street, lane; 26. *ngog* to ramble, stroll.

27. *kǔng* to speak, explain: 28. *kók*, *kóg* to tell.

29. *kǎng* to change, alter: 30. *kǎg* to change, alter.

31. *k'iók* leaven, yeast: 32. *kóg* leaven, yeast.

33. *ngǎng* to meet, go out to meet: 34. *ngǎk* to go out to meet, go against, oppose.

35. *k'áng* husk of grain: 36. *k'ǎk* husk of grain: 37. *kuk* (husked things:) grain.

38. *ǎng* a swelling, tumour: 39. *ǎng* carbuncle, ulcer.

40. *g'ung* wild goose, wild swan: 41. *kók* snow-goose, swan.

42. *giuk* to bathe; 43. *ók* to soak, moisten: 44. *ǎk* to soak.

A

45. *g'ien*g contour, shape, form: 46. *giwěng* to draw a plan, to plan (to build etc.); 47. *·iǎng* form, image, shadow.

48. *k'áng* violent: 49. *g'iwang* violent, mad, furious: 50. *g'ǎng* to be violent, quarrel.

51. *g'ǎng* yoke of an ox, horizontal bar of a balance, cross-wise: 52. *g'wǎng* cross-wise, horizontal: 53. *kiweng* door-bar, bolt: 54. *kǔng* cross-bar.

55. *kieng* underground stream: 56. *kieng* to flow: 57. *g'iweng* rivulet: 58. *giwěng* rivulet: 59. *kǔng* river: 60. *g'wǎng* accumulated water: 61. *g'ien*g watery expanse: 62. *giwěng* to wade in water: 63. *k'óng* flood, inundation: 64. *g'ung* flood, inundation: 65. *g'óg* expanse of water: 66. *xung* expanse of water: 67. *xiung* to flow, rushing water: 68. *·iang* to flow, float: 69. *·wǎng* expanse of water: 70. *·ung* to flow, float.

71. *k'wáng* to hate, be annoyed at, abhor: 72. *g'ien*g to be annoyed.

73. *k'ǎng* frightened: 74. *k'ǎng* (to frighten:) to warn: 75. *k'ǎng* (awed:) respectful: 76. *g'wǎng* frightened: 77. *k'iwang* frightened: 78. *k'ǎng* frightened, respectful: 79. *k'ung* respectful: 80. *k'ung* to be frightened, fear: 81. *k'iwak* frightened looks: 82. *k'iwak* startled: 83. *g'eg* frightened: 84. *g'ǐəg* to fear: 85, 86. *ngák* to scare, scared: 87. *xwǎng* troubled: 88. *xiung* frightened: 89. *ǎk* frightened: 90. *ǎk* frightened.

91. *kǎng* mountain ridge: 92. *g'ǎng* to lift: 93. *g'ien*g cliff, precipitous: 94. *g'wěng* lofty: 95. *kǔng* to lift: 96. *k'ǐəg* to raise oneself on the toes: 97. *k'ǐəg* to lift, rise: 98. *kog* high: 99. *k'iog* to raise oneself on the toes: 101. *g'iog* to lift: 102. *k'ióg* hill: 103, 104. *ngǎng* high, to raise: 105. *ngiang* to raise the eyes, lift the face, look upwards: 106. *ngák* cliff, hill-side, edge: 107. *ngǎk* top of the head, forehead: 108. *ngók* mountain, peak: 109. *ngǐg* cliff, hill-side, edge: 110. *ngiŭg* high, precipitous, dangerous: 111. *ngog* haughty: 112. *ngiog* high, lofty: 113. *ngiog* high, precipitous: 114. *xiang* to lift, raise.

115. *xiang* fragrant: 116. *xiang* musk: 117. *xieng* fragrant.

118. *k'áng* to dry: 119. *k'wáng* sunburnt, desolate, waste: 120. *g'ák* to dry up: 121. *g'ók* a spring drying up, become dry: 122. *kog* straw: 123. *kog* dry, withered, rotten: 124. *kōg* dried grass: 125. *k'ióg* parched rice, dry provisions: 126. *ngog* to dry, roast: 127. *xung* to burn, roast: 128. *xok* hot, burning.

129. *k'ǐəng* neck, throat: 130. *kieng* to cut the neck, behead: 131. *g'ǔng* neck.

132. *g'ien*g shin-bone, shank: 133. *g'eg* shin-bone, shank.

134. *k'áng* rich year, prosperity: 135. *k'ǎng* felicity, blessings, to felicitate: 136. *g'ěng* luck, fortunate: 137. *g'ǐəg* felicity: 138. *xiəg* joy, to rejoice: 139. *xóg* to find pleasure in, to love.

A

140. *kīwang* to deceive, cheat: 141. *g'īwang* to deceive, lie: 142. *g'wək* deceive, mislead, doubt: 143. *kwǝg* deceitful, crafty: 144. *kwǝg* to deceive: 145. *kīwǝg* deceitful, to deceive: 146. *kweg* bewildered, astonished: 147. *k'īæg* to deceive, to cheat: 148. *kīwæg* traitor: 149. *kog* crafty: 150. *kīog* to feign; 151. *ngīæg* doubt, to be in doubt; 152. *χwǝng* to lie.

153 梗 154 颞 155 荊 156 疆 157 耕 158 穎 159 刑 160 研 161 獲 162 刻 163 棘
 164 戩 165 剗 166 鉸 167 鋸 168 疆 169 竟 170 境 171 垌 172 互 173 蔚 174
 郭 175 極 176 國 177 域 178 囿 179 永 180 詠 181 恆 182 久 183 疚 184
 綱 185 綆 186 韁 187 韁 188 經 189 紘 190 鞅 191 望 192 羅 193 繫 194 系 195
 係 196 絞 197 敲 198 繳 199 糾 200 繯 201 鞅 202 縲 203 策 204 縊 205 約 206
 坑 207 塋 208 陸 209 礪 210 磬 211 磬 212 望 213 孔 214 空 215 腔 216 望
 217 谷 218 壞 219 臼 220 齧 221 窖 222 胸 223 整 224 洩 225 枵 226 腐 227 臄
 228 慷 229 廣 230 宏 231 弘 232 擴 233 廓 234 吳 235 呼 236 瓊 237 絳
 238 紅 239 頤 240 僵 241 傾 242 降 243 航 244 匡 245 公 246 考 247
 舊 248 舅 249 翁 250 剛 251 銅 252 僵 253 強 254 競 255 勁 256 確 257 硬
 258 凝 259 垠 260 缸 261 銅 262 盜 263 嬰 264 嬰 265 寶 266 枉 267 肱
 268 弓 269 鏗 270 曲 271 踣 272 屮 273 尫 274 奧 275 澳 276 皇 277 王
 278 獲 279 搜 280 勾 281 拘 282 據 283 攜 284 右 285 有 286 握 287 擁 288
 厄 289 扼 290 阨 291 軼 292 噬 293 抑 294 隘 295 戰 296 擊 297 摑 298 殛
 299 蹙 300 考 301 敲 302 縞 303 皓 304 皤 305 皤 306 皎

153. *kǝng* spinous tree, thorny: 154. *kǝng* fish-bones, pricking: 155. *kǝng* bramble, thorn: 156. *kwǝng* awn of wheat etc: 157. *kǝng* plough-bill: 158. *g'īwǝng* awn, tip, sharp: 159. *g'īeng* (to cut:) to amputate, punish: 160. *g'īeng* (sharpen:) whetstone: 161. *g'wǝk* to cut the grain: 162. *k'ǝk* to cut: 163. *kǝk* brambles, thorny: 164. *kwǝk* cut off the ear (of an enemy, trophy): 165. *k'īweg* to stab, to cut: 166. *kog* sharp point, scissors; 167. *ngǝk* point, edge of a blade.

168. *kǝng* limit, boundary, frontier: 169. *kǝng* limit, end, finish: 170. *kǝng* limit, boundary, region: 171. *kiweng* border regions, frontier area: 172. *kǝng* extreme, limit: 173. *g'īǝng* extreme, limit, end, go to the extreme, exhaust, poor: 174. *kwǝk* outer wall of a city (its boundary): 175. *g'īǝk* extreme, end: 176. *kwǝk* (delimited, boundaried area:) state, country: 177. *g'īwǝk* boundary, region, state: 178. *g'īǝg*, *g'īǝk* (fenced area:) park.

179. *g'īwǝng* long, distant, continuous, eternal: 180. *g'īwǝng* (to draw out the words:) recite, sing: 181. *g'ǝng* perpetual, constant: 182. *kǝg* a long time, long: 183. *kǝg* chronic disease.

A

184. *kǎng* string, tie, bond: 185. *kǎng* a long rope: 186. *kǎng* reins, bridle: 187. *kǎng* strings by which to wrap a child and carry it on the back: 188. *kieng* threads of a warp: 189. *g'weng* hat string, to tie: 190. *k'ung* bridle, halter: 191. *k'ung* to tie with a leather strap: 192. *k'ieŋ* halter: 193. *g'ieg* to tie, bind: 194, 195. *g'ieg* to bind, connect: 196. *kog* to wrap, to wind around, strangle: 197. *k'og* to tie: 198. *k'og* to wind around, bind: 199. *k'og* threefold cord, to tie: 200. *χwǎk* string: 201. *·iang* halter: 202. *·iēng* cap string, tassel: 203. *·iwēng* entwine: 204. *·iek* to strangle: 205. *·iok* to bind.

206. *k'ǎng* hole, pit, moat, canal: 207. *k'wǎng* grave, tomb, vault: 208. *g'wǎng* empty, city moat: 209. *kwǎng* a mine: 210. *k'ieŋ* empty, hollow: 211. *k'ieŋ* (hollow stone:) instrument of sonorous stone: 212. *g'iwēng* grave: 213. *k'ung* hole: 214. *k'ung* hollow, empty: 215. *k'ǎng* chest: 216. *k'ung* eyehole in axe: 217. *kuk* ravine, valley: 218. *g'og* moat: 219. *g'igog* mortar: 220. *k'ioŋ* hole: 221. *k'og* cave: 222. *χiung* chest: 223. *χǎk* ravine, gully: 224. *χiwǎk* moat, canal: 225. *χioŋ* hollow tree, hollow, empty: 226. *·iēng* chest, breast: 227. *·iək* breast.

228. *k'áng* wide-hearted, magnanimous: 229. *kwǎng* wide, broad, vast: 230. *g'weng* spacious, large hall: 231. *g'wǎng* vast, liberal: 232. *k'wǎk* to widen, enlarge, extend: 233. *k'wǎk* wide, vast: 234. *g'óg* vast (as the sky): 235. *χioŋ* vast.

236. *g'iwēng* red-coloured precious stone: 237. *k'ong* strong red, purple: 238. *g'ung* red.

239. *g'áng* to go down in flying (birds): 240. *k'iang* to fall down, prostrate: 241. *k'iwēng* to fall down, tumble over: 242. *k'ong* to descend, go down, throw down.

243. *g'áng* square raft, two boats lashed together so as to form a square: 244. *k'iwang* square.

245. *kung* »old man», father: 246. *k'og* old: 247. *g'igog* ancient, old: 248. *g'igog* (»old man»:) uncle: 249. *·ung*, father, old man. Possibly *χiwǎng*, Mand. »hiung» 'elder brother' belongs here too.

250. *kǎng* hard: 251. *kǎng* steel: 252. *k'iang* stiff, rigid: 253. *g'iang* strong: 254. *g'ǎng* strong (see Tso chuan, Hi, 7th year): 255. *k'ieŋ* strong, vigorous: 256. *k'ók* solid, hard: 257. *ngǎng* hard: 258. *ng'iang* (frozen water:) to become hard, solid, congeal.

259. *k'ung* jar: 260. *g'ǎng* jar: 261. *g'ieŋ* soup-tureen: 262. *·áng* bowl, basin: 263, 264. *·ēng* jar: 265. *·ung* jar.

266. *g'iwang* to bend, crooked: 267. *kwǎng* (bending part:) elbow: 268. *k'üŋ* a bow: 269. *k'iwak* hook: 270. *k'üuk* to bend, crooked: 271. *g'üuk* crooked, cramped: 272. *k'igog* hook: 273. *·wǎng* crooked leg, lame: 274. *·og* angle, corner of the house: 275. *·iok* (»hook») the concave side of a bend in a meandering river.

A

276. *g'wǎng* emperor, imperial: 277. *giwang* king, royal.

278. *g'wǎk* to catch, seize: 279. *kɿwak* to grasp, seize: 280. *kɿók* a handful: 281. *kɿók* to hold in both hands: 282. *kɿwag* to lay hands on, seize, hold: 283. *g'ieg* to hold by the hand, lead: 284. *giǔg* right hand: 285. *giǔg* to hold, have; 286. *·ūk* to grasp, seize.

287. *·iung* to press: 288. *·ėk* narrow, straits: 289. *·ėk* to press, throttle, grasp firmly: 290. *·ėk* narrow pass, defile: 291. *·ėk* yoke, to restrain: 292. *·iėk* throat: 293. *·iək* to press down: 294. *·ėg* a pass, straits, narrow.

295. *kɿak* long lance: 296. *kiek* to beat, strike, kill: 297. *kwek* to beat: 298. *kɿək* to kill: 299. *kóg* large drum (which is beaten): 300. *k'óg* to beat: 301. *k'og* to beat.

302. *kog* white silk: 303, 304. *g'óg* white: 305. *kǐög* white: 306. *kiog* white.

307 鞠 308 鞠 ○ 309 盪 310 盪 311 霍 312 瘡 ○ 313 脛 314 鞋 ○ 315 陳 316
 卻 317 邇 318 隔 319 隔 320 解 321 異 ○ 322 殃 323 惡 324 亞 325 虐 ○ 326
 冀 327 欲 328 求 329 要 ○ 330 鞞 331 革 ○ 332 頰 333 頤 ○ 334 跪 335 跽
 ○ 336 蟹 337 基 ○ 338 蟹 339 學 340 校 341 效 342 敦 343 學 344 巧 345 考
 ○ 346 豪 347 驍 348 趙 ○ 349 奇 350 幽 351 黠 352 竈 ○ 353 嬌 354 夭 355
 妖 356 么 ○ 357 交 358 交 ○ 359 綯 360 球 361 毬 ○ 362 鳥 363 鴉 364 鴿
 365 毫 366 裴 367 九 368 逵 ○ 369 覺 ○

307. *kɿók* to rear, nourish: 308. *χɿók* to rear, nourish.

309. *kɿak* haste, urgent: 310. *g'ɿwag* sudden, rapid: 311. *χwák* sudden: 312. *χwák* (sudden illness:) cholera.

313. *kɿak* leg, foot: 314. *g'ėg* boot.

315. *k'ɿak* rift, crack: 316. *k'ɿăk* (to separate oneself from:) reject, decline: 317. *k'wák* rift in the clouds, weather clearing: 318. *kėk* to separate, partition: 319. *kėk* diaphragm: 320. *kėg* to separate, divide, dissolve: 321. *giəg* separate, different.

322. *·iang* misfortune, calamity, to destroy: 323. *·ăk* bad, vicious, wicked: 323. *·ăg* to hate: 324. *·ăg* inferior; 325. *ngɿok* cruel, wicked, to maltreat, destroy.

326. *kɿək* to wish, hope for: 327. *giuk* to wish, desire: 328. *g'ɿóg* to seek for, pray for, entreat, aim at: 329. *·iog* to wish, seek for.

330. *k'wák* leather: 331. *kėk* hide, skin, to flay.

332. *g'əg* chin, jaw: 333. *giəg* chin, jaw.

334. *g'ɿwǔg* to kneel: 335. *g'ɿəg* to kneel.

336. *g'ėg* crab: 337. *g'ɿəg* small crab.

338. *g'wǎng* school: 339. *g'ók* to study, learn, a school: 340. *kög* school: 340. *g'ög*

A

to study, examine, compare: 341. *g'ōg* (to learn:) to imitate: 342. *kōg* to teach: 343. *g'ōg* to teach: 344. *k'ōg* (trained:) skilled: 345. *k'ōg* (to study): to examine.

346. *g'og* vigorous, martial, brave: 347. *kiog* vigorous: 348. *kiōg* vigorous, courageous.

349. *·iog* sundown, darkness: 350, 351. *·iōg* dark, black: 352. *·iōg* secluded.

353. *kiog* delicate, beautiful; 354. *·iog* young, delicate, tender, fresh, beautiful:

355. *·iog* to die young: 356. *·iōg* small, tender.

357. *kōg* to cross, entwine: 358. *g'ōg* to cross.

359. *kiōk* foot-ball: 360. *g'iōg* jade ball: 361. *g'iōg* ball.

362. *kiōg* owl: 363. *giog* (so in Ts'ie yün) owl (the dialects point to an Arch. *χiog*); 364. *χiōg* owl.

365. *g'og* hair: 366. *g'iōg* fur.

367. *k'üŋ* nine: 368. *g'iwəg* point where nine roads meet.

369. *kōk* to wake up: 369. *kōg* to wake up.

B. WORDS OF TYPE T-NG.

1 償 2 貽 3 賜 4 贈 5 賞 6 賜 7 正 8 政 9 整 10 征 11 懲
 12 董 13 職 14 飭 15 勅 16 帝 17 治 18 則 19 司 20 正 21 直
 22 植 23 置 24 蒔 25 栽 26 章 27 程 28 稱 29 度 30 尺 31 測 32 商
 33 升 34 通 35 桶 36 筩 37 筒 38 衙 39 衙 40 銃 41 潰 42 發
 43 窗 44 撐 45 撐 46 板 47 璋 48 杖 49 丈 50 珽 51 挺 52 筵 53 挺
 54 楨 55 楹 56 橋 57 杙 58 支 59 枝 60 肢 61 條 62 橋 63 棚 64 上
 65 尚 66 揚 67 賜 68 頂 69 登 70 乘 71 棟 72 冢 73 塚 74 戴
 75 陟 76 卓 77 提 78 臺 79 擡 80 崇 81 截 82 穎 83 昇 84 嵩 85 盈
 86 贏 87 贏 88 盛 89 充 90 容 91 勇 92 阻 93 獎 94 壯 95 勝
 96 龍 97 憚 98 祉 99 怡 100 悰 101 偵 102 瞪 103 督 104 覲 105 矚
 106 眙 107 眇 108 相 109 悞 110 省 111 伺 112 打 113 鉦 114 沖 115 撞 116 鐘
 117 衝 118 鏞 119 柝 120 鐸 121 櫨 122 策 123 極 124 錫 125 觸 126 咎 127 擣
 128 舂 129 膛 130 宕 131 洞 132 井 133 井 134 中 135 仲 136 堂 137 廷
 138 庭 139 寺 140 宗 141 脹 142 漲 143 腫 144 瘡 145 瘡 146 成
 148 終 149 己

1. *diang* to give compensation: 2. *diəg* to give, bequeath: 3. *iōg* to give; 4. *dz'əng* to give, bestow; 5. *siang* to give, bestow, reward: 6. *sieŋ* to give, bestow.

7. *iēng* straight, correct: 8. *iēng* (correcting, regulating, adjustment:) government, administration: 9. *iēng* (to make straight:) to adjust: 10. *iēng* (to correct:)

B

to punish: 11. *d'iang* to correct, punish: 12. *tung* to correct, govern: 13. *iək* to direct, govern, office, official: 14, 15. *t'ək* to direct, to order: 16. *tiæg* ruler, emperor: 17. *d'iaæg* to govern; 18. *tsək* law, rule; 19. *siæg* to direct, govern, manage.

20. (cf. the preceding) *iëng* straight: 21. *d'ək* straight, upright: 22. *diək* door-post, to set upright, to erect, to plant: 23. *tiæg* to put up, establish, place: 24. *diæg* to erect, to plant; 25. *tsæg* to plant.

26. *iang* a measure, norm, rule: 27. *d'icng* measure, to measure (weight, length, volume): 28. *i'iang* to weigh, steelyard: 29. *d'ák* to measure, *d'ág* a measure: 30. *i'iak* a measure of ten inches; 31. *ts'iek* to measure, to fathom; 32. *siang* to measure, to appreciate, to deliberate: 33. *sieng* a pint.

34. *t'ung* to pass through, communicate (all through, all etc.): 35. *t'ung* (tube-formed:) tub, barrel: 36, 37. *d'ung* tube, pipe: 38. *d'ung* passage, connecting lane: 39. *i'ung* passage, connecting lane: 40. *i'iong* hole through the head of an axe: 41. *d'uk* sluice, drain, gutter, ditch: 42. *d'üg* sluice, drain, gutter, ditch; 43. *ts'ung* vent, flue, window.

44, 45. *t'äng* a post, pole, to pole: 46. *d'äng* a prop, post: 47. *iang* sceptre: 48. *d'iang* staff, pole: 49. *d'iang* a length of ten feet: 50. *i'eng* sceptre, baton: 51. *i'eng*, *d'ieng* to stick out, stiff: 52. *d'ieng* stalk, straw, small beam: 53. *d'ieng* stalk, stick, staff: 54. *tiëng* pole: 55. *diëng* pillar, column: 56. *tüng* pole, post, stake: 57. *diək* stake: 58, 59. *i'icg* branch: 60. *i'icg* limb: 61. *d'ióg* branch, stick; 62. *dz'iang* boom, spar: 63. *ts'ek* bars, fence, palissade.

64. *liang* above, on top, high, ascend: 65. *liang* high: 66. *liang* to raise, lift: 67. *liang* tossed up by the wind: 68. *tieng* top of the head, to carry on the head, summit: 69. *täng* to ascend, rise, mount: 70. *d'iang* to ascend, to mount, ride on: 71. *tung* ridge-pole, the top: 72. *tiung* peak, lofty, mound, tumulus: 73. *tiung* tumulus, tomb: 74. *tag* to carry on the head: 75. *tiək* to ascend, rise: 76. *tök* high, lofty: 77. *d'ieg* to raise high, lift: 78. *d'æg* elevated platform, a look-out, high: 79. *d'æg* to raise high, lift, carry; 80. *dz'ióng* high, lofty: 81. *tsæg* (put on top:) to load (as a car); 82. *säng* (top of the head:) forehead: 83. *sieng* to ascend, rise: 84. *sióng* lofty.

85. *diëng* full: 86. *diëng* full, ample, surplus: 87. *diëng* abundant, surplus: 88. *diëng* to be full of, hold, contain; abundant, overflow: 89. *i'iong* full, to fill: 90. *diung* (to be full of:) contain, hold.

91. *diung* vigorous, brave; 92. *tsäng* strong horse: 93. *tsiang* to encourage: 94. *tsiang* strong, vigorous, robust; 95. *sieng* to have the force for doing, capable; (to be the stronger:) to conquer.

B

96. *t'jung* to find pleasure in; be in favour: 97. *djak* pleased, happy, joy: 98. *t'jæg* happy, happiness: 99. *djæg* pleased, joy; 100. *dz'óng* pleased, glad, joy.

101. *t'jæng* to spy: 102 *d'eng* to stare: 103. *tók* to supervise, inspect, examine: 104. *d'ók* to see: 105. *ijuk* to look, stare: 106. *t'jæg* to stare: 107. *t'ioğ* to stare; 108. *siang* to gaze, look, consider, regard: 109. *sieng* (discerning, mentally clear-sighted:) intelligent, to understand: 110. *siæng* to watch, look, examine: 111. *siæg* to watch, spy.

112. *tieng* to beat: 113. *ijæng* a small bell (which is struck): 114. *d'jóng* to dash against: 115. *d'üng* to strike, beat, knock against: 116. *ijung* a bell: 117. *ijung* rush against: 118. *djung* a big bell: 119. *t'ák* »beater», wooden knocker used by night guards: 120. *d'ák* bell with clapper: 121. *d'ók* (beater, propeller:) oar, scull: 122. *tjók* to ram, pound, build: 123. *tük* to beat, strike: 124. *d'ük* small bell: 125. *ijuk* to butt, knock against, rush against, strike: 126. *t'jæg* to beat, flog: 127. *tóg* to pound; 128. *sjung* to pound, to ram.

129. *t'áng* (the dialects indicate a *d'áng*) cavity, hollow, chest, palate: 130. *d'áng* cave, cavern, grotto: 131. *d'ung* cave, hole, pit, ravine: 132. *tsjæng* a well: 133. *dz'jæng* pit, pit-fall, hole.

134. *tjóng* middle, centre, interior, inside: 135. *d'jóng* (the middle one:) second of three (or four) brothers, second of three months.

136. *d'áng* hall: 137. *d'jeng* court: 138. *d'jeng* hall, court; 139. *dzjæg* hall: 140. *tsóng* ancestral hall, temple (the last, however, more probably to gr. 542 below).

141. *tjang* to swell, swelled abdomen, dropsical: 142. *tjang* swelling water, to flood, to rise: 143. *ijung* to swell, tumefy, tumour: 144, 145. *djung* dropsy of leg.

147. *djæng* to complete, finish, achieve: 148. *ijóng* to finish, end; 149. *zjæg* to finish, end.

150. *t'äk* to cleave, split: 151. *ijak* to hew, chop: 152. *tiək* barb of arrow: 153. *tek* (to prick:) to blame, criticize: 154. *t'iek* to cut asunder: 155. *tük* to cut, chop, hew: 156. *tük* to cut gems: 157. *ijæg* a spinous orange tree, thorn: 158. *ijæg* tooth, esp. front tooth: 159. *tog* knife: 160, 161. *tjóg* to cut gems, carve; 162. *dz'jang* to wound, kill: 163. *ts'jang* a wound, sore: 164. *ts'jang* to wound: 165. *tsjak* to cut off: 166. *ts'ek* thorn: 167. *tsək* to blame, criticize: 168. *ts'jək*, *ts'jæg* thorn, to sting, blame, criticize: 169. *dz'ək* a cut-throat, bandit; to hurt, to wound: 170. *ts'jak* a sharp plough share: 171. *dz'ák* chisel, bore: 172. *tsuk* barb of arrow: 173. *ts'ük* to spear fish, pierce, stab: 174. *ts'jæg* thorn: 175. *ts'jæg* to blame, criticize: 176. *tsæg* to slaughter: 177. *dz'æg* to cut: 178. *tsjæg* a hoe: 179. *tsjæg* to cut, slice, mince: 180. *tsjæg* to stab: 181. *dzjæg* plough-share: 182. *tsóg* jujube (thorny): 183. *tsjog* to

B

150	坏	151	斫	152	鏑	153	誦	154	剔	155	斲	156	琢	157	杙	158	齒	159	刀	160	珣
161	彫	162	戕	163	瘡	164	創	165	斲	166	萊	167	責	168	刺	169	賊	170	罍	171	鑿
172	鏃	173	擗	174	刺	175	諫	176	宰	177	裁	178	鉞	179	戢	180	剗	181	起	182	夷
183	刺	184	傷	185	整	186	析	187	削	188	斯	189	撕	○	190	張	191	長	192	昶	193
腸	194	塲	195	掌	196	常	197	敞	○	198	棠	199	同	200	銅	201	調	202	相	○	203
等	204	嶺	205	增	206	層	207	阡	○	208	腺	209	腥	○	210	糖	211	棠	212	橙	213
蔗	214	飴	215	錫	○	216	疼	217	痛	218	恫	○	219	葬	220	賊	221	倉	222	藏	○
223	湧	224	溶	225	濤	226	滔	227	漾	○	228	澤	229	滴	230	涿	231	汜	232	漬	233
滋	234	液	○	235	濯	236	滌	237	洮	238	澡	239	浙	240	漱	241	涓	○	242	陽	243
昌	244	曠	245	的	246	燭	247	熾	248	朝	249	潮	250	耀	251	耀	252	煖	253	昭	254
照	255	晝	256	晶	257	晴	258	晴	259	燭	260	星	261	晰	262	燦	○	263	陽	264	湯
265	煬	266	鼎	267	蒸	268	蒸	269	融	270	鎔	271	揮	272	炙	273	炙	274	粥	275	熟
276	匄	277	陶	278	鑄	279	涸	280	壑	281	壑	282	鼎	283	炒	284	竈	285	焦	286	湘
287	鬲	288	腊	289	鏢	290	燥	291	燒	292	脩	293	銷	294	消	○	295	精	296	清	297
淨	298	激	299	澄	300	湜	301	淑	○												

cut off; 184. *xiang* to wound, injure: 185. *xiak* to sting: 186. *xiək* to cut asunder, cleave: 187. *sio*k to slice off, cut off, to pare: 188, 189. *siə*g to cleave.

190. *tiang* to extend, draw out, spread, expanse, surface: 191. *tiang* become long, grow up: 192. *t'iang* a long day: 191. *d'iang* extended, long: 193. *d'iang* (the long ones:) bowels: 194. *d'iang* area, arena, open space: 195. *tiang* palm of the hand: 196. *diang* (long:) constant: 197. *t'iang* plateau, high open space.

198. *tang* league, union, party, faction: 199. *d'ung* join, together: 200. *d'ung* alloy, bronze: 201. *d'io*g to join, mix; 202. *siang* joined, together, mutual.

203. *tang* steps of staircase, ledge, section, degree: 204. *tang* ledges, tiers of a hill; 205. *tsang* (to place layer on layer:) to pile up, accumulate, add, increase: 206. *dz'ang* layer, stratum, storey, degree: 207. *dz'ag* staircase.

208. *sog* fat, smell of raw meat, rancid: 209. *sieng* fat, smell of raw meat, rancid.

210. *d'ang* sugar: 211. *d'ang* sweet apple: 212. *d'eng* orange (cf. *kan* orange = *kan* the sweet fruit): 213. *liag* sugar cane: 214. *diog* sugar, sweet; 215. *dz'ie*ng sugar, sweets.

216. *d'ong* pain, to ache: 217. *t'ung* pain, to ache: 218. *t'ung* pained, to suffer, moan.

219. *tsang* to hide, bury: 220. *tsang* hidden goods: 221. *ts'ang* store-room, granary: 222. *dz'ang* to hide, to store, store-room.

223. *djung* to well up (as water): 224. *djung* overflowing water: 225. *d'og* great waves: 226. *t'og* swelling and rushing water: 227. *ziang* moving water, waves.

B

228. *d'āk* to soak: 229. *tiēk* to drop, drip: 230. *tūk* to drip, trickle; 231. *dz'ūk* to soak: 232. *dz'ięg* to soak: 233. *tsięg* to soak: 234. *ziak* fluid, juice.

235. *d'ōk*, *d'ōg* to wash, rinse: 236. *d'ōk* to wash, scour: 237. *t'og* to wash, rinse; 238. *tsog* to bathe; 239. *siek* to wash rice: 240. *sīōg* to rinse: 241. *sīōg* water in which rice has been washed.

242. *diang* light and heat of the sun: 243. *i'iang* sunshine, shining: 244. *d'ung* sun just rising: 245. *tiok* bright: 246. *i'uk* torch: 247. *i'ięg* to blaze: 248. *tiog* dawn, morning: 249. *d'iog* morning ceremony, audience etc.: 249. *d'iog* morning tide: 250, 251. *dīog* bright: 252. *dīog* bright, to enlighten: 253. *i'iog* bright, brilliant: 254. *i'iog* to shine upon, enlighten: 255. *tiōg* day-light, day-time (as opp. to night); 256. *tsięng* bright, clear, chrystal: 257. *tsięng* iris of the eye: 258. *dz'ięng* clearing sky, light: 259. *tsiok* torch, to light; *sieng* star: 261. *siek* bright: 262. *sīok* bright, flashing.

263. (cf. the preceding group) *diang* light and heat of the sun: 264. *t'áng* hot liquid: 265. *diang* to roast, to heat, to fuse: 266. *tieng* (boiler:) tripod: 267. *i'ęng* to steam: 268. *i'ęng* firewood, to smoke: 269. *dīōng* to steam, to heat: 270. *dīung* to fuse metal: 271. *t'ák* withered leaves: 272. *i'iak*, *i'iaq* to roast: 273. *i'ok* to roast, burn: 274. *i'ōk* to boil rice, gruel: 275. *dīōk* heated through, well cooked, ripe: 276, 277. *d'ōg* (burner, burnt goods:) kiln; pottery: 278. *i'ōg* to fuse, cast metal: 279. *tiōg* fade, wither: 280. *dīog* kiln: 281. *dīog* jar (burnt piece); 282. *tsięg* a kind of tripod (boiler): 283. *ts'ōg* to fry, roast: 284. *tsōg* stove, furnace: 285. *tsiog* to scorch, burn; 286. *sīang* to boil: 287. *sīang* to boil: 288. *sīak* dried meat: 289. *sīok* to melt; bright: 290. *sog* to dry: 291. *sīog* to burn, bake: 292. *sīōg* dried meat: 293, 294. *sīog* to melt (metal).

295. *tsięng* pure, fine, essence: 296. *ts'ięng* pure, clean, limpid: 297. *dz'ięng* clean, pure, cleanse; 298, 299. *d'ięng* clear, pure, limpid: 300. *dīak* pure, clean water: 301. *dīōk* pure.

302, 303, 304. *dz'ięng* quiet, still: 305. *dz'iek* quiet, still.

306. *t'iang* dejected, disappointed: 307. *i'iang* dejected: 308. *t'ióng* grieved: 309. *d'ieg* to wail: 310. *tog* grieved: 311. *d'og* grieved: 312. *t'iog* grieved: 313. *dīog* distressed: 314. *t'ióg* dejected: 315. *d'ióg* sorry; 316. *ts'iang* grieved: 317. *ts'iak* grief, pity: 318. *ts'og* sad: 319. *ts'iog* grief, sorry: 320. *dz'iog* distressed: 321. *dz'ióg* grief; 322. *sáng* mourning: 323. *sīak* pained, grief, pity.

324. *t'iek* to fear, respectful; 325. *dz'ák* bashful; 326, 327. *sīung* terrified, respectful: 328. *sīōk* terrified, respectful: 329. *sīęg* to fear.

330. *t'ięng* red: 331. *d'ięng* naked (red, flesh-coloured): 332. *d'ōng* red: 333. *i'iak*

B

302 靖 303 諄 304 靜 305 寂 ○ 306 悵 307 悄 308 忡 309 啼 310 忉 311 淖 312
 詔 213 惶 314 惆 315 怵 316 愴 317 惻 318 悵 319 悄 320 悵 321 愁 322 喪 323
 惜 ○ 324 惕 325 悵 326 竦 327 悵 328 肅 329 蕙 ○ 330 頽 331 程 332 彤 333
 赤 334 駢 335 頽 336 舊 337 寇 ○ 338 蒼 339 青 ○ 340 醒 341 耐 342 耐 343
 洒 344 滴 ○ 345 碇 346 訂 347 定 348 亭 停 停 ○ 349 孕 350 育 351 畜 352
 胎 353 子 354 孽 字 355 生 356 姓 357 性 ○ 358 虹 359 蜓 360 蜻 ○ 361 倉
 362 逋 363 仵 364 蹙 365 促 366 蹙 367 造 368 倏 369 倏 370 速 ○ 371 羊 372
 牂 ○ 373 洋 374 瀛 ○ 375 聽 376 聰 ○ 377 頑 378 臧 379 祥 380 祿 ○ 381
 像 382 似 383 俏 384 肖 385 猶 ○ 386 童 387 犖 388 犖 ○ 389 冬 390 凍 391
 滄 392 清 393 霜 ○ 394 知 395 智 396 志 397 誌 398 悠 399 想 400 諒 401 思
 402 猶 ○ 403 擋 404 塘 405 塘 406 帳 407 障 408 障 409 廠 410 城 411 壁 412
 町 413 幃 414 墉 415 場 416 遮 417 幃 418 晴 419 綢 420 侑 421 幃 422 育 ○
 423 攪 424 撥 425 由 426 抽 ○ 427 當 428 食 429 饒 430 饒 431 飼 ○

red; 334. *siəng* red horse: 335. *siak* red: 336. *siək* a red flower: 337. *siōg* Rubia, red-colouring stuff.

338. *ts'ang* green, blue: 339. *ts'ien* green, blue.

340. *d'ieng* drunk: 341. *tieng* drunk: 342. *d'iōg* fine spirits; 343. *tsiōg* spirits: 344. *dz'iōg* spirits.

345. *tieng* (a «fixer»:) anchor: 346. *t'ieng* (to settle:) to judge, decide: 347. *d'ieng* to fix, settle: 348 a. *d'ieng* to settle, to stop: 348 b. *d'ieng* stopping place, resting place, pavilion: 348 c. *d'ieng* stagnant water.

349. *dieng* pregnant: 350. *diōk* to rear and foster children: 351. *l'iōk* to rear, nourish, feed: 352. *t'ag* womb, pregnant; 353. *tsiōg* (progeny:) son: 354. *dz'iag* to breed, rear, bear, nurture; 355. *seng* to bear, beget; live: 356. *siəng* (maternity:) clan (issuing from the same mother): 357. *siəng* (innate qualities:) nature, natural disposition.

358. *tieng* dragon fly: 359. *d'ieng* dragon fly; 360. *ts'ien* dragon fly.

361. *ts'ang* hurried: 362. *ts'ung* hurried: 363. *tsəg* suddenly, abruptly: 364. *tsiōk* to urge, urgent: 365. *ts'juk* to urge: 366. *tsog* to hasten, quickly: 367. *ts'ōg* to hasten: 368, 369. *siōk* suddenly, quickly: 370. *suk* to hurry, quickly.

371. *ziang* sheep (male and female): 372. *tsang* sheep (female).

373. *ziang* ocean: 374. *dieng* ocean.

375. *t'ieng* to hear: 376. *ts'ung* acute of hearing.

377. *tieng* auspicious; 378. *tsang* auspicious: 379. *dziang* auspicious: 380. *dz'ag* auspicious.

B

381. *dziang* image, shape, form, like: 382. *dziæg* like, resembling: 383. *ts'ioŋ* like, resembling: 384. *sioŋ* like, resembling: 385. *ziôg* like, resembling.

386. *d'ung* youngster, young boy or girl: 387. *d'ung* hornless calf: 388. *d'uk* calf.

389. *tông* winter: 390. *tung* to freeze: 391. *ts'iang* cold: 392. *ts'ieŋ* cold: 393. *siang* frost.

394. *tiæg* (to have in the mind:) to know: 395. *tiæg* knowledge, wisdom: 396. *liæg* mind, thought, will, resolution: 397. *liæg* (to have in the mind:) to remember; a record: 398. *diôg* to think (of): 399. *siang* to think (of): 400. *siæk* to know: 401. *siæg* to think (of): 402. *ziôg* a counsel, a plan.

403. *tâng* to screen, protect, prevent, resist: 404. *d'âng* to obstruct, parry: 405. *d'âng* dyke: 406. *tiang* curtain: 407. *liang* dyke, bank: 408. *liang* dyke, screen, barricade, partition: 409. *li'iang* cover, shelter, shed: 410. *li'ieŋ* city wall: 411. *tang* umbrella: 412. *ti'eng* raised path (dyke) between fields: 413. *d'ũng* a plume fan, to screen, curtain: 414. *diung* a wall: 415. *diæk* dyke between fields, boundary: 416. *liæg* to screen, protect, cover, hide: 417. *d'ôg* a covering, *d'iôg* a screen, curtain: 418. *d'iôg* dyke between fields, field: 419. *d'iôg* coverlet, curtain: 420. *tiôg* to veil, conceal: 421. *dz'iang* wall: 422. *seng* film over the eye; obstruction, calamity.

423. *d'ôk* to pull out: 424. *l'ôg* to draw out, pull out: 425. *diôg* come out, out from, from: 426. *l'iôg* to draw out, pull out.

427. *liang* to taste: 428. *d'iæk* to eat: 429. *li'æg* nourishment, food and drink: 430. *l'og* to eat gluttonously: 431. *dziæg* food.

432 邇 433 緯 434 紹 435 縱 436 釋 437 赦 438 寯 439 瀉 ○ 440 踴 441 倒 ○
 442 育 443 苞 ○ 444 暉 445 曷 446 錫 447 遞 448 代 449 斲 ○ 450 夕 451 汨
 452 宿 453 夙 454 夜 455 舍 456 寯 ○ 457 贖 458 籍 459 冊 460 策 ○ 461 策
 462 借 463 債 ○ 464 等 465 待 466 侍 467 俟 468 厠 469 伺 ○ 470 膝 471 繩
 472 統 473 繹 474 繪 475 繳 476 勒 477 屬 478 蜘蛛 479 締 480 紉 481 係 482 紹
 483 紉 484 倭 485 紂 486 綏 487 綏 488 綏 489 綜 490 總 491 席 492 績 493 績
 494 攀 495 緇 496 緇 497 囚 498 素 499 束 500 縮 501 絲 502 縲 503 蛸 504 繡
 ○ 505 腰 506 踰 507 踰 508 迪 509 躅 510 踰 511 踰 512 踰 513 踰 514 之 515
 踰 516 踰 517 從 518 踰 519 踰 520 跡 521 踰 522 踰 523 足 524 踰 525 踰 526
 踰 527 踰 ○ 528 承 529 擇 530 拈 531 拈 532 持 533 受 534 投 535 將 536 捉
 537 采 538 揀 539 稽 540 以 541 收 ○ 542 烝 543 衆 544 氏 545 僑 546 宗 547
 叢 548 積 549 族 550 曹 ○ 551 稍 552 小 553 少 ○ 554 甄 555 律 556 匱 557
 韜 558 鞘 ○ 559 鷄 560 鷄 561 鷄 ○ 562 笛 563 飢 564 麓 565 鳴 566 蕭 ○
 567 旌 568 幟 ○ 569 得 570 迨 571 到 ○ 572 即 573 就 ○ 574 作 575 造 ○

B

432. *d'ag* remiss, careless: 433. *l'ïok* to slacken, indulgent: 434. *l'ïog* to unbend a bow, to slacken, release: 435. *tsiung* to slacken the rein, lax, loose; 436. *sïak* to loosen, let go: 437. *sïag* (to let off:) to pardon, amnesty: 438. *sïag* to let loose, to ease, release: 439. *sïag* (to let loose water:) to drain.

440. *d'ang* to fall, slip: 441. *tog* to fall.

442. *tsiung* leek flower: 443. *ts'ung* leek, onion.

444. *dïak* relay of horses, post: 445. *dïek* to change, *dïeg* (changeable, mobile:) easy: 446. *dïek* chameleon, lizard: 447. *d'ieg* to substitute, take the place of, change: 448. *d'ag* to substitute, take the place of, change; 449. *siek* chameleon, lizard.

450. *dziak* evening: 451. *dziak* evening tide; 452. *sïök* to pass the night: 453. *sïök* early morning (before dawn): 454. *ziag* night: 455. *sïag* (place for passing the night:) hotel, lodging, house: 456. *sïog* night, darkness.

457. *d'uk* writing tablet, document; 458. *dz'ïak* writing tablet, list, register, record: 459, 460. *ts'ek* writing tablet, list, register, record.

461. *t'ag* to lend, to loan (on interest); 462. *tsïak*, *tsïag* to lend, to loan: 463. *tsëg* debt, to owe money.

464. *tang* to wait: 465. *d'ag* to wait, to wait upon, to treat: 466. *dïag* to wait upon; 467. *dë'ïag* to wait; 468. *sïeg* waiter, attendant: 469. *sïag* to wait, to wait upon.

470. *d'ang* string, cord, to tie, to bind: 471. *dïang* string, cord: 472. *t'ung* a thread, to attach, join: 473. *dïak* to draw out threads (from cocoons), unravel: 474. *l'ïak* to weave: 475. *l'ïok* silk string (tied to arrow): 476. *tiok* reins: 477. *l'ïuk* tied to, attached to; *dïuk* attached, subordinated: 478. *tïeg* («weaver») spider: 479. *d'ieg* to tie, knot: 480. *d'ôg* to twist, bind: 481. *t'ôg* twisted silk cord: 482. *dïog* («anknüpfen») to join on, continue: 483. *d'ïôg* thread, to weave, woven silk: 484. *d'ïôg* leather reins: 485. *d'ïôg* crupper: 486. *d'ïôg* to bind, tie: 487. *dïôg* seal ribbon; 488. *tsang* woven silk: 489. *tsông* warp, to weave: 490. *tsung* to tie together, collect: 491. *dziak* (plaited, woven thing:) mat: 492. *tsiek* to twist, spin: 493. *dziuk* («anknüpfen») to join on, continue: 494. *tsïôg* to bind, tie together: 495, 496. *ts'ïôg* crupper: 497. *dziôg* to tie, bind, captivate, a captive; 498. *sdk* cord: 499. *sïuk* to tie together, bundle: 500. *sïuk* to bind, bind together, restrict etc.: 501. *sïag* silk thread: 502. *sog* to draw out threads (from cocoons), unravel: 503. *sëg* («weaver») spider: 504. *siôg* spider.

505. *dïang* to follow, escort: 506. *l'ïung* heel, to follow: 507. *l'ïak* to tread, trample, foot-sole: 508. *d'ïök* to tread, walk: 509. *d'ïuk* to tread, foot-print: 510. *d'ieg* to tread on, kick: 511. *d'ieg* hoof: 512. *d'ieg* leather shoes: 513. *l'ïag* foot, toes: 514

B

liæg to go to: 515. *d'óg* to trample: 516. *tsiung* footsteps, to follow: 517. *dz'jung* to follow: 518. *tsiak* to walk carefully: 519. *tsiak* to walk with small step: 520. *tsiék* foot-prints, traces: 521. *tsiök* to walk carefully: 522. *ts'iók* to trample, kick: 523. *tsiuk* foot: 524. *dziæg* to follow after, succeed: 525. *siaik* slipper: 526. *siék* to go to: 527. *siök* to walk carefully.

528. *diæg* to hold, receive, present: 529. *d'ák* to pick, select: 530. *liak* to pick up, take: 531. *ték* to take, to pick: 532. *d'ieg* to grasp, hold: 533. *diög* to receive (534. *diog* to pass from hand to hand, transmit, same word); 535. *tsiang* to take, bring: 536. *tsük* to catch, seize: 537. *ts'æg* to take, pick: 538. *ts'og* to grasp, hold: 539. *siaik* to harvest: 540. *ziæg* to take: 541. *siög* to take, collect, gather, harvest.

542. *liæg* all: 543. *liung* a crowd, many, all: 544. *diæg* (a group:) a clan, family: 545. *d'ióg* a group, flock, party; comrade: 546. *tsóng* a clan; ancestors etc.: 547. *dz'ung* to crowd, crowded: 548. *tsiék* to accumulate, collect, many; *tsiæg* provisions: 549. *dz'uk* bundle, crowd, group, clan: 550. *dz'óg* group, flock, partner, plural mark.

551. *šög* small quantity, little: 552. *siog* small: 553. *siog* little, few.

554. *i'iang* bow case: 555. *t'ák* sheath of bamboo: 556. *d'uk* a case, sheath: 557. *t'óg* sword case, sheath: 558. *šög* sword case, sheath.

559. *diog* bird of prey, kite, falcon: 560. *tiög* eagle: 561. *dz'ióg* eagle, vulture.

562. *d'iók* flute: 563, 564. *d'ieg* flute: 565. *siög* to blow, whistle: 566. *siög* flute.

567. *tsiæng* flag, banner: 568. *liak* flag, banner.

569. *tak* to reach, obtain: 570. *d'æg* to reach: 571. *tog* to reach.

572. *tsiak* to go to: 573. *dz'ióg* to go to.

574. *tsák* to make, do, act: 575. *dz'óg* to make.

576 勺 577 酌 578 匙 579 匙 580 州 581 酢 582 挑 583 酹 584 醺 ○ 585 斥
586 逐 ○ 587 止 588 島 589 州 590 洲 ○ 591 逖 592 逖 593 迢 ○ 594 箭 595
宋 596 贖 597 市 598 售 599 翟 ○ 600 帚 601 婦 602 掃 ○ 603 騰 604 踊 605
躍 606 跳 607 超 ○ 608 蚤 609 蟲 ○ 610 請 611 祝 612 咒 613 祝 614 衛 615
討 ○ 616 軸 617 周 618 週 ○ 619 啖 620 嘲 621 譙 622 笑 ○ 623 嫫 624 醜
○ 625 伺 626 遼 627 慙 628 癡 629 蚩 ○ 630 是 631 之 632 茲 633 斯 ○ 634
傲 635 肇 636 始 ○ 637 賈 638 袋 ○ 639 箱 640 筍 ○ 641 孀 642 敵 643 擊
○ 644 漳 645 糟 ○ 646 塗 647 緇 648 朔 649 早 650 皂 ○ 651 稻 652 櫻 653
栗 ○ 654 再 655 雙 ○ 656 恥 657 羞 ○ 658 役 659 徭 ○ 660 碩 661 變 ○
662 射 663 弋 664 矰 ○ 665 撻 666 爪 667 蚤 668 撻 669 騷 ○ 670 肘 671 袖
672 手 ○ 673 昨 674 昔 ○ 675 則 676 即 ○ 677 招 678 召 679 誘 ○ 680 證
681 徵 ○ 682 孰 683 曙 ○ 684 植 685 獨 ○ 686 勳 687 惶 688 掉 689 搖 ○
690 駢 691 逃 692 盜 693 趙 ○

B

576. *điok* spoon, ladle: 577. *l̃iok* (to ladle a cup full:) to fill a cup of wine, to pledge: 578. *điẽg* spoon, ladle: 579. *điog* ladle: 580. *đióg* to fill a cup, to pledge (a host pledging a guest); 581. *dz'ák* to fill a cup, to pledge (a guest pledging the host): 582. *t'iog* big ladle; 583. *tsiog* to fill a cup and drink it: 584. *tsiog* (to ladle out wine:) libation sacrifice.

585. *l̃'iak* to drive, expel: 586. *d'iók* to drive, expel.

587. *l̃iæg* islet: 588. *tog* island: 589, 590. *l̃ióg* islet.

591. *t'iek* far off, distant: 592. *điog* far off, distant: 593. *d'iog* far off, distant.

594. *điók* to sell: 595. *d'iok* to buy grain: 596. *điuk* to trade, to deal: 597. *điæg* market, fair: 598. *đióg* to sell, to buy: 599. *t'iog* to sell grain.

600. *l̃ióg* broom: 601, 602. *sóg* to sweep.

603. *d'əng* to jump: 604. *đung* to jump, leap: 605. *điok* to jump, skip, leap: 606. *d'iog* to jump, skip, leap: 607. *t'iog* to leap, skip.

608. *l̃ióng* grasshopper: 609. *d'ióng* insects, vermin.

610. *ts'ięng* to request, pray, seek; 611. *l̃iók* to pray to: 612, 613. *l̃ióg* imprecation: 614. *tóg* to pray: 615. *t'óg* to ask for, to seek.

616. *d'iók* a roller, a pivot: 617. *l̃ióg* a circle, all round: 618. *l̃ióg* a circle, a revolution, a year.

619. *l̃'iæg* to laugh: 620. *tóg* to deride, to ridicule; 621. *dz'iog* to ridicule, to scold: 622. *sioğ* to laugh.

623. *l̃'iæg* ugly: 624. *l̃'ióg* ugly.

625. *t'ung* stupid: 626. *t'ũng* stupid: 627. *tũng* stupid: 628. *t'ięg* stupid: 629. *l̃'ięg* stupid.

630. *điẽg* this: 631. *l̃iæg* him, her, it; 632. *tsiæg* this; 633. *sĩẽg* this.

634. *l̃'iók* to begin: 635. *d'iog* to begin; 636. *sĩæg* to begin.

637. *t'ák* bag, sack: 638. *d'æg* bag, sack.

639. *sĩang* hack of a car, box: 640. *sĩæg* square box.

641. *tiék* (a *vis-à-vis*, a partner, an equal:) principal wife: 642. *d'iek* (a *vis-à-vis*;) adversary, opponent, enemy: 643. *đióg* (a *vis-à-vis*;) adversary, opponent, enemy; to answer back.

644. *tsiæg* dregs: 645. *tsóg* dregs.

646. *d'æg* to blacken the eyebrows: 647. *tsiæg* black; 648. *şák* 1st day of the moon; North (the fundamental sense, therefore, must be 'dark'): 649. *tsóg* early morning: 650. *dz'óg* black.

651. *d'óg* rice; 652. *tsiək* millet: 653. *sĩuk* millet, grain (generally).

654. *tsæg* twice, a second time: 655. *şung* two, a pair, both.

B

656. *t'iaq* shame: 657. *siôg* shame.
 658. *diək* forced labour, expedition: 659. *diog* forced labour, expedition.
 660. *diak* great: 661. *diək* great.
 662. *d'iak*, *d'iaq* to shoot with bow and arrow: 663. *diək* arrow with string attached; 664. *tsəng* arrow with string attached.
 665. *d'iek* to scratch; 666. *tsŏg* nail, claw: 667. *tsôg* to scratch; a flea; 668. *sôg* to scratch with the nails: 669. *sôg* to rub a horse.
 670. *tiôg* arm, elbow; 671. *dziôg* sleeve: 672. *siôg* hand, arm.
 673. *dz'ák* yesterday: 674. *siak* yesterday, previously, long ago.
 675. *tsək* then, thereupon: 676. *tsiak* then, thereupon.
 677. *liog* to call, beckon: 678. *d'iog* to call, summon; 679. *dziôg* to call, allure, entice.
 680. *liəng* to verify, prove, testify, evidence: 681. *tiəng* to verify, prove, testify, evidence.
 682. *diók* who, which: 683. *d'ióg* who?
 684. *d'ək* alone, single: 685. *d'uk* alone, single.
 686. *d'ung* to move: 687. *d'ung* moved, excited: 688. *d'iog* to shake: 689. *diog* to shake.
 690. *t'ieŋg* to run, gallop: 691. *d'og* to run, to run away: 692. *d'ôg* a run-away, a bandit: 693. *d'iog* to run, hasten.

C. WORDS OF TYPE N-NG.

1 溺 2 理 ○ 3 壤 4 農 ○ 5 諾 6 苦 ○ 7 聆 8 耳 ○ 9 攘
 10 掠 11 搨 12 撈 13 揀 ○ 14 量 15 料 ○ 16 朗 17 亮 18 烙 19 璫
 20 瞭 21 燎 ○ 22 力 23 勞 ○ 24 輓 25 輓 26 較 27 輓 28 輓 ○ 29
 樂 30 徠 ○ 31 苓 32 籠 33 麓 ○ 34 撓 35 撓 36 繞 ○ 37 滾 38 零
 39 瀝 40 漉 41 露 42 雷 43 溜 ○ 44 圖 45 牢 ○ 46 戮 47 劉 ○ 48
 繞 49 扭 50 紐 51 縲 ○ 52 校 53 菱 ○ 54 能 55 耐 ○ 56 量 57 仍
 ○ 58 良 59 令 60 靈 ○ 61 陵 62 壘 ○ 63 涼 64 冷 65 凌 ○ 66 膿
 67 濃 68 醞 ○ 69 浪 70 潦 71 流 ○ 72 乃 73 而 ○

1. *lak* veins in stone and mineral: 2. *liəg* veins in stone and mineral.
 3. *niəng* fertile and arable soil, worked soil: 4. *nông* to work the soil, agriculture.
 5. *nāk* (like that, thus, sic:) yes, to say yes: 6. *niak* to resemble, like, as, according to.
 7. *liəng* to hear: 8. *niəg* ear.

C

9. *ńiang* to grasp, seize, snatch: 10. *gliang*, *gliak* to grasp, seize, snatch: 11. *něk* to grasp, seize: 12. *log* to seize, take out: 13. *liog* to grasp, take.

14. *liang* to measure: 15. *liog* to measure.

16. *láng* bright, clear: 17. *gliang* bright, clear: 18. *glák* to burn: 19. *liok* lustre, brilliancy (of gems): 20. *liog* bright, clear (said of eyes): 21. *liog* torch, to blaze, burn, shine, bright.

22. *liək* force, strength: 23. *log* to use force, to toil.

24. *liang* a car (the explanation »a pair« i. e. two-wheeled car is a folk etymology, as shown by the existence of cognate words »car«): 25. *gliang* a car where there is place to lie down: 26. *liəng* to crush under car wheels: 27. *liok* to crush under car wheels: 28. *gląg* chariot.

29. *lāk* joy: 30. *luk* happiness.

31. *lieng* basket: 32. *lung* basket: 33. *luk* basket.

34. *nōg* to disturb, to trouble: 35. *ńiog* to disturb, to trouble: 36. *niog* to trouble, molest.

37. *ńiang* to drip, dew: 38. *lieng* dropping rain, to drip: 39. *liek* to drip, soak: 40. *luk* to drip, soak: 41. *gląg* dew: 42. *lióg* opening in the roof (of the ancient house) for allowing rain water to drip down; to drip: 43. *lióg* to drip.

44. *lieng* prison: 45. *lóg* a fold, enclosure, prison.

46. *gliók* to kill: 47. *lióg* to kill, slay.

48. *ńiog* to tie, wind round: 49. *ńióg* to tie, twist: 50. *ńióg* to tie, a knot: 51. *liog* to tie, twist, coil.

52. *ləng* square piece of wood, angle, angular, edge, corner, lozenge-shaped: 53. *liəng* (the lozenge-shaped fruit:) water-chestnut.

54. *nəng* be able to, can: 55. *nəg* capable of, to endure, bear (55. sometimes used for 54., e. g. in Li ki: Li yün).

56. *nāng* of old, anciently: 57. *ńiəng* as of old, as before.

58. *liang* good: 59. *liəng* good: 60. *lieng* wonderful, divine.

61. *liang* mound, tumulus: 62. *liung* mound, tumulus.

63. *gliang* cold: 64. *lieng* cold (the dialects point to an Arch. *ləng*): 65. *liəng* ice.

66. *nóng* (thick fluid:) pus, matter: 67. *ńiung* thick, rich (fluid): 68. *ńiung* strong, rich (wine).

69. *láng* waves: 70. *log* torrent, flood: 71. *lióg* to flow.

72. *nəg* then, thereupon: 73. *ńiəg* then, thereupon.¹⁾

¹⁾ The word 72. Anc. *nái*, Arch. *nəg* I have studied in my article »The pronoun *kūe* in the Shu king« (Göteborgs Högskolas Årsskrift 1933). In classical script 72. serves regularly for two words,

D. WORDS OF TYPE P -NG.

1 棚 2 棚 3 蓬 4 覆 5 幘 6 盲 7 氓 8 冥 9 瞑 10 瞢 11 蒙
 12 矇 13 夢 14 幪 15 幕 16 幙 17 塤 18 幕 19 膜 20 眈 21 昌 22 帽
 23 霧 24 默 25 墨 26 黑 27 煤 28 溟 29 溼 30 第 31 旁 32
 傍 33 膀 34 傍 35 螃 36 房 37 虻 38 井 39 竝 40 朋 41 副 42 陪 43 明
 44 望 45 烹 46 炳 47 病 48 烽 49 爆 50 曝 51 卜 52 累 53
 炮 54 庖 55 燠 56 炸 57 白 58 帛 59 伯 60 目 61 眸 62 兵
 63 鋒 64 蜂 65 峯 66 剝 67 芒 68 鉞 69 蟲 70 萌 71 苗 72 彫 73
 旆 74 毛 75 旄 76 芒 77 博 78 丕 79 茫 80 滂 81 龐 82 蓀 83
 防 84 屏 85 軒 86 邦 87 封 88 障 89 擄 90 棒 91 搏 92 拍 93 擗
 94 撲 95 秉 96 柄 97 捧 98 奉 99 倭 100 搏 101 拏 102 拏 103 摸 104 擘
 105 擘 106 擘 107 派 108 剖 109 瓶 110 缶 111 甌 112 杯 113 觥
 114 紡 115 縛 116 保 117 包 118 袍 119 袍 120 經 121 北 122 背 123 背 124
 信 125 負 126 脩 127 平 128 杆 129 牌 130 碑 131 鈞 132 崩 133 匄 134
 踞 135 伏 136 服 137 仆 138 馱 139 馱 140 馱 141 馱 142 飄 143 颯 144
 蛟 145 胚 146 胞 147 駁 148 利 149 彪 150 脬 151 鯢

1. *b'äng* awning, shelter, shed: 2. *pěng* shelter, awning: 3. *b'ung* mat cover, awning, sail: 4. *p'îók* to cover: 5. *b'îuk* cover for the head, head-towel, turban, cap; 6. *mǎng* blind: 7. *mǎng* (»the stupid ones», *yü min*.) the common people: 8. *mieng* (sun covered:) darkness, dark: 9. *mieng* to shut the eyes, bad sight: 10. *mäng* darkened, stupid, ignorant: 11. *mung* to cover; ignorant: 12. *mung* a film over the pupil, blind, ignorant: 13. *mung* (shut eyes:) to sleep, dream: 14. *mung* a cover, head-towel: 15. *miek* to cover with a towel: 16. *miek* to cover, a veil: 17. *miek* to cover with plaster: 18. *mág* (a cover:) veil, curtain: 19. *mág* membrane covering a muscle: 20. *mog* dim-sighted: 21. *móg* to cover, veil, cap: 22. *móg* a cap: 23. *müj* fog, mist.

24. *mək* black, dark, silent: 25. *mək* ink (made from soot): 26. *χmək* black: 27. *mwəg* soot-black, ink; coal.

Anc. *nái* 'then, thereupon' and Anc. *nái* 'your'. In the former sense the bronze inscriptions mostly have another character, yet 72. sometimes serves also in this sense of 'then' in Archaic script as well, e. g. in the O hou ting inscr. (K'i ku shi ki kin wen shu k 2, p. 7).

B. Schindler, in *Asia Major* 1933, has published a kind of «criticism» of my above-mentioned article. It is really discouraging to find that he studies Arch. grammatical particles by adducing examples right and left from all kinds of texts without discrimination and without suspecting an important fact which I proved a decennium ago (On the authenticity and nature of the Tso chuan): that the use of the grammatical particles was not at all the same in all Arch. texts but that marked differences existed, based on different dialects. Moreover, a good part of his Shu king examples are drawn from the *spurious* Shu king chapters!

D

28. *mieng* drizzle: 29. *mung*: drizzle.

30. *pwǎng* (be at the side of:) to assist, help: 31, 32. *b'wǎng* side, beside: 33. *b'wǎng* sides of the body, loins: 34. *b'wǎng* to walk beside, accompany: 35. *b'wǎng* («side-walker») crab: 36. *b'iwang* side-room: 37. *b'ǔng* crab: 38. *piěng* side by side, two together, abreast: 39. *b'ien* side by side, two together, abreast: 40. *b'ang* companion, associate, friend; a pair: 41. *p'ǐk* an associate, assistant: 42. *b'wəg* (to be at the side of:) to accompany, to second, aid.

43. *mǎng* bright, luminous, clear: 44. *mǐwang* full moon.

45. *p'ǎng* to boil, fry: 46. *piǎng* to blaze, flame, luminous: 47. *b'ǎng* (fever:) sickness: 48. *p'jung* to burn, to blaze, beacon: 49. *b'ok* to burn: 50. *b'ok* to dry in the sun: 51. *puk* to burn tortoise shell for divination: 52. *b'og* scorching heat: 53. *b'og* to fry: 54. *b'og* kitchen: 55. *piog* leaping flames: 56. *b'ig* to steam, to heat.

57. *b'āk* white: 58. *b'āk* white silk: 59. *pāk* (the white one, white-haired:) old man, eldest, eldest brother etc.

60. *mǐok* eye: 61. *mǐog* pupil of the eye.

62. *piǎng* a pointed weapon: 63. *p'jung* point of a weapon, sharp point: 64. *p'jung* wasp, bee, hornet: 65. *p'jung* point of a mountain, peak: 66. *p'og* sharp point, to pierce: 67. *mǎng* point of grass, awn: 68. *mǐwang* sharp point of a weapon: 69. *mǎng* gadfly: 70. *mǎng* sprouts: 71. *mǐog* sprouts.

72. *piog* long hair; 73. *mǔng* long-haired: 74. *mog* hair: 75. *mog* yak's tail used as a flag: 76. *mog* grass covering the earth (like hair), herbs, edible herbs etc.

77. *pāk* wide, vast: 78. *p'ig* wide, vast, great; 79, 80. *mǎng* wide, vast: 81. *mǔng* wide, vast: 82. *mǐog* vast as sea.

83. *b'iwang* rampart, embankment, to guard, protect: 84. *piěng* to screen, protect, *b'ien* protecting wall, screen: 85. *b'ien* screened carriage: 86. *pǔng* (walled territory:) state, country: 87. *piung* frontier wall, boundary, fief: 88. *b'ig* parapet, wall.

89. *b'ǔng* to beat, fustigate: 90. *b'ǔng* cudgel, to beat: 91. *pāk* to beat: 92. *p'āk* to beat: 93. *b'iek* to beat the breast: 94. *p'uk* to beat.

95. *piǎng* to grasp, hold: 96. *piǎng* a handle: 97. *p'jung* to hold with both hands: 98. *b'jung* to hold in both hands, receive, present: 99. *b'jung* (what is received:) salary: 100. *pāk* to grasp: 101, 102. *b'ǔg* to grasp; 103. *māk* to take in the hand, grasp, feel.

104. *pək* to cleave, split: 105. *p'iek* to cleave, split: 106. *b'iek* to open: 107. *p'eg* (to cleave:) to bifurcate, to branch (flowing water): 108. *p'üg* to cleave, split.

D

109. *b'ien* jug, vase, bottle: 110. *p'ôg* earthenware jar, jug, pot, bottle: 111. *p'ûg* jar, pot: 112. *pwæg* cup. (The Fang yen defines 110. by 111. and 109. as a small 111.).

113. *peng* to tie, bind, wrap, swaddle: 114. *p'iwang* to bind, twist, spin: 115. *b'iwak* to tie, bind: 116. *pôg* (to swaddle:) to protect, guard (the same char. enlarged by rad. 145 means 'swaddling clothes'): 117. *pôg* to wrap: 118. *b'ôg* (a wrapper:) a long robe: 119. *b'ôg* to enfold, embrace; 120. *mæk* to bind, a cord.

121. *pæk* (back side:) North: 122. *pwæg* back; behind: 123, 124. *b'wæg* to turn the back, turn round: 125. *b'ïü*g to carry on the back.

126. *pâng* flat board, tablet: 127. *b'ïäng* level, flat, even: 128. *b'ïäng* to plane: 129. *b'æg* flat board, tablet: 130. *p'ïeg* stone tablet, stele: 131. *b'ôg* to plane.

132. *pæng* mountain collapsing, land-slip, to collapse: 133, 134. *b'wæk* to fall to the ground, fall prostrate: 135. *b'ïük* fall to the ground, fall prostrate: 136. *b'ïük* to throw down, subdue, throw oneself down, submit: 137. *p'ïug* to fall prostrate.

138, 139. *p'ïog* whirl-wind: 140, 141. *p'ïog* whirl-wind: 142. *b'ïog* whirl-wind (these readings in the Ts'ie yün ms. from Tun-huang, Stein coll., London).

143. *mäng* toad: 144. *mäg* frog, toad.

145. *p'wæg* foetus, embryo: 146. *p'ôg* placenta.

147. *pök* streaked horse, streaked: 148. *pög* (striped animal:) leopard: 149. *p'ïog* stripes of a tiger, stripes, streaked.

150. *p'ôg* bladder: 151. *b'ïog* fish's air bladder.

152 匏 153 瓢 ○ 154 便 155 覆 156 報 ○ 157 步 158 葬 ○ 159 頁 160 葉 161 倅 ○ 162 模 163 摹 164 貌 165 描 ○ 166 芥 167 馥 ○ 168 洒 169 迫 170 逼 ○ 171 複 172 倍 ○ 173 買 174 賣 175 貿 ○ 176 福 177 富 ○ 178 謨 179 媒 180 謀 ○ 181 漂 182 浮 ○ 183 訪 184 聘 ○ 185 豐 186 華 187 丰 ○ 188 培

152. *b'ôg* gourd: 153. *b'ïog* gourd (cognate to 150, 151: bladder-shaped?)

154. *b'ïök* to turn, return, go back: 155. *p'ïök* to turn over, capsize: 156. *pôg* (to return:) to give back, requite, recompense, report etc.: 154. *b'ïôg* (reverting:) again.

157. *mwek* wheat, barley: 158. *m'ïog* barley.

159. *miek* to strive for, to seek: 160. *mäg* to long for, to seek: 161. *m'ïog* to covet.

162. *mäg* model, pattern: 163. *mäg* follow a pattern: 164. *mög* form, shape: 165. *m'ïog* to trace according to a pattern, to depict.

166. *p'iwang* fragrant: 167. *b'ïök* fragrant.

D

168. *pák* to press: 169. *pāk* to press: 170. *pjak* to press.
 171. *pjók* double: 172. *b'wəg* double.
 173. *məg* to buy: 174. *məg* to sell: 175. *mug* to buy and sell, trade.
 176. *pjūk* happiness, prosperity: 177. *pjüg* wealth.
 178. *māg* to scheme, to plan: 179. *mwəg* (schemer, planner:) a marriage go-between: 180. *mjüg* to scheme, to plan.
 181. *p'ioḡ* to float: 182. *b'ioḡ* to float.
 183. *p'iwang* to enquire: 184. *p'iəng* to enquire.
 185. *p'ioṅg* luxuriant, rich, ample, fine: 186. *pung* luxuriant, rich (growth): 187. *p'iung* luxuriant, fine.
 188. *b'ūg* a small mound: 188. *b'wəg* to earth up a plant (make a small mound around).

E. WORDS OF TYPE K -N.

1 乾	2 九	3 還	4 圍	5 寰	6 盤	7 環	8 卷	9 圈	10 桮	11 拳
12 圓	13 員	14 環	15 圓	16 環	17 鉉	18 困	19 袞	20 鞫	21 困	22 軍
23 運	24 罩	25 衛	26 回	27 迴	28 歸	29 圍	30 尫	31 盤	32 乾	33 乾
34 早	35 晁	36 燿	37 炫	38 煨	39 渴	40 煇	41 煥	42 煥	43 煥	44 煥
45 煥	46 晞	47 火	48 燬	49 輝	50 溫	51 燭	52 煨	53 煨	54 冠	55 泰
56 純	57 朝	58 聚	59 絃	60 弦	61 緄	62 網	63 網	64 結	65 髻	66 繼
67 絹	68 純	69 緇	70 裏	71 緯	72 綱	73 訖	74 居	75 率	76 既	77 暨
78 迄	79 奸	80 姦	81 幻	82 猾	83 譎	84 獮	85 諼	86 質	87 罵	88 駭
89 快	90 遠	91 越	92 還	93 昌	94 豈	95 安	96 焉	97 寬	98 闊	99 懇
100 悃	101 慨	102 乞	103 勾	104 祈	105 挺	106 皓	107 翠	108 搗	109 掀	110 獻
111 齧	112 齧	113 齧	114 齧	115 吠	116 闕	117 掘	118 穴	119 窟	120 捐	121 剋
122 空	123 扶	124 官	125 宦	126 吾	127 尹	128 看	129 見	130 見	131 現	132 觀
133 眷	134 覲	135 顯	136 睇	137 眼	138 云	139 曰	140 話	141 謂	142 言	143 諺
144 謁	145 怒	146 鬱	147 畏	148 威	149 管	150 筧	151 蒯	152 蒯	153 匱	154 挽
155 堅	156 鑿	157 濬	158 宛	159 純	160 宛	161 委	162 隈			

1. *kwān* a crank, to turn round, to wind: 2. *g'wān* rolling object, ball: 3. *g'wan* revolve, return: 4. *g'wan* revolve, surround: 5. *g'wan* surrounding wall: 6. *g'wan* to wind the hair into a knot: 7. *g'wan* a ring, to surround: 8. *kiwan* a roll, a scroll: 9. *g'iwan* enclosure, *k'iwan* circle, encircle: 10. *k'iwan* round wooden bowl: 8. *g'iwan* curved: 11. *g'iwān* (hand made round:) fist: 12. *giwan* round: 13. *giwan*, *giwān* jade

E

ring: 14. *giwǎn* (enclosure:) garden, park: 15. *g'iwen* to encircle, wind round, bind: 16. *g'iwen* tripod handle rings: 17. *k'wən* surrounded, enclosed, pressed: 18. *kwən* dress embroidered with curled dragons: 19. *kwən* to revolve smoothly (a wheel): 20. *k'iwen* round granary: 21. *kiwən* troupe, army (escorting footmen surrounding the war chariot; so the character): 22. *giwən* to revolve: 23. *giwən* a halo round the sun or the moon: 24. *giwad* (to surround:) to escort, guard: 25. *g'wər* to revolve, a circuit, come back, return, a turn: 26. *g'wər* eddy: 27. *kiwər* to return: 28. *giwər* to surround; 29. *ngwǎn* to cut round (cut off corners so as to make a thing round); 30. *·wǎn* round bowl: 31. *·wǎt* a crank, to turn round, to wind.

32. *kǎn* dry, *g'ian* (sunshine:) Heaven: 33. *g'ǎn* drought, dry: 34. *g'ian* dawn, beginning sunshine: 35. *kwǎn* to make fire by a burning mirror: 36. *g'iwen* light, dazzling: 37. *g'wən* light, fire, flames: 38. *k'ǎt* (dry:) thirsty: 39. *giwər* blazing, red as fire; 40. *ǎn* dry, hot, parched: 41. *xiwǎn* to dry in the sun, heat, light: 42. *xiən* dawn, beginning sun-shine: 43. *xiən* brightness, heat: 44. *xiwar* to make fire by a burning mirror: 45. *xiər* dawn, to dry in the sun: 46. *xwər*, *xwǎr*, fire: 47. *xiwər*, *xiwar*, *xwǎr* fire, to blaze, burn: 48. *xiwər* bright, blazing; 49. *·wən* warm: 50. *·ǎt* to suffer from heat: 51. *·wər* fire in a brazier.

52. *k'ian* to attach: 53. *kwǎn* band holding together the hair (Shuo wen), cap: 54. *kiwǎn* string which ties up a sleeve: 55. *k'iwǎn* to attach: 56. *g'iwən* bindings of a harness: 57. *kiən* to tie tightly: 58. *g'ien* string: 59. *g'ien* bow string: 60. *kwən* cord, girdle: 61, 62. *k'wən* to tie together: 63. *kwǎt* to bind: 64. *kiet* to tie, a knot: 65. *kied* top-knot of hair: 66. *kied* (»anknüpfen«) to connect, continue: 67. *kwət* to tie: 68. *g'wət* tassel: 69. *giwet* well-rope: 70. *kwǎr* to bind: 71. *giwər* woof thread.

72. *g'iat* to exhaust, finish: 73. *kjət* to finish, end, stop, reach to: 74. *kəd* to end, finish, reach to: 75. *kjəd* end of a season, last: 76. *kjəd* to end, finished: 77. *kjəd* to end, finish, reach to; 78. *xiət* to reach to.

79. *kǎn* fornication, to deceive: 80. *kan* fornication, to deceive: 81. *g'wǎn* to cheat, trick, magic: 82. *g'wet* deceitful: 83. *kiwat* deceitful: 84. *kwǎd* deceitful; 85. *xiwǎn* deceitful; 86. *ngan* counterfeit, false: 87. *ngjən* deceitful.

88. *kiwat* swift horse: 89. *k'wad* swift, quick.

90. *giwǎn* distant, go away: 91. *giwǎt* pass away, pass over, transgress: 92. *giwər* distant, pass away, transgress.

93. *g'át* how, what, why?: 94. *k'jər* how?: 95. *·ǎn* how?: 96. *·ian* how?

97. *k'wǎn* broad, wide, magnanimous: 98. *k'wǎt* broad, wide, magnanimous.

99. *k'ən* intense feelings, earnestly: 100. *k'wən* intense feelings, earnestly: 101. *kəd* intense feelings, suppressed feelings.

E

-102. *k'iat* to beg: 103. *kād* to beg: 104. *g'iar* to beg, to pray.

105. *g'ian* to lift: 106. *kat* to tuck up the skirt: 107. *k'iat* to lift: 108. *k'iat*, *g'iat*, *k'iad* to lift: 109. *xiān* to lift: 110. *xiān* to lift up and present.

111. *k'en* to gnaw: 112. *g'wat* to gnaw: 113. *ngiat* to gnaw.

114. *k'wān* a hole: 115. *kiwan* a dug ditch, canal: 116. *k'iwāt* a hole in a wall, opening, gate: 117. *g'iwāt* to make a hole, *g'iwat* to dig, excavate: 118. *g'iwet* hole, pit, cave: 119. *k'wat* hole, pit, cave: 120. *kwat* to dig, make a pit: 121. *·wān* to scoop out, dig: 122. *·wat* to dig, excavate: 123. *·iwat* to dig out (e. g. the eyes of a person).

124. *kwān* officer, official: 125. *g'wan* officer, official: 126. *k'iwān* chief, prince: 127. *giwen* officer, governor.

128. *k'ān* to look, see: 129. *kian* to look, see: 130, 131. *g'ian* become visible, appear: 132. *kwān* to look, regard: 133. *k'iwān* to regard, love etc.: 134. *g'ian* (to be seen, appear:) have audience; 135. *xian* visible, manifest, display: 136. *xiar* to regard; 137. *ngen* eye.

138. *giwān* to say, to have said: 139. *giwāt* to say: 140. *g'wad* talk, speech, words: 141. *giwad* to say; 142. *ngiān* to talk: 143. *ngian* a saying, proverb; 144. *·iāt* to tell, report.

145. *·ien* oppressed, anxious, zealous: 146. *·iwat* oppressed, anxious: 147. *·iwār* awe, to fear: 148. *·iwār* awe-inspiring.

149. *kwān* reed, pipe, flute: 150. *g'wān* sedge: 151. *k'wad* reed: 152. *giwār* reed.

153. *·iān* bend, bend down: 154, 155. *·wān* the wrist: 156. *·wan* to bend, curve: 157. *·wan* a bend, a bay: 158, 159. *·iwān* (bending:) soft, yielding, docile: 160. *·iwān* crooked, a wrong: 161. *·iwar* to bend down, crooked: 162. *·wār* a bay.

163, 164, 165. *g'wān* confused, turbid: 166. *kwat* confused, turbid.

167. *giwān* a fragrant herb; 168. *xiwān* fragrant: 169. *xiwān* onions and other strong-smelling food stuffs: 170. *xiwān* reeking of liquor: 171. *xiwān* smoke: 172. *g'iwad* fragrant orchid; 173. *·ien* smoke.

174. *k'iad* to stop, rest: 175. *xiāt* to stop, rest: 176, 177. *·āt* to stop, rest.

178. *·ān* peace, tranquillity: 179. *·an* peace, tranquillity: 180, 181. *·ian* at ease, feast.

182. *g'en* threshold: 183. *k'wān* threshold: 184. *k'iwāt* threshold.

185. *g'ān* bar, bolt: 186. *g'iān* bar, bolt: 187. *kwan* bar, bolt.

188. *g'wan* market wall: 189. *giwān* wall; 190. *ngiwān* walled-in premises, courtyard; 191. *·iān* embankment, dyke: 192. *·iwān* (fenced-in area:) park: 193. *·ien* dyke, to dam.

E

163 混	164 涸	165 渾	166 滑	167 芒	168 蕪	169 葦	170 醺	171 熏	172 蕙	173
烟	174 悵	175 歇	176 遏	177 闕	178 安	179 晏	180 燕	181 宴	182 限	
183 相	184 板	185 閑	186 捷	187 關	188 闌	189 垣	190 院	191 堙	192 苑	
193 亘	194 閑	195 間	196 閒	197 闕	198 闕	199 架	200 希	201 稀		
202 壑	203 閑	204 閨	205 啓	206 權	207 勻	208 均	209 鈞	210 計	211 撰	
	212 閑	213 訓	214 困	215 茵	216 隱	217 依	218 隱	219 衣	220 辰	221
醫	222 醫	223 諱	224 報	225 聖	226 侃	227 悍	228 個	229 攔	230 健	231
劫	232 倡	233 傑	234 估	235 仇	236 活	237 滑	238 壘	239 灌	240 課	241
浣	242 澣	243 澣	244 潔	245 溉	246 醴	247 介	248 鎧	249 憫	250 歡	251
欣	252 按	253 印	254 軋	255 抑	256 尉	257 尉	258 慰	259 苑	260 蘊	261
苑	262 藹	263 袁	264 裙	265 氤	266 氲	267 飲	268 餛	269 餛	270 餓	
	271 干	272 杆	273 訐	274 稽	275 鍾	276 飢	277 饑	278 恨	279 忤	
悵	280 悵	281 怨	282 慍	283 恩	284 愛	285 毀	286 壞	287 咽	288 噎	
皆	289 諧	290 摻	291 攀	292 攀	293 援	294 輾	295 牽	296 緯	297 據	298 摺
握	299 握	300 升	301 旁	302 虔	303 痕	304 斤	305 割	306 戛	307 刮	308 鉞
	309 鉞	310 創	311 害	312 肱	313 創	314 乞	315 劇	316 梨	317 契	318 剗
	319 剗	320 刈	321 剗	322 近	323 筵	324 幾	325 捐	326 拈	327 棄	328 遺
	329 燕	330	331 箭	332 腴	333 賈	334 撰				

194. *kǎn* interstice, interval: 195. *g'ǎn* (interval in time:) leisure: 196, 197. *kǎn* (to peep through a crevice:) to peep, spy; 198. *xiēn* crevice, interstice, disaccord, feud.

199. *xān* rare, sparse, few: 200. *xiār* rare, sparse, few: 201. *xiār* sparse, thin.

202. *k'ān* to open up new soil: 203. *k'ār* to open: 204. *k'ār* to open: 205. *k'iar* to open.

206. *g'iwan* steelyard, to weigh with a steelyard: 207. *giwēn* equal, even: 208, 209. *kiwēn* equal, equalize, in balance.

210. *kiār* to reckon, calculate: 211. *g'iwēr* to reckon, calculate, measure.

212. *g'ǎn* to train, teach, exercise: 213. *xiwān* to teach.

214. *·iēn* to base oneself upon, rely on, because of etc.: 215. *·iēn* support, cushion: 216. *·iān* to lean upon: 217. *·iār* to lean upon, rely upon.

218. *·iān* to conceal, hide, cover, screen: 219. *·iār* (covering:) clothes: 220. *·iār* a screen: 221. *·iār* a screen: 222. *·iār* cataract over eye; 223. *xiwār* to conceal, hide, taboo.

224. *kēn* hard-broken soil, hard, difficult: 225. *kien* hard, firm.

226. *k'ān* strong-minded, bold, straightforward: 227. *g'ān* energetic, courageous: 228, 229. *g'ǎn* strong, energetic, martial: 230. *g'iǎn* strong, robust: 231. *k'at*

E

energetic: 232, 233. *g'iat* strong, martial, heroic: 234. *g'iet* strong, robust; 235. *ngiat* strong, martial.

236. *g'wät* moist (as opp. to dry), living, to live etc.: 237. *g'wet* slippery.

238. *kwän* (to pour water over:) to wash the hands: 239. *kwän* to pour water, to water, to wash: 240. *kwän* to pour out wine, libation: 241, 242. *g'wän* to wash: 243. *kiwen* clean: 244. *kiat* clean, to cleanse: 245. *käd* to pour water, wash; 246. *xwäd* to wash the face.

247. *käd* scales, armour: 248. *k'er* armour.

249. *g'än* glad, pleased; 250. *xwän* glad, pleased: 251. *xiän* glad, pleased.

252. *än* to press down, lay hand on: 253. *'ien* (to press down:) to seal, a seal: 254. *'at* to press down under the wheels, to crush: 255. *'iet* to press down, repress (*'iet* as shown by Shī rimes; the same character is also used for a synonymous word *'iek*): 256. *'iwät* (to press:) to iron linen: 257. *'iwäd* to press down, subdue, pacify: 258. *'iwäd* to pacify, soothe, to comfort.

259. *'iwän* rich vegetation, luxuriant: 260. *'iwän* rich vegetation, luxuriant: 261. *'iwät* rich vegetation, luxuriant: 262. *'äd* rich vegetation, luxuriant.

263. *giwän* long robe: 264. *giwän* skirt.

265. *'ien* aura: 266. *'iwän* aura.

267. *xäd* spoilt tainted food; 268. *'iad* spoilt tainted food: 269. *'ied* spoilt tainted food: 270. *'iwäd* spoilt tainted food.

271. *kän* shield: 272. *g'dän* to ward off, protect.

273. *k'iet* to investigate, examine: 274. *kiär* to investigate, examine.

275. *g'ien* famine: 276. *kier* famine: 277. *kier* famine.

278. *g'en* to hate, angry: 279. *ngiat* aversion; 280. *'iwan* angry: 281. *'iwän* to hate, resentment: 282. *'iwän* to hate, resentment.

283. *'en* love: 284. *'äd* love.

285. *xiwar* to destroy: 286. *g'wer* to destroy.

287. *'ien* throat, to swallow: 288. *'iet* food sticking in the throat, to choke.

289. *ker* all: 290. *g'er* all together, in union, harmony.

291, 292. *g'ian* to pull out, seize: 293. *giwän* to pull out, draw, drag, seize: 294. *giwän* (»pullers») cart shafts: 295. *k'ien* to haul, drag, lead: 296. *k'ien* tow-rope, to haul: 297, 298. *kiwän* to take, pick up; 299. *'at* to pull up.

300. *k'dän* to cut: 301. *k'iwän* (a cut notch:) notch, tally, bond, deed: 302. *g'ian* to cut, kill (char. used *kia tsie*): 303. *g'en* (a cut:) scar: 304. *kiän* (cutter:) axe: 305. *kät* to cut: 306. *ket* lance: 307. *kwät* cut off, scrape off: 308. *giwät* halberd: 309. *k'iat* sickle, to cut: 310. *giwät* to incise: 311. *g'äd* (to cut:) to injure: 312. *kwäd*

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to cut meat to pieces, to mince: 313. *kwád* cut off: 314. *kwád*, *kjəd* to cut, to wound: 315. *kjwád* sharp, to cut, to wound: 316, 317. *k'iad* to cut, a notch: 318. *kər*, *ngər* scythe, sharp, to sharpen; 319. *ngwăt*, *ngjwăt*, *ngwət* to amputate, cut off the feet: 320. *ngjäd* to mow, cut: 321. *ngjäd* to amputate, cut off the nose.

322. *g'ien* close, near: 323. *g'jar* (close quarters:) Royal domain proper: 324. *kjar* near (in the common expression »*ki hu*«).

325. *giwăn* to throw away: 326. *giwən* to drop, to lose: 327. *k'jed* to throw away: 328. *giwer* to leave behind, lose.

329. *ian* swallow: 330. *at* swallow.

331. *kjan* sinew: 332. *g'jăn* ends of a sinew.

333. *kwán* to perforate, pass through: 334. *g'wan* to perforate, pass through.

F. WORDS OF TYPE T-N.

1 緣 2 紕 3 紕 4 紕 5 紕 6 紕 7 紕 8 紕 9 紕 10 紕 11 紕
 12 紕 13 紕 14 紕 15 紕 16 紕 17 紕 18 紕 19 紕 20 紕 21 紕 22 紕
 23 紕 24 紕 25 紕 26 紕 27 紕 28 紕 29 紕 30 紕 31 紕 32 紕 33 紕
 34 紕 35 紕 36 紕 37 紕 38 紕 39 紕 40 紕 41 紕 42 紕 43 紕
 44 紕 45 紕 46 紕 47 紕 48 紕 49 紕 50 紕 51 紕 52 紕 53 紕
 54 紕 55 紕 56 紕 57 紕 58 紕 59 紕 60 紕 61 紕 62 紕 63 紕
 64 紕 65 紕 66 紕 67 紕 68 紕 69 紕 70 紕 71 紕 72 紕 73 紕
 74 紕 75 紕 76 紕 77 紕 78 紕 79 紕 80 紕 81 紕 82 紕 83 紕
 84 紕 85 紕 86 紕 87 紕 88 紕 89 紕 90 紕 91 紕 92 紕 93 紕
 94 紕 95 紕 96 紕 97 紕 98 紕 99 紕 100 紕 101 紕 102 紕 103 紕
 104 紕 105 紕 106 紕 107 紕 108 紕 109 紕 110 紕 111 紕 112 紕 113 紕
 114 紕 115 紕 116 紕 117 紕 118 紕 119 紕 120 紕 121 紕 122 紕 123 紕
 124 紕 125 紕 126 紕 127 紕 128 紕 129 紕 130 紕 131 紕 132 紕 133 紕
 134 紕 135 紕 136 紕 137 紕 138 紕 139 紕 140 紕 141 紕 142 紕 143 紕
 144 紕 145 紕 146 紕 147 紕 148 紕 149 紕 150 紕 151 紕 152 紕 153 紕
 154 紕 155 紕 156 紕 157 紕 158 紕 159 紕 160 紕 161 紕 162 紕 163 紕
 164 紕 165 紕 166 紕 167 紕 168 紕 169 紕 170 紕 171 紕 172 紕 173 紕
 174 紕 175 紕 176 紕 177 紕 178 紕 179 紕 180 紕 181 紕 182 紕 183 紕
 184 紕 185 紕 186 紕 187 紕 188 紕 189 紕 190 紕 191 紕 192 紕 193 紕
 194 紕 195 紕 196 紕 197 紕 198 紕 199 紕 200 紕 201 紕 202 紕 203 紕
 204 紕 205 紕 206 紕 207 紕 208 紕 209 紕 210 紕 211 紕 212 紕 213 紕
 214 紕 215 紕 216 紕 217 紕 218 紕 219 紕 220 紕 221 紕 222 紕 223 紕
 224 紕 225 紕 226 紕 227 紕 228 紕 229 紕 230 紕 231 紕 232 紕 233 紕
 234 紕 235 紕 236 紕 237 紕 238 紕 239 紕 240 紕 241 紕 242 紕 243 紕
 244 紕 245 紕 246 紕 247 紕 248 紕 249 紕 250 紕 251 紕 252 紕 253 紕
 254 紕 255 紕 256 紕 257 紕 258 紕 259 紕 260 紕 261 紕 262 紕 263 紕
 264 紕 265 紕 266 紕 267 紕 268 紕 269 紕 270 紕 271 紕 272 紕 273 紕
 274 紕 275 紕 276 紕 277 紕 278 紕 279 紕 280 紕 281 紕 282 紕 283 紕
 284 紕 285 紕 286 紕 287 紕 288 紕 289 紕 290 紕 291 紕 292 紕 293 紕
 294 紕 295 紕 296 紕 297 紕 298 紕 299 紕 300 紕 301 紕 302 紕 303 紕
 304 紕 305 紕 306 紕 307 紕 308 紕 309 紕 310 紕 311 紕 312 紕 313 紕
 314 紕 315 紕 316 紕 317 紕 318 紕 319 紕 320 紕 321 紕 322 紕 323 紕
 324 紕 325 紕 326 紕 327 紕 328 紕 329 紕 330 紕 331 紕 332 紕 333 紕
 334 紕 335 紕 336 紕 337 紕 338 紕 339 紕 340 紕 341 紕 342 紕 343 紕
 344 紕 345 紕 346 紕 347 紕 348 紕 349 紕 350 紕 351 紕 352 紕 353 紕
 354 紕 355 紕 356 紕 357 紕 358 紕 359 紕 360 紕 361 紕 362 紕 363 紕
 364 紕 365 紕 366 紕 367 紕 368 紕 369 紕 370 紕 371 紕 372 紕 373 紕
 374 紕 375 紕 376 紕 377 紕 378 紕 379 紕 380 紕 381 紕 382 紕 383 紕
 384 紕 385 紕 386 紕 387 紕 388 紕 389 紕 390 紕 391 紕 392 紕 393 紕
 394 紕 395 紕 396 紕 397 紕 398 紕 399 紕 400 紕 401 紕 402 紕 403 紕
 404 紕 405 紕 406 紕 407 紕 408 紕 409 紕 410 紕 411 紕 412 紕 413 紕
 414 紕 415 紕 416 紕 417 紕 418 紕 419 紕 420 紕 421 紕 422 紕 423 紕
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 824 紕 825 紕 826 紕 827 紕 828 紕 829 紕 830 紕 831 紕 832 紕 833 紕
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 894 紕 895 紕 896 紕 897 紕 898 紕 899 紕 900 紕 901 紕 902 紕 903 紕
 904 紕 905 紕 906 紕 907 紕 908 紕 909 紕 910 紕 911 紕 912 紕 913 紕
 914 紕 915 紕 916 紕 917 紕 918 紕 919 紕 920 紕 921 紕 922 紕 923 紕
 924 紕 925 紕 926 紕 927 紕 928 紕 929 紕 930 紕 931 紕 932 紕 933 紕
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 974 紕 975 紕 976 紕 977 紕 978 紕 979 紕 980 紕 981 紕 982 紕 983 紕
 984 紕 985 紕 986 紕 987 紕 988 紕 989 紕 990 紕 991 紕 992 紕 993 紕
 994 紕 995 紕 996 紕 997 紕 998 紕 999 紕 1000 紕

1. *djwan* border band (on dress), along the edge, along, sequel, cause: 2. *djwan* silk thread: 3. *d'iet* band of hemp worn on mourning dress: 4. *tjwat*, *tjwad* to tie, fasten, sew; (»anknüpfen«): connect, continue: 5. *tád* band round the waist, girdle: 6. *djwar* a string, to tie: 7. *tswán* red string, to tie together, compile: 8. *tswán*

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(«anknüpfen»:) continue: 9. *dziwən* silk cord, to bind: 10, 11. *sjan* cord, thread: 12. *sien* band round the waist, girdle: 13, 14. *sjat* to tie, fasten.

15. *twán* to cut to pieces: 16. *twán* (cut short:) short: 17. *twán*, *d'wán* to cut off: 18. *d'wán* (a cut-off piece:) section, piece, slice: 19. *d'wán*, *liwan* to cut to pieces: 20. *diwan* to mince meat: 21. *tiēt* sickle, to cut grain: 22. *liad* to cut, trim, restrain, regulate: 23. *liad* to cut: 24. *t'iar* to cut hair: 25. *t'iar* to cut grass; 26. *ts'án* to cut, cut off: 27, 28. *tsian* to cut, clip: 29. *ts'wən* to cut to pieces: 30. *dz'iat* to cut off: 31. *dz'iwat* to cut off: 32. *tsiet* section: 33. *ts'iet* to cut: 34. *tsiar* to mince: 35. *dz'iar* to cut, to trim: 36. *dz'iar* to cut grain; 37. *san* to cut.

38. *twán* point, tip, end: 39. *diwad* pointed, sharp: 40. *liwər* pointed, sharp, awl, tip; 41. *tsian* arrow: 42. *tsian* a hoe: 43. *tsiwan* to pierce, engrave, incise; 44. *siwən* (points:) bamboo shoots: 45. *sjar* arrow. (This group may be cognate to the preceding).

46. *lian*, *sjan* pregnant: 47. *san* to bear.

48. *sjar* to die: 49. *sjar* corpse.

50, 51. *t'ien* to laugh: 52. *sien* to laugh: 53. *d'iet* to laugh.

54. *tán* true, sincere: 55. *lien* true, sincere: 56. *sien* true, believe, faith: 57. *siwən* true, sincere.

58. *diwat* to examine: 59. *ts'at* to examine.

60. *ts'ien* close, intimate, love etc.: 61. *ts'ien* inner coffin (closest to the body): 62. *ts'ien* inner garments.

63. *tswən* high as a mountain, lofty: 64. *dz'iwat*, *tsiwəd* mountain peak: 65. *ts'wər*, *dz'wər* high, lofty, precipitous: 66. *sün* mountain: 67. *siwən* high, steep, lofty.

68. *d'ien* to destroy, annihilate: 69. *dz'án* to murder, kill, destroy.

70. *sán* to scatter: 71. *sát* to scatter.

72. *tsän* wine cup: 73. *dz'án* libation cup.

74. *d'án* to strip, doff, take off clothes, lay bare: 75. *tian* pellicule, to peel off: 76. *t'wát* to peel off, strip, doff, undress; 77. *siwad* cast off skin (as reptiles).

78. *li'wan* to bore through, to pierce: 79. *tswán* to bore, awl.

80, 81. *twən* mound, heap: 82. *twər* mound, heap.

83. *t'ien* to reach to, catch up: 84. *d'əd* to reach, catch up: 85. *liəd* to arrive, reach to, go to: 86. *tiəd* to cause to go (causative to 85.); 87. *tsien* to arrive, reach to.

88. *dz'ian* to advance, precede, before: 89. *tsien* to advance, bring forward: 90. *tsien* to advance: 91, 92. *dz'ien* to bring forward as a present, a gift: 93. *tsiad* to bring forward in sacrifice, to sacrifice: 94. *tsiar* to bring forward as a present, a gift; 95. *sien* to advance, precede, before.

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96. *l'iwət* to come out, bring out, out: 97. *t'iwət* to eliminate, degrade.
 98. *tsjən* to ford, a ford: 99. *tsiər* to ford, a ford.
 100. *t'wən* to swallow, gulp down, devour: 101. *ts'án* to swallow, gulp down, devour: 102. *ts'wad* to devour (swallow a whole slice of meat in one mouthful).
 103. *d'iat* to seize: 104. *l'iat* to grasp, lay hold of: 105. *twát* to pluck, take, grasp: 106. *d'wát* to snatch, carry off, take by force; 107. *ts'wan* to take by force, grab: 108. *tswən* to pinch, lay hand on: 109. *ts'wát* to pinch, pick, gather: 110. *ts'iat* to snatch, steal: 111. *dz'wət* to grasp, seize.
 112. *tjan* to tread, trample: 113. *djwan* heel, to trample; 114. *dz'jan* to tread, trample: 115. *dz'wən* to kick, trample: 116. *dz'wət* to trample: 117. *ts'iar* to tread, trample.
 118. *d'iwən* to follow, obey, accord with: 119. *d'iwət* to follow: 120. *tjwər* to follow after; 121. *dzjwən* to follow in death: 122. *tsiwən* to follow, obey, accord with: 123. *dzjwən* to follow, accord with: 124. *dzjwən* (obeying:) docile, tamed (horse): 125. *dzjwəd* to follow; 126. *swən* (follower:) grandson: 127, 128. *šjwət* to follow; (cause to follow:) to lead: 127, 128. *šjwəd* leader.
 129. *t'wər* to push; 130. *dz'wər* to push, press, thrust: 131. *ts'wər* to push, to urge.
 132. *djwan* quickly, to hasten: 133. *d'wət* quickly, suddenly, rush out: 134. *t'wád* fleet horse, to run swiftly; 135. *tsiwən* fleet horse: 136. *tsiwən* quick-witted, intelligent, gifted: 137. *ts'wət* abruptly; 138. *sjən* quick.
 139. *d'wən* pig: 140. *t'wət* pork fat.
 141, 142. *djət* to let loose, escape, relieve, leisure; 143. *tsjər* to let loose, unrestrained, licentious; 144. *sjat* to let out water, leak, disperse: 145. *sjat* to let out water, leak, relieve, disperse; *zjad* to disperse: 146. *zjad* to let out water, let loose, relieve, leisure.
 147. *t'iar* dripping from the eye, tears: 148. *t'iar*, *djər* dripping from the nose, nasal mucus; 149. *sjəd* nasal mucus.
 150. *d'wən* to collect, to mass; a group of soldiers, a camp, to camp: 151. *d'wəd* a group of soldiers, a regiment: 152. *dz'wád* to collect, to assemble, accumulate: 153. *dz'iwəd* to collect, numerous, dense, thicket.
 154. *l'jan* property; 155. *tsjər* property.
 156. *dzjwad* broom; 157. *šjwat* to brush, to scrape clean: 158. *šjwad*, *šjwət* towel, kerchief: 159. *šjwət* towel, kerchief.
 160. *tán* grieved, distressed: 161. *d'wán* grieved, distressed: 162. *d'wən* grieved, sad: 163. *tát* grieved, distressed: 164. *tjwat* grieved, sad: 165. *l'jwar* sad, anxious;

F

166. *ts'wət*, *dz'iwəd* grieved, sad: 167. *ts'iar* grieved, distressed; 168. *s'iwət* pained, grieved, to pity.

169 紕 170 尙 ○ 171 戰 172 鍛 173 頓 174 摸 175 捷 176 扶 177 抵 178 抵 179
 挺 180 碓 181 挫 182 桂 183 椎 184 碓 ○ 185 攤 186 誕 187 闌 188 挺 189 延
 190 筵 191 演 192 陳 193 陣 194 引 195 示 196 申 197 伸 198 肆 ○ 199 躡 200
 跌 201 躡 202 躡 203 躡 ○ 204 腦 205 腎 206 腿 ○ 207 燁 208 燁 209 前 210
 211 餵 212 餵 213 餵 214 煖 215 燄 ○ 216 敦 217 諄 ○ 218 炭 219 衫 220
 221 田 222 眇 ○ 223 銓 224 忖 225 寸 226 揣 227 算 ○ 228 戰 229 顫
 230 振 231 震 232 電 ○ 233 灌 234 浚 235 濬 236 邃 237 秩 238 第 239 第 240
 241 梯 242 次 ○ 243 係 244 輟 245 竣 246 卒 ○ 247 撤 248 逝 249 逝 250
 251 獸 252 醉 ○ 253 纘 254 纘 ○ 255 賤 256 仝 257 肩 258 細 ○ 259
 260 辰 261 輟 262 纏 263 纏 264 適 265 團 266 擣 267 轉 268 傳 269 棟 270
 271 旋 272 旋 273 波 274 蛟 275 藩 276 徇 277 旬 278 巡 279 昨 280 宣 281
 282 歲 ○ 283 坦 284 壇 285 堦 ○ 286 咒 287 屏 ○ 288 迭 289 替 ○ 290
 291 水 ○ 292 大 293 大 294 泰 ○ 295 蟬 296 螭 297 蛭 ○ 298 單 299 禪 300
 301 禮 302 擅 303 專 ○ 304 滿 305 汎 306 洗 307 洒 ○ 308 損 309 衰 ○
 310 旦 311 晨 312 暎 ○ 313 瑱 314 瑱 315 室 ○ 316 達 317 徹 ○ 318 丹 319
 320 緇 321 茜 322 績 323 紫 ○ 324 齊 325 妻 326 儕 ○ 327 術 328 隧 ○
 329 遁 330 趣 331 退 332 遠 333 遜 ○ 334 鮮 335 新 ○ 336 折 337 脬 338 碎
 ○ 339 舌 340 誓 341 說 ○ 342 墜 343 墮 ○ 344 斂 345 鐫 ○ 346 鈇 347 杙
 ○ 348 瘡 349 疾 350 瘵 351 瘵 ○ 352 盡 ○ 353 軼 ○

169. *d'iet* to sew: 170. *t'iar* to sew, embroider.

171. *t'ian* to fight, battle: 172. *tuán* to hammer, to forge: 173. *tuán* to beat the head against the floor: 174. *d'ien* to beat, to drum: 175. *t'át* to beat, flog: 176. *t'iet* to beat, flog: 177. *t'iar* to butt, resist: 178. *t'iar* to knock against, resist: 179. *tuár* to throw: 180. *tuár* pestle: 181, 182. *t'iwár* to beat: 183. *d'iwár* to beat, hammer, pestle.

185. *t'án* to extend, spread: 186. *d'án* to extend, enlarge, large, exaggerate: 187. *t'ian* to enlarge, open out: 188. *t'ian* drawn out, long: 189. *dian* to extend, spread out, prolong etc.: 190. *dian* (what is spread out:) mat: 191. *dian* to expand, extend: 192. *d'ien* to spread out in a row, arrange: 193. *d'ien* a troupe spread out in a row, array: 194. *dien* to draw out, stretch; lead etc.: 195. *d'iar* (to spread out:) make known, exhibit, proclaim, announcement, presage; 196. *s'ien* to extend, expand, make known, repeat etc.: 197. *s'ien* to extend, stretch out: 198. *s'iar* to extend, spread out, display etc.

F

199. *tien* to stumble: 200. *d'iet* to stumble: 201. *t'át*, 202, 203. *tjær* to stumble.
 204. *djwan* calf of the leg: 205. *d'wæn* rump, buttocks: 206. *t'wær* leg, thigh, ham.
 207. *t'jan* to burn, to cook: 208. *t'wæn* colour of fire, *d'wæn* torch for burning divination shell; 209. *tsjan* to roast, fry: 210. *ts'wân* to make fire, burn, cook: 211, 212. *dz'wan* cooked food: 213. *tsjwæn* fire-prepared, cooked food: 214. *tswæn*, *tsiwæn*, *ts'wæt* to make fire, burn: 215. *dziwäd* to ignite, draw fire.
 216. *twæn* firm, solid: 217. *îjwæn* (to make solid:) inculcate.
 218. *t'ân* coal: 219. *îjæn* black clothes: 220. *îjæn* black hair.
 221. *d'ien* cultivated field: 222. *îjæn* raised borders of the fields.
 223. *ts'jwan*: 224. *ts'wæn* to calculate, measure: 225. *ts'wæn* inch: 226. *ts'jwar* to measure, estimate: 227. *swân* to calculate, estimate, reckon.
 228, 229. *îjan* to tremble, shiver, shake: 230. *îjæn* to shake: 231. *îjæn* shock of thunder, shake: 232. *d'ien* (vibration in atmosphere:) lightning, thunder.
 233. *ts'wær* deep: 234, 235. *sijwæn* deep: 236. *sijwäd* deep.
 237. *d'jæt* order, series, degree, rank: 238. *d'jær* order, series, sequence, degree: 239. *d'jær* (the next one, the second as opp. to the first:) younger brother: 240. *d'jær* younger sister: 241. *t'jær* steps (in sequence), stairs: 242. *ts'jær* order, sequence, the next one, the second.
 243. *t'jad* to stop, finish (Ch'u dialect): 244. *tjwat* to stop, finish: 245. *ts'jwæn* to stop, finish: 246. *tsjwät* to stop, finish, die.
 247. *t'jat*, *d'jat* to remove: 248. *d'jad* to go away, pass away, die: 249. *d'iad* to go away: 250. *ts'jan* 'to go away, remove'.
 251. *îjwat* to drink: 252. *tsjwäd* drunk.
 253. *îjæn* fine and tight textile: 254. *d'jäd* fine and tight textile.
 255. *dz'jan* petty, cheap, mean: 256. *ts'iar* small, petty: 257. *sjat* small, trifling, petty: 258. *sjar* small, fine, thin.
 259. *d'ân* bullet, to shoot with bullet: 260. *tjan* to unroll, open out, turn round: 261. *tjan* to turn round: 262. *d'jan* to wind round, wrap up: 263. *d'jan* to go round, revolution of a celestial body, path etc.: 264. *d'jan* to turn round (Ch'u dialect): 265. *d'wân* round: 266. *d'wân* to roll round: 267. *tjwan* to revolve, turn round: 268. *d'jwan* (to circulate:) to hand over from one to another, transmit: 269. *d'jwan* round spars, rafters (r o u n d, see Tso chuan, Huan 14th year, comm.): 270. *îjæn* to twist, revolve: 271. *dziwan* to turn round, revolve: 272. *dziwan* whirl-wind: 273. *dziwan* circling water: 274. *dziwan* snail: 275. *dziwan* sphere: 276. *dziwæn* all round: 277. *dziwæn* a cycle of 10 days: 278. *dziwæn* to go round, patrol: 279. *tswäd* (a com-

F

plete cycle:) a complete year: 280. *siwan* to revolve, go everywhere, to pass round, circulate, proclaim: 281. *siən* to whirl, pirouette: 282. *siwad* a cycle, a year.

283. *t'ân* level, even, flat: 284. *d'ân* altar: 285. *đian* cleared (levelled, flat) open area (for sacrifices).

286. *dziər* rhinoceros: 287. *siər* rhinoceros.

288. *d'iet* (to substitute one for another:) to alternate, change: 289. *t'ied* to substitute, change, instead of.

290. *tiwən* a water level: 291. *siwər* water.

292. *d'ád* great: 293, 294. *t'ád* great.

295. *đian* cicada; 296. *dz'ien* small cicada: 297. *tsat* small cicada.

298. *tân* single, alone, only: 299. *tân* single, unlined dress: 300. *d'ân* single, alone, only: 301. *tian* single (not double, e. g. mat): 302. *đian* to act alone, on one's own responsibility, dictatorial: 303. *tiwan* single, alone, special, particular.

304. *tsian* to wash: 305. *siən* to throw water over, sprinkle: 306. *siən* to wash the feet: 306, 307. *siər* to wash: 307. *ser* to throw water over, sprinkle.

308. *swən* to diminish: 309. *siwər* to diminish.

310. *tân* dawn, morning: 311. *đian*, *d'ien* dawn, morning: 312. *t'wən* sun just rising.

313. *d'ien* to block, fill up, stop up: 314. *t'ien*, *tiən* ear-stopper: 315. *tiət*, *tiet* to block, fill up, stop up.

316. *d'ât* to penetrate, pass through, communicate, everywhere: 317. *d'iat*, *t'iat* to penetrate, pass through, communicate, everywhere.

318. *tân* cinnabar, red: 319. *tiän* red banner: 320. *tsiən* light red silk, red: 321. *ts'ien* Rubia, red-colouring stuff: 322. *ts'ien* red-coloured silk: 323. *tsiar* purple silk, purple.

324. *dz'ier* uniform, equal, even: 325. *ts'ier* wife of first rank (equal, mate): 326. *dz'er* group of equals, category, class, company.

327. *d'iwət* road, way; method etc.: 328. *dziwəd* road, way; tunnel.

329, 330. *d'wən* to draw back, retire, skulk, flee: 331. *t'wər* to draw back, withdraw, retire: 332. *ts'iwən* to draw back, shrink: 333. *swən* to recede, give way, yield.

334. *sian* fresh, new: 335. *siən* new.

336. *tiät* to break, snap, *d'iat* fracture: 337. *ts'iwəd* easy to break, brittle: 338. *swəd* to break into pieces, fragments.

339. *d'iat* tongue: 340. *điad* a speech, pronouncement, declaration: 341. *siwat* to speak, explain, *siwad* to speak to.

F

342. *d'iwəd* to fall down, collapse: 343. *d'wər* to fall down, collapse.
 344. *d'wər* metal butt on spear shaft: 345. *dz'wən* metal butt on spear shaft.
 346. *d'iad* fetters on the feet: 347. *ḷiət* fetters on the feet.
 348. *tien* sickness, ill; 349. *dz'iet* sickness, ill: 350. *tsəd* sickness, ill: 351. *tsiər* sickness, ill.
 352. *tsiən* to exhaust: 352. *dz'ien* exhausted, empty.
 353. *d'iet*, *ḍiət* a car passing another.

G. WORDS OF TYPE N- N.

1 赧 2 然 3 燃 4 烜 5 煖 6 煖 7 熬 8 熬 9 日 10 泥 11 爛
 ○ 12 鍊 13 煉 14 烈 ○ 15 聯 16 連 17 綸 18 倫 19 緯 20 累 21 纍
 22 縲 23 紉 ○ 24 縲 25 刺 26 棘 27 刃 28 裂 29 厲 30 礪 31 利 32
 犁 33 耒 34 刃 ○ 35 賴 36 利 ○ 37 踳 38 躪 39 履 ○ 40 鄰 41 和
 42 昵 43 暱 44 尼 45 邇 ○ 46 輓 47 嬖 ○ 48 孺 49 孺 50 冽 51 漂

1. *nan* (hot in the face:) to blush, ashamed: 2, 3. *nian* to burn, roast: 4, 5. *nuán* hot, warm: 6. *nuán* hot water: 7. *níat* hot, heat: 8. *níwat* to burn, to heat: 9. *níet* sun, sun heat: 10. *níər* (hot in the face:) to blush, ashamed; 11. *glán* heated through, well-cooked etc.: 12, 13. *glian* to smelt, refine: 14. *liat* burning, flaming, bright.

15, 16. *lian* to string together, connect, join: 17. *liwən* silk strings: 18. *liwən* ties, human relationships: 19. *siwat* rope, cord: 20, 21. *liwər* to tie, bind, string: 22. *liwər* to tie, bind, string; 23. *nien* to thread, string.

24. *liwan* (*bliwan*?) to slice, slice: 25. *lát* to cut: 26. *lát* (cutting:) sharp, pungent: 27. *liat* (to cut apart:) to divide, distribute: 28. *liat* to cleave, split: 29, 30. *liəd* (sharpeners) whetstone: 31. *liəd* sharp: 32. *liər* («cleaver, cutter») plough: 33. *lwər* plough; 34. *nien* edge, blade.

35. *lād* profit, benefit: 36. *liəd* profit (probably same stem as the preceding group: the cuttings, the harvest).

37, 38. *liən* to tread, trample: 39. *liər* to tread, trample, a shoe.

40. *liən* near, neighbour: 41. *níet* clothes nearest to the body: 42, 43. *níat* near: 44. *níər* near: 45. *níar* near.

46. *níwan* weak, soft: 47. *nuwən* weak, soft.

48. *liəd* oyster: 49. *liəd* oyster.

50. *liat* cold: 51. *liət* cold.

H. WORDS OF TYPE P -N.

1 翻	2 扇	3 奮	4 扮	5 飛	6 蹣	7 蹣	8 般	9 拚	10 撥
11 潑	12 發	13 被	14 廢	15 費	16 播	17 駢	18 嬖	19 匹	20 比
21 妣	22 嬖	23 妃	24 配	25 北	26 采	27 辨	28 辯	29 片	30 篇
31 半	32 判	33 胖	34 班	35 板	36 版	37 扁	38 分	39 別	40 伐
41 刑	42 仞	43 判	44 勿	45 辦	46 舛	47 編	48 幘	49 紵	50 紵
51 紵	52 紵	53 紵	54 坊	55 粉	56 叁	57 拂	58 悖	59 譯	60 聞
61 味	62 貝	63 拜	64 憤	65 悲	66 悶	67 逆	68 閑	69 兇	70 怖
71 扮	72 捫	73 繁	74 纒	75 紡	76 素	77 芥	78 菜	79 酵	80 苾
81 緋	82 蜚	83 不	84 尤	85 喙	86 匪	87 非	88 末	89 茂	90 勿
91 未	92 慢	93 愧	94 瞞	95 惛	96 惛	97 民	98 眠	99 昏	100 暗
101 惛	102 惛	103 殍	104 昧	105 忽	106 昧	107 昧	108 疾	109 昧	110 迷
111 輶	112 藩	113 臆	114 閑	115 韞	116 筆	117 市	118 弔	119 款	120 第
121 第	122 敵	123 閑	124 閑	125 庀	126 弔	127 扉	128 扉	129 悞	130 墮
131 門	132 閤	133 幘	134 密	135 本	136 苾	137 拔	138 蹢	139 肺	140 鼻
141 泌	142 溺	143 敗	144 敝	145 弊	146 斃	147 畢	148 墻	149 墳	150 蕪
151 肥	152 邊	153 偏	154 瀆	155 曼					

1. *p'iwǎn* to fly: 2. *p'ien* to fly: 3, 4. *piwǎn* to fly: 5. *piwǎr* to fly.

6. *b'iwǎn* animal's foot, paw: 7. *b'wǎt* foot, heel, trample, march.

8. *puǎn* to transport away, distribute etc.: 9. *p'wǎn* throw away (dialectal word): 10. *puǎt* to spread, distribute, scatter, transfer, throw away, dismiss: 11. *p'wǎt* to throw, spill: 12. *piwǎt* to throw out, launch, send out, issue: 13. *p'iwǎt* to throw out bad influences, expel: 14. *p'iwǎd* to throw away: 15. *p'iwǎd* (to throw away:) to spend, to waste: 16. *puǎr* to spread out, sow; throw away.

17. *b'ien* two horses abreast, a pair: 18. *b'ien* mate, wife: 19. *p'iet* mate: 20. *b'iet*, *b'ier*, *piar* (a pair, two together:) to bring together, compare etc.: 21. *piar* (mate of the father:) deceased mother: 22. *p'ier* mate: 23. *p'iwǎr* mate, wife: 24. *p'wǎr* to mate, match.

25. *b'ien* female (of animals): 25. *b'ier* female (of animals) — possibly cognate to the preceding group: the mate of the male animal.

26. *b'ǎn* to separate, distinguish, discriminate: 27. *b'ǎn*, *b'ian* to divide, distinguish, discriminate: 28. *b'ian* to distinguish, discriminate, argue: 29. *p'ian* cleft wood, splint, slice, slip, board, tablet: 30. *p'ian* (cleft wood:) writing tablet: 31. *puǎn* divide in half, half: 32. *p'wǎn* to cleave, divide, discriminate: 33. *p'wǎn* one half of a victim divided in two parts: 34. *puan* to distribute: 35, 36. *puan* board:

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37. *pīen* board, tablet, flat: 38. *pīwən* to divide, *b'īwən* a part: 39. *pīat*, *b'īat* to cleave, separate, divide: 40. *b'īwāt* to cut, attack: 41. *b'īwət* to cut, attack: 42. *p'īər* to separate, part: 43. *b'īwər* to cut off the feet, amputate; 44. *mīwən* to cut, cut the throat.

45. *b'īan* to twist cord: 46. *pwān* (to tie:) fetter, hobble: 47. *pīen* to tie together: 48. *b'īwən* cords on horse's bit: 49, 50. *pīwət* rope: 51. *pīwət* silk band: 52. *pīər* (cords:) reins, bridle: 53. *mīən* cord, string.

54. *b'īwən* dust: 55. *pīwən* (powdered:) flour: 56. *pīwən* to dust: 57. *p'īwət* to dust.

58, 59. *b'wād*, *b'wət* disorderly, rebellious.

60. *mīwən* to hear, to smell (perceive by the s e n s e s): 61. *mīwəd* to taste, taste:

62. *pād* cowry shell, valuables: 63. *pwād* (to bring cowry shells as presents, present valuables:) to salute (this primary sense of »pai« is revealed by the earliest bronze inscriptions which show a man presenting strings of cowries).

64. *b'īwən* to grieve: 65. *pīər* to grieve: 66, 67. *mwən* to grieve: 68. *mīən* to grieve: 69. *mīwən* mourning hair dress: 70. *b'īwət* distressed.

71. *b'īwən* to grasp, seize: 72. *mwən* to grasp, seize.

73. *b'īwān* mixed and numerous, multitudinous: 74. *p'īən* tangled, confused, multitudinous: 75. *pīwən* tangled, confused, multitudinous; 76. *mīwən* tangled, confused.

77. *p'īwən* strong-smelling, fragrant: 78. *b'īwən* fragrant wood, aromatic wood: 79. *b'wət* fragrant: 80. *b'īət* fragrant: 81. *pīwər* fragrant: 82. *b'īwər* stinking insects.

83. *pwət* not (no such reading in the ancient dictionaries, but indicated by all the dialects): 84. *pīwət* not (with the modal sense of 'not be able to, not be willing to'): 85. *b'īwət* to say no, contradict: 86. *pīwər* not: 87. *pīwər* is not; 88. *mwāt* not (char. used *kia tsie*): 89. *miat* not (char. used *kia tsie*): 90. *mīwət* don't: 91. *mīwəd* not yet.

92. *man* careless, forgetful, negligent: 93. *mwən* careless, forgetful.

94. *mwān* closed eyes, darkened sight: 95, 96. *mīən* darkened intelligence, stupid: 97. *mīən* the common people (the »darkened, stupid ones«, »*yü min*« stupid people): 98. *mīen* closed eyes, to sleep: 99. *χmwən* darkness, darkened, darkened intelligence, stupid: 100, 101, 102, 103. *χmwən* dim sight, darkened intelligence: 104. *mwāt* troubled sight: 105. *χmwət* not discerning, stupid, careless: 106. *mwād* troubled sight: 107. *mwād* darkness before dawn: 108. *mīəd* to sleep: 109. *mīər* something in the eye, troubled sight: 110. *mīər* (to blind:) to confuse, lead astray.

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111. *p'iwǎn* (a cover:) carriage screen: 112. *p'iwǎn* (cover:) hedge: 113. *b'ǐēn* knee cap (bone which is like a cover, a cap): 114. *piat* to shut a door, close, cover: 115. *piēt* knee cover: 116. *piēt* (cover, screen:) wicker hedge, shutter, small door made of branches: 117, 118, 119. *piwət* knee cover: 120, 121. *piwət* carriage screen: 122. *piad* to cover, conceal: 123. *piad* to shut a door, close, cover: 124. *piəd* to shut a door, shut, covered, hidden, secret: 125. *piəd* to cover, screen: 126. *piwəd* covering, shady: 127. *piwər* (a shutter:) door-leaf: 128. *b'iwər* covered, screened, secret: 129. *mwán* screen, curtain: 130. *mwán* (covering:) plaster on a wall: 131. *mwən* (shutters:) door-leaves, door, gate: 132. *χmwən* gate-keeper: 133. *miat* a cover on the front part of a carriage: 134. *mǐēt* shut off, covered, concealed, solitary, secret, silent.

135. *pwən* lower part of a tree or a plant, stem, root (as opp. to top, branches, ear of grain): 136. *pwát* lower part of a tree or a plant, stem, root; straw (as opp. to ear): 137. *b'wat* to uproot, pull up: 138. *b'wát* lowest part of a vertical object, base, foot, heel, to trample (cf. »*ken*» which means both 'stem, root' and 'heel').

139. *piwǎd* (breathers:) lungs: 140. *b'iad* (breather:) nose.

141. *piəd* to gush forth (as a source): 142. *piwəd* to gush forth.

143. *b'wad* to destroy, ruin, spoil, defeat: 144. *b'iad* to spoil, ruin, wear out: 145. *b'iad* spoiled, bad: 146. *b'iad* (destruction:) death: 147. *piēt* (destroy:) make an end to, to end, finish.

148. *b'iwǎn* grave-mound: 149. *b'iwən* grave-mound.

150. *piwən* fertilizer, ordure, manure: 151. *b'iwər* fat, rich, fertile.

152. *pian* side, edge, border: 153. *p'ian* on one side, oblique: 154. *piēn* (side, edge of the water:) shore, beach.

155. *mwán* long, extended: 155. *mǐwǎn* long, extended.

I. WORDS OF TYPE K-M.

1 闕 2 黔 3 龔 4 閏 5 閏 6 蓋 7 蓋 8 掩 9 掩 10 奄 11 閏
 12 黔 13 龔 14 閏 15 閏 16 閏 17 暗 18 黔 19 蔭 20 陰 21 頤 22
 函 23 街 24 領 25 龔 26 珍 27 頰 28 領 29 拊 30 鉗 31 夾 32 狹
 33 狎 34 陟 35 鈇 36 扶 37 扶 38 肱 39 脇 40 鹽 41 鹹 42 醃
 43 飲 44 吸 45 喻 46 飲 47 坎 48 陌 49 珍 50 壓 51 摩 52
 鉅 53 甲 54 檻 55 輶 56 桺 57 陰 58 龔 59 匿 60 匣 61 爰
 62 岵 63 岵 64 巖 65 陳 66 峯 67 險 68 瞰 69 監 70 鑑 71 覽
 72 兼 73 咸 74 協 75 洽 76 裕 77 合 78 會 79 會 80 翕 81 飲 82
 謙 83 歉 84 欠 85 感 86 撼 87 劍 88 戡

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1. *g'iam* (shutter:) gate: 2. *g'iem* dark, black: 3. *k'iem* (covering:) coverlet: 4. *g'áp* (shutter:) door, to close: 5. *káp* (shutter:) inner door (to the harem): 6, 7. *g'áp*, *káb* covering, to cover: 8. *ám* covered, obscured, dark: 9. *iam* to cover, conceal: 10. *iam* to cover, all-covering: 11. *iam* (the man who covers, shuts, keeps the doors of the harem shut, cf. 132 above:) door-keeper, eunuch: 12. *em* dark, black: 13, 14. *iem* to cover, conceal: 15. *em* to cover with earth, bury: 16. *em* to shut a door, shut, closed, covered, dark: 17. *em* darkened sun, dark, darkness: 18. *em* black: 19. *iem* (covering:) shade: 20. *iem* darkness, dark, shade, North side etc. For the connection between 'to cover' and 'dark': 'to shut': 'gate', cf. group H 111—134 above.

21. *g'am* jaws, jowl: 22. *g'am* to hold in the mouth: 23. *g'em* (what is »jawed», bitten:) horse's bit, to hold in the mouth: 24. *g'em* jaws, jowl: 25. *g'em* to hold in the mouth, contain: 26. *g'em* objects put in a corpse's mouth: 27. *kiap* jaws, jowl, cheeks: 28. *g'ap* jaws, jowl (possibly cognate to next group: »pincher» = jaw?).

29. *g'iam* to press from both sides, pinch: 30. *g'iam* (pincher:) iron collar; gyves; pincers: 31. *kap* to grasp from both sides, squeeze, pinch: 32. *g'ap* (pressing from both sides:) narrow: 33. *g'ap* (pressing:) close; intimate: 34. *g'ap* (narrow, pressing from both sides:) chasm, pass: 35. *kiap* pincers: 36. *kiap* pincers, chopsticks: 37. *g'iap* to press from both sides, hold under the arms, clasp under the arm to support; to press, coerce: 38. *k'ĩp* to open the sides, to rifle; *k'iwǎb* sides of the body, flanks: 39. *χĩp* sides of the body, flanks; to press from both sides, to press, coerce.

40. *g'iam* salt: 41. *g'em* salty, salted: 42. *ngĩm* soda (drawn from salty soil).

43. *χap* to suck in, draw in, swallow, drink: 44, 45. *χĩp* to suck in, draw in, swallow: 46. *iem* to swallow, drink.

47. *k'am* pit: 48. *g'am* pit, throw or fall into a pit: 49. *k'iem* pit.

50. *ap* to press down: 51. *iap* put down the finger on.

52. *g'am* mail: 53. *kap* scales, mail.

54. *g'am* cage (for wild animals etc.): 55. *g'am* car with cage on: 56. *g'ap* cage (for wild animals etc.): 57. *k'iwǎb* pen, enclosure for wild animals.

58. *k'em* to hold, contain, receptacle, box: 59. *k'iap* box, satchel: 60. *g'ap* case, box: 61. *g'iep*, *g'ĩep* satchel, book box.

62. *k'em* cliff: 63. *k'iem* high, precipitous: 64. *ngam* cliff: 65. *ngiam* high bank: 66. *ngĩm* high ridge of cliff: 67. *χiam* high, precipitous, difficult of access, perilous.

I

68. *k'ám* (*k'lám*?) to see, watch: 69. *klam* to see, examine, inspect: 70. *klam* mirror: 71. *glám* to see, watch.

72. *kiam* (*kliam*?) to join two or several things, bring together, combine, unite, both: 73. *g'em* (several united:) all: 74. *g'iap* to unite, accord, harmony: 75. *g'ep* accord: 76. *g'ep* collective sacrifice (to all ancestors): 77. *g'ap* to join, bring together, unite, accord: 78. *g'wəb* (> *Shī g'wəd* > *Anc. yuái*) to come together, bring together, unite, accord: 79. *kwəb* (to add up:) to make up accounts: 80. *xiəp* to unite, accord, harmony.

81. *k'ám* (to make oneself defective, imperfect:) to humble oneself: 82. *k'iam* to humble oneself: 83. *k'am*, *k'iam* insufficient food, insufficient: 84. *k'iam* deficient (char. used *kia tsie*).

85. *kəm* to move, be moved: 86. *g'am* to move, shake.

87. *k'iam* sword: 88. *k'am* to stab, cut down.

K. WORDS OF TYPES *T-M*, *N-M*, *P-M*.

1 鑿	2 鏡	3 刻	4 掘	5 鐵	6 尖	7 錯	8 簪	9 箴	10 鍼	11 鉈
12 插	13 鉞	14 斬	15 芟	16 疊	17 鑿	18 褶	19 鑿	20 攝	21	
22 惛	23 惛	24 熱	25 織	26 撈	27 港	28 潛	29 潭	30 沈	31 墊	
32 添	33 袋	34 憺	35 恬	36 寔	37 駁	38 捷	39 駁	40 憺		
41 炎	42 煙	43 炮	44 疔	45 濕	46 隕	47 繫	48 緝	49 瞻	50	
51 覘	52 執	53 拾	54 擊	55 捷	56 攝	57 談	58 詹	59 譚	60	
61 諧	62 喋	63 沾	64 漸	65 濺	66 浸	67 滂	68 涇	69 滄	70	
71 忒	72 譏	73 譌	74 三	75 參	76 參	77 僂	78 參	79		
80 逕	81 接	82 雜	83 集	84 輯	85 戢	86 答	87 對	88 納	89 內	90
91 入	92 範	93 法								

1. *tsám* awl, chisel, to pierce: 2. *dz'am* awl, chisel: 3. *diam* sharp-pointed, sharp: 4. *liam* to pierce: 5. *tsiam* awl, sharp: 6. *tsiam* sharp point, sharp: 7. *tsəm* needle: 8. *tsiəm* hair pin: 9, 10. *liəm* needle: 11. *ts'ap* long needle: 12. *ts'ap* to put a spit through, pierce: 15. *səp* chisel, to engrave.

14. *tsam* to cut off, decapitate: 15. *sam* to cut grass, to mow.

16. *d'iap* layer, to fold: 17, 18. *d'iap* dress with two layers of stuff, lined, double; put in layers, to fold: 19. *dziəp* dress with two layers of stuff, lined, double; put in layers, to fold.

20. *liap* to fear, afraid: 21. *d'iap* afraid: 22. *liəp* to fear, afraid: 23. *liəp* to fear, afraid.

K

24. *siam* slender, thin, fine, small: 25. *sem* slender, delicate.
26. *d'em* to sink in water, deep water: 27. *dz'iem* to go down in the deep, go down in water, lie hidden: 28. *d'am* deep, deep water, abyss: 29. *d'iem* to go down in the deep, sink in water: 30. *d'iam* down in the depths of earth, deep down: 31. *s'iem* deep water, deep.
- 32, 33. *d'am* quiet, peaceful: 34. *d'iam* quiet, peaceful: 35. *t'iap* quiet, peaceful.
36. *tsam* swift: 37. *ts'iem*, *ts'iem* to gallop, run swiftly: 38. *dz'iap* swift: 39. *sap* to gallop, run swiftly.
40. *d'am* to burn: 41. *diam* to blaze, flame: 42. *d'am*, *dz'iem* to burn, to heat: 43. *d'iam* stove: 44. *siam* fever.
45. *s'iap* moist, wet, damp: 46. *dz'iap* marsh.
47. *tiap* to bind, tie: 48. *ts'iap* to bind together.
49. *liam* to look, see: 50. *t'iam* to look, see.
51. *liap* to catch, grasp, seize: 52. *d'iap* to seize, grasp, pick up: 53. *liab* (a »catcher«): bird of prey: 54. *liab* to grasp: 55. *dz'iap* (to catch:) to take booty, quarry, game: 56. *s'iap* to grasp, hold.
57. *d'am* to talk, chat: 58. *liam* to chatter: 59. *d'am* to talk, chat: 60. *d'ap* to chatter: 61. *d'iap* to chatter.
- 62, 63. *tiam* to soak: 64, 65. *tsiam* to soak: 66. *ts'iem* to soak; 67. *s'iem* to soak: 68, 69. *z'iem* soaking rain.
70. *t'iam* to put to shame, disgrace: 71. *dz'am* shame, to be ashamed.
72. *dz'am* to slander: 73. *dz'iem* to slander.
74. *sam* three: 75. *ts'am* three (persons or things) together, a triad: 76. *s'iem* (the three-star constellation:) Orion.
77. *dz'am* of unequal length: 78. *ts'iem* of unequal length.
79. *d'ap* mixed: 80. *tsiap* to bring together, join: 81. *dz'ap* collected, mixed: 82. *dz'iap* to bring together, gather, collect, mix: 83. *dz'iap* to bring together, gather, harmony: 84. *ts'iap* to collect.
85. *tap* to answer: 86. *twab* (> Shī *twad* > Anc. *tudi*) to answer.
- 87, 88. *nep* (to cause to enter, causative to 90:) to bring in, to present: 89. *nwab* (> Shī *nwad* > Anc. *nudi*) interior, inside, in: 90. *iiap* to enter.
91. *b'iwam* pattern, rule, law, norm: 92. *piwāp* pattern, rule law, norm.

* * *

LAWS OF ALTERNATIONS.

At a first glance at the word families established above the reader may well ask: is it not extremely farfetched to imagine an etymological connection between words which are so strongly dissimilar phonetically as many of the cases proposed? Is it not unreasonable to combine e. g. A 213 *k'ung*: 225 *χiog*; B. 64 *điang*: 76 *tōk*: 79 *d'æg*; E 93 *g'át*: 94 *k'iar*; E 142 *ngiăn*: 140 *g'wad*; F. 66 *săn*: 65 *dz'wər* — when in all such cases there is n o t o n e consonant or vowel common to the two members of the combination? This objection may seem quite justified. Yet a rash judgment of the kind is not just; all the families proposed must be seen in the light of the general laws of alternations which can be derived from the materials adduced. In the following résumé I shall pick out a number of comparatively safe and convincing examples of those various laws.

The final consonants.

Here we have three undeniable series of alternations:

1. *ng* ~ *k* ~ *g*;
2. *n* ~ *t* ~ *d* ~ *r*;
3. *m* ~ *p* ~ *b*.

ng ~ *k*:

A 33. *ngiăng*: 34. *ngiăk*; 103. *ngăng*: 106. *ngák*; 118. *k'áng*: 120. *g'ák*; 155. *k'ăng*: 163. *k'ăk*; 172. *kəng*: 175. *g'ăk*; 226. *·iəng*: 227. *·iək*; 229. *kwăng*: 232. *k'wák*; 268. *k'jung*: 270. *k'juk*, 271. *g'juk*; B 69. *təng*, 70. *đ'əng*: 75. *t'ək*; 115. *d'ũng*: 123. *tũk*; 260. *sieng*: 261. *siək*; 269. *điông*: 275. *điók*; 298. *d'əng*: 300. *điək*; 302. *dz'jəng*: 305. *dz'iek*; 322. *săng*: 323. *s'jak*; 387. *d'ung*: 388. *d'uk*; 516. *tsjung*: 523. *tsjuk*; 547. *dz'ung*: 549. *dz'uk*; C 10. *gliang* and *gliak*; 32. *lung*: 33. *luk*; 38. *lieng*: 39. *liek*; D 8,9. *mieng*: 15—17. *miek*; 114. *p'iwang*: 115. *b'iwak*.

ng ~ *g*:

A 29. *kăng*: 30. *kæg*; 63. *kông*: 65. *g'óg*; 73. *k'ăng*: 84. *g'jæg*; 92. *g'jăng*: 97. *k'jæg*; 135. *k'jăng*: 137. *g'jæg*; 132. *g'ien*: 133. *g'eg*; B 69. *təng*: 79. *d'æg*; 210. *d'áng*: 213. *l'jag*; 308. *l'jông*: 314. *l'jóg*; 378. *tsăng*: 380. *dz'ág*; 408. *l'jag*: 416. *l'jag*; 464. *təng*: 465. *d'æg*; 528. *điəng*: 532. *d'jæg*; C 25. *gliang*: 28. *glág*; 54. *nəng*: 55. *næg*; D 84. *p'jəng*, *b'ien*: 88. *b'jæg*; 143. *măng*: 144. *mäg*.

k ~ *g*:

A 28. *kók* and *kóg*; 31. *k'ók*: 32. *kóg*; 279. *k'iwak*: 282. *k'iwag*; 290. *·ək*: 294. *·jg*; 318. *k'ək*: 320. *k'jg*; 323. *·ák* and *·ág*; 339. *g'ók*: 343. *g'óg*, 342. *kóg*, 340. *k'óg*, *g'jg*; 369. *k'ók* and *k'óg*; 22. *điək*: 24. *điæg*; B 29. *d'ák* and *d'ág*; 41. *d'uk*: 42. *d'ũg*; 169. *dz'ək*: 177. *dz'æg*; 168. *ts'jək* and *ts'jæg*; 170. *ts'jək*: 181. *dz'jæg*; 235. *d'ók* and *d'óg*; 272. *l'jak* and *l'jag*; 289. *s'jók*: 291. *s'jog*, 293. *s'jog*; 436. *s'jak*: 437. *s'jag*; 445. *d'jək* and *d'jæg*;

450. *dziak*: 454. *ziag*; 452. *siók*: 456. *siog*; 462. *tsiak* and *tsiag*; 548. *tsiэк* and *tsiэг*; 565. *siók*: 566. *sióg*; 569. *tək*: 570. *d'æg*; 576. *ɖiok*: 579. *djog*; 595. *d'io:k*: 599. *t'io:g*; 594. *djók*: 598. *djóg*; 605. *djok*: 606. *d'io:g*, 607. *t'io:g*; 611. *liók*: 612. *lióg*; 662. *d'iak* and *d'ia:g* 682. *djók*: 683. *d'jóg*; C 1. *lək*: 2. *liэг*; D 24,25. *mək*: 27. *mwæg*; 49. *b'ok*: 52. *b'og*; 60. *mjók*: 61. *mjóg*; 104. *pək*: 107. *p'эг*; 121. *pək*: 122. *pwæg*, 123, 124. *b'wæg*; 135. *b'jūk*: 137. *p'jüg*; 147. *pők*: 148. *pög*; 154. *b'jók* and *b'jóg*; 176. *pjūk*: 177. *pjüg*.

n ~ t:

E 1. *kwán* and *'wát*; 32. *kán*: 38. *k'át*; 53. *kwán*: 63. *kwát*; 90. *giwán*: 91. *giwát*; 97. *k'wán*: 98. *k'wát*; 105. *g'ian*: 108. *g'iat*; 121. *'wán*: 122. *'wat*; 138. *giwán*: 139. *giwát*; 163. *g'wán*: 166. *kwát*; 226. *k'án*: 231. *k'at*; 230. *g'jǎn*: 232, 233. *g'iat*; 252. *án*: 254. *'at*; 253. *'jén*: 255. *'jět*; 260. *'jwán*: 261. *'jwát*; 287. *'ien*: 288. *'iet*; 300. *k'án*: 305. *kát*; 329. *'ian*: 330. *'at*; F 10. *sian*: 13. *sjat*; 27. *tsian*: 30. *dz'iat*; 70. *sán*: 71. *sát*; 108. *tswán*: 111. *dz'wát*; 115. *dz'wán*: 116. *dz'wát*; 118. *d'jwán*: 119. *d'jwát*; 139. *d'wán*: 140. *t'wát*; 160. *tán* and *tát*; 214. *tswán* and *ts'wát*; 245. *ts'jwán*: 246. *tsjwát*; 314. *tjén*: 315. *tjět*; 313. *d'ien*: 315. *tiet*; G 2, 3. *ńian*: 7. *ńiat*; 4, 5. *nuán*: 8. *ńiwat*; H 27. *b'ian*: 39. *b'iat*; 48. *b'jwán*: 49—51. *pjwát*; 56. *pjwán*, 54. *b'jwán*: 57. *p'jwát*; 64. *b'jwán*: 70. *b'jwát*; 78. *b'jwán*: 79. *b'wát*; 94. *muán*: 104. *muát*; 99—103. *ɣmwán*: 105. *ɣmwát*; 113. *b'jén*: 115. *pjět*.

n ~ d:

E 12. *giwan*: 24. *giwad*; 105. *g'ian*: 108. *k'iad*; 138. *giwán*: 141. *giwəd*; 149. *kwán*, 150. *g'wán*: 151. *k'wad*; 167. *giwán*: 172. *g'iwəd*; 283. *'ən*: 284. *'əd*; 300. *k'án*: 316, 317. *k'iad*; F. 123. *dziwán*: 125. *dziwəd*; 150. *d'wán*: 151. *d'wad*; 214. *tsjwán*: 215. *dziwəd*; 234. *sjwán*: 236. *sjwad*; H 94. *muán*: 106. *muád*; 98. *mien*: 108. *mjəd*.

n ~ r:

A series of examples of this alternation has already been given on pp. 20, 27 above. We have furthermore: E 19. *kwén*, 22. *giwán*: 25. *g'wér*, 27. *kjwér*, 28. *giwér*; 37. *g'wán*: 39. *giwér*; 42. *ɣjén*: 45. *ɣjér*; 49. *'wán*: 51. *'wér*; 53. *kwán*: 70. *kwár*; 158, 160. *'jwán*: 161. *'jwér*; 202. *k'én*: 203, 204. *k'ér*; F. 2. *djwán*: 6. *djwér*; 80. *twán*: 82. *twér*; 89, 90. *tsjén*: 94. *tsjér*; 98. *tsjén*: 99. *tsjér*; 118. *d'jwán*: 120. *tjwér*; 205. *d'wán*: 206. *t'wér*; 223. *ts'jwán*: 226. *ts'jwér*; 306. *sién* and *siér*; 308. *swán*: 309. *sjwér*; 329. *d'wán*: 331. *t'wér*; 345. *dz'wán*: 344. *d'wér*; G 17. *ljwán*: 22. *ljwér*; 37, 38. *ljén*: 39. *ljér*; H 77. *p'jwán*, 78. *b'jwán*: 81. *pjwér*, 82. *b'jwér*; 95, 96. *mjén*: 110. *miér*; 150. *pjwán*: 151. *b'jwér*.

l ~ d:

E 64. *kiet*: 65, 66. *kied*; 73. *kjət*: 76, 77. *kjəd*; 83. *kiwat*: 84. *kwád*; 88. *kiwat*: 89. *k'wad*; 108. *kjǎt*, *g'iat* and *k'iad*; 107. *k'iat*: 108. *k'iad*; 139. *giwát*: 140. *g'wad*, 141. *giwəd*; 256. *'jwát*: 257. *'jwəd*; 305. *kát*: 311. *g'ád*; 307. *kwát*: 313. *kwád*; 308. *giwát*: 315. *kjwád*; 309. *k'iat*: 316, 317. *k'iad*; 319. *ngjwát*: 320. *ngjád*; F 4. *tjwat* and *tjwad*; 64. *dz'jwát* and *tsjwád*; 127, 128. *sjwát* and *sjwád*; 144, 145. *sjat*: 146. *zjad*; 157. *sjwat*: 158. *sjwad*; 166. *ts'wát* and *dz'jwád*; 341. *sjwat* and *sjwad*; G 25. *lát*, 27. *ljat*: 29, 30. *ljad*; H 12. *pjwát*, 11. *p'wát*: 14. *p'jwád*; 13. *p'jwát*: 15. *p'jwád*; 58, 59. *b'wát*

and *b'wǎd*; 90. *mǐwət*: 91. *mǐwəd*; 104. *mǔd*: 106. *mǔd*; 114. *piat*: 123. *piad*; 116. *pǐt*: 124. *pǐd*; 120, 121. *pǐwət*: 126. *pǐwəd*.

t ~ r:

E 63. *kwǎt*: 70. *kwǎr*; 102. *k'ǐət*: 104. *g'ǐər*; 146. *ǐwət*: 147, 148. *ǐwər*; F. 64. *dz'ǐwət*: 65. *dz'wər*; 119. *d'ǐwət*: 120. *tǐwər*; 164. *tǐwat*: 165. *ǐiwar*; G 42, 43. *nǐət*: 44. *nǐər*; H 10. *pwǎt*: 16. *pwǎr*; 20. *b'ǐət* and *b'ǐər*; 41. *b'ǐwət*: 43. *b'ǐwər*; 79. *b'wət*: 81. *pǐwər*, 82. *b'ǐwər*; 84. *pǐwət*: 86, 87. *pǐwər*; 120, 121. *pǐwət*: 128. *b'ǐwər*.

d ~ r:

E 24. *gǐwad*: 28. *gǐwər*; 247. *kǎd*: 248. *k'ər*; F. 64. *tsǐwad*: 65. *ts'wər*; 236. *sǐwad*: 233. *ts'wər*; G 31. *lǐəd*: 32. *liər*; H. 126. *pǐwad*: 128. *b'ǐwər*.

m ~ p:

I 1. *gǐam* 4. *g'ǎp*; 21, 22. *g'ám*: 27. *kiap*; 24—26. *g'əm*: 28. *g'əp*; 29, 30. *g'ǐam*: 37. *g'ǐap*; 52. *g'ám*: 53. *kap*; 54. *g'am*: 56. *g'ap*; 72. *kiam*: 74. *g'ǐap*; 73. *g'em*: 75, 76. *g'əp*; K 34. *d'iam*: 35. *t'ǐap*; 36. *tsám*: 38. *dz'ǐap*; 37. *ts'ǐəm*: 39. *səp*; 57. *d'ám*: 60. *d'ǎp*.

p ~ b:

I 38. *k'ǐǎp* and *k'ǐwǎb*; 77. *g'əp*: 78. *g'wəb*; K. 51. *ǐǐəp*: 53, 54. *ǐǐəb*; 87. *nəp*: 89. *mwəb*; 85. *təp*: 86. *twəb*.

For further examples of these alternations of final consonants see pp. 107—110 below.

The initial consonants.

Here we have first four great series of consonants, the members of which interchange freely in the formation of words from one and the same stem:

1. *k ~ k' ~ g ~ g'*;
2. *t ~ t' ~ d ~ d' ~ ǐ ~ ǐ' ~ ǎ ~ ǎ'*;
3. *ts ~ ts' ~ dz ~ dz' ~ tʂ ~ tʂ' ~ dʒ' ~ s ~ ʂ ~ ʒ ~ z*;
4. *p ~ p' ~ b'*.

Examples:

k ~ k':

A 32. *kôg*: 31. *k'ǐôk*; 37. *kuk*: 36. *k'ǔk*; 79. *kǐung*: 80. *k'ǐung*; 148. *kǐwəg*: 147. *k'ǐəg*; 191. *kǐung*: 190. *k'ung*; 268. *kǐǔng*: 270. *k'ǐuk*; 331. *kek*: 330. *k'wák*; 342. *kôg*: 344. *k'ôg*; E 120. *kwət*: 119. *k'wət*; 129. *kian*, 132. *kwán*: 128. *k'án*; 149. *kwán*: 151. *k'wad*.

k ~ g:

A 3. *kwáng*: 6. *gǐwang*; 156. *kwǎng*: 158. *gǐwǎng*; 176. *kwək*: 177. *gǐwək*; 209. *kwǎng*: 212. *gǐwǎng*; 362. *kiôg*: 363. *gǐog*; E 19. *kwən*: 22. *gǐwən*; 208. *kǐwǎn*: 207. *gǐwǎn*; E 315. *kǐwǎd*: 308. *gǐwǎt*.

k ~ g':

A 3. *kwáng*: 4, 5. *g'wǎng*; 98. *kog*: 100. *g'ǐog*; 140. *kǐwang*: 141. *g'ǐwang*; 172. *kəng*:

175. *g'ǎk*; 237. *kōng*; 238. *g'ung*; 250. *kāng*; 253. *g'iang*; 259. *kūng*; 260. *g'ǔng*; 268. *kǐūng*; 271. *g'ǐuk*; 279. *kǐwak*; 278. *g'wǎk*; 302. *kog*; 303. *g'óg*; 340. *kǒg* and *g'ǒg*; 342. *kǒg*; 343. *g'óg*; 347. *kiog*; 346. *g'og*; 357. *kǒg*; 358. *g'óg*; E 1. *kwán*: 2. *g'wán*, 3—7. *g'wan*; 8. *kǐwan* and *g'ǐwan*; 32. *kán* and *g'ǐan*; 32. *kán*: 33. *g'án*; 57. *kǐēn*: 58. *g'ien*; 67. *kwət*: 68. *g'wət*; 108. *kǐǎt* and *g'ǐat*; 124. *kwán*: 125. *g'wan*; 129. *kian*: 130. *g'ian*; 149. *kwán*: 150. *g'wán*; 166. *kwət*: 163. *g'wən*; 187. *kwan*: 185. *g'ǎn*; 195. *kǎn* and *g'ǎn*; 238. *kwán*: 241, 242. *g'wán*; 271. *kán*: 272. *g'án*; 276. *kǐer*: 275. *g'ien*; 289. *ker*: 290. *g'er*; 305. *kát*: 311. *g'ád*; 333. *kwán*: 334. *g'wan*; I 31. *kap*: 32—34. *g'ap*; 35, 36. *kiap*: 37. *g'iap*; 72. *kiam*: 74. *g'iap*; 79. *kwəb*: 78. *g'wəb*; 85. *kəm*: 86. *g'əm*.

k' ~ g:

E 9. *k'ǐwan*: 12. *gǐwan*; 301. *k'ǐwǎn*: 310. *gǐwət*.

k' ~ g':

A 77. *k'ǐwang*: 76. *g'wǎng*; 207. *k'wǎng*: 208. *g'wǎng*; 246. *k'óg*: 247. *g'ǐóg*; 270. *k'ǐuk*: 271. *g'ǐuk*; E 9. *k'ǐwan* and *g'ǐwan*; 10. *k'ǐwan*: 11. *g'ǐwan*; 102. *k'ǐət*: 104. *g'ǐər*; 116. *k'ǐwǎt*: 117. *g'ǐwǎt*; 119. *k'wət*: 117. *g'ǐwət*; 151. *k'wad*: 150. *g'wán*; 226. *k'án*: 227. *g'án*; 300. *k'án*: 302. *g'ǐan*; I 47. *k'ám*: 48. *g'ǎm*; 59. *k'iap*: 60. *g'ap*.

g ~ g':

A. 6. *gǐwang*: 4. *g'wǎng*; 22, 23. *gǐwang*: 21. *g'wǎng*; 58. *gǐwǎng*: 57. *g'iwǎng*; 277. *gǐwang*: 276. *g'wǎng*; 333. *gǐəg*: 332. *g'əg*; E 39. *gǐwər*: 37. *g'wən*; I 40. *gǐam*: 41. *g'əm*.

t ~ t':

B 190, 191. *tǐang*: 192. *t'ǐang*; 310. *tog*: 312. *t'ǐog*; 614. *tóg*: 615. *t'óg*; 627. *tǔng*: 626. *t'ǔng*.

t ~ d:

B. 248. *tǐog*: 250. *dǐog*; 560. *tǐóg*: 559. *dǐog*.

t ~ d':

B. 69. *təng*: 79. *d'əg*; 103. *tók*: 104. *d'íók*; 123. *tūk*: 124. *d'ǔk*; 134. *tǐóng*: 135. *d'ǐóng*; 191. *tǐang* and *d'ǐang*; 248. *tǐog*: 249. *d'ǐog*; 310. *tog*: 311. *d'og*; 358. *tieng*: 359. *d'ien*; 403. *táng*: 404, 405. *d'áng*; 420. *tǐóg*: 417, 419. *d'ǐóg*; 531. *tək*: 529. *d'ǎk*; 569. *tək*: 570. *d'əg*; 641. *tiək*: 642. *d'iek*; F. 17. *twán* and *d'wán*; 180. *twər*: 183. *t'wər*; 261. *tǐan*: 262, 263. *d'ǐan*; 267. *tǐwan*: 268. *d'ǐwan*; 298. *tán*: 300. *d'án*; 314. *tǐēn*: 313. *d'ien*.

t ~ î:

B. 190. *tǐang*: 195. *îiang*; 248. *tǐog*: 253. *îiog*; 588. *tog*: 589. *îíóg*; 614. *tóg*: 613. *îíóg*; 681. *tǐəng*: 680. *îiəng*; F. 86. *tǐəd*: 85. *îiəd*; 164. *tǐwat*: 165. *îiwar*; 216. *twən*: 217. *îiwən*; K 4. *tǐəm*: 9. *îiəm*.

t ~ î':

B. 123. *tūk*: 125. *î'ǐuk*; 190. *tǐang*: 197. *î'ǐang*.

t ~ d:

B. 23. *tǐəg*: 24. *dǐəg*; 464. *təng*: 466. *dǐəg*; F. 112. *tǐan*: 113. *dǐwan*; 301. *tǐan*: 302. *dǐan*.

$t \sim d'$:

B. 69. *təng*, 75. *t̃ək*: 70. *d'̃əng*.

$t' \sim d$:

B. 98. *t'̃əg*: 99. *d̃̃əg*; 312. *t'̃iog*: 313. *d̃̃iog*; 351. *t'̃iók*: 350. *d̃̃iók*; 426. *t'̃ióg*: 425. *d̃̃ióg*; 582. *t'̃iog*: 579. *d̃̃iog*; F. 148. *t'̃iər* and *d̃̃iər*; 188. *t'̃ian*: 189. *d̃̃ian*.

$t' \sim d'$:

B. 34. *t'ung*: 38. *d'ung*; 44. *t'ǎng*: 46. *d'ǎng*; 51. *t'̃ieng* and *d'̃ieng*; 119. *t'ák*: 120. *d'ák*; 192. *t'iang*: 191. *d'iang*; 217. *t'ung*: 216. *d'óng*; 226. *t'óg*: 225. *d'óg*; 237. *t'og*: 235. *d'ög*; 314. *t'̃ióg*: 315. *d'̃ióg*; 330. *t'̃iəng*: 331. *d'̃iəng*; 346. *t'̃ieng*: 347. *d'̃ieng*; 481. *t'óg*: 480. *d'óg*; 599. *t'̃iog*: 595. *d'̃iok*; 607. *t'̃iog*: 606. *d'̃iog*; F. 140. *t'wat*: 139. *d'wən*; 185. *t'án*: 186. *d'án*; 206. *t'wər*: 205. *d'wən*; 208. *t'wən* and *d'wən*; 241. *t'̃iər*: 238. *d'̃iər*; 247. *t'̃iat* and *d'̃iat*; 283. *t'án*: 284. *d'án*; 314. *t'ien*: 313. *d'ien*; 317. *t'̃iat* and *d'̃iat*; 331. *t'wər*: 329. *d'wən*.

$t' \sim t̃$:

B. 14. *t'̃ək*: 13. *t̃̃ək*; K. 50. *t'̃iam*: 49. *t̃̃iam*.

$t' \sim t̃'$:

B. 306. *t'̃iang*: 307. *t̃̃iang*; 628. *t'̃iəg*: 629. *t̃̃iəg*; F. 97. *t'̃iwət*: 96. *t̃̃iwət*.

$t' \sim d$:

B. 599. *t'̃iog*: 598. *d̃̃ióg*; F. 283. *t'án*: 285. *d̃̃ian*.

$t' \sim d'$:

F. 312. *t'wən*: 311. *d'̃iən*.

$d \sim d'$:

B. 280. *d̃̃iog*: 277. *d'óg*; 445. *d̃̃iək*: 447. *d'̃ieg*; 592. *d̃̃iog*: 593. *d'̃iog*; 605. *d̃̃iok*: 606. *d'̃iog*; 689. *d̃̃iog*: 688. *d'̃iog*; F. 189. *d̃̃ian*: 186. *d'án*; 194. *d̃̃iən*: 192. *d'̃iən*; 353. *d̃̃iet* and *d'̃iet*.

$d \sim t̃$:

B. 118. *d̃̃iung*: 116. *t̃̃iung*; 250. *d̃̃iog*: 253. *t̃̃iog*.

$d \sim t̃'$:

B. 242. *d̃̃iang*: 243. *t̃̃iang*; F. 189. *d̃̃ian*: 187. *t̃̃ian*.

$d \sim d̃$:

B. 66. *d̃̃iang*: 64. *d̃̃iang*; 85. *d̃̃iəng*: 88. *d̃̃iəng*; 269. *d̃̃ióng*: 275. *d̃̃iók*: 594. *d̃̃iók*: 596. *d̃̃iuk*, 598. *d̃̃ióg*.

$d \sim d'$:

B. 66. *d̃̃iang*: 70. *d'̃iəng*; 663. *d̃̃iək*: 672. *d'̃iak*.

$d' \sim t̃$:

B. 609. *d'̃ióng*: 608. *t̃̃ióng*; 678. *d'̃iog*: 677. *t̃̃iog*; F. 19. *d'wán* and *t̃̃iwan*; 183. *d'̃iwər*: 181. *t̃̃iwər*; K. 21. *d'̃iap*: 20. *t̃̃iap*.

$d' \sim t̃'$:

B. 29. *d'ák*: 30. *t̃̃iak*; 635. *d'̃iog*: 634. *t̃̃iók*.

d' ~ d:

B. 21. *d'ɿək*; 22. *d̥ɿək*; 298. *d'ɿəŋg*; 300. *d̥ɿək*; 465. *d'əg*; 466. *d̥ɿəg*; 470. *d'əŋg*; 471. *d̥ɿəŋg*; 532. *d'ɿəg*; 528. *d̥ɿəŋg*; 683. *d'ɿók*; 682. *d̥ɿók*; F. 284. *d'án*; 285. *d̥ian*.

d' ~ d':

B. 79. *d'əg*; 70. *d̥'ɿəŋg*.

ī ~ i':

B. 116. *īiung*; 117. *i'īiung*.

ī ~ d:

B. 143. *īiung*; 144. *d̥iung*; 274. *īók*; 275. *d̥ók*; 577. *īók*; 576. *d̥ók*; F. 19. *īiwan*; 20. *d̥iwan*; K. 51. *īəp*; 52. *d̥əp*.

ts ~ ts':

B. 219, 220. *tsáng*; 221. *ts'áng*; 295. *tsǐəŋg*; 296. *ts'ǐəŋg*; 366. *tsog*; 367. *ts'óg*; F. 32. *tsiet*; 33. *ts'iet*; 214. *tswən* and *ts'wət*; 246. *tsiəwət*; 245. *ts'īwən*.

ts ~ dz:

B. 178. *tsjəg*; 181. *dzjəg*; 378. *tsáng*; 379. *dzjəŋg*; F. 89. *tsǐən*; 91. *dzǐən*; 122. *tsiəwən*; 123. *dzjəwən*; 214. *tsiəwən*; 215. *dzjəwəd*.

ts ~ dz':

B. 132. *tsǐəŋg*; 133. *dz'ǐəŋg*; 176. *tsəg*; 177. *dz'əg*; 205. *tsəŋg*; 206. *dz'əŋg*; 219. *tsáng*; 222. *dz'áng*; 256. *tsǐəŋg*; 258. *dz'ǐəŋg*; 295. *tsǐəŋg*; 297. *dz'ǐəŋg*; 343. *tsók*; 344. *dz'ók*; 353. *tsjəg*; 354. *dz'jəg*; 378. *tsáng*; 380. *dz'ág*; 516. *tsiung*; 517. *dz'jəŋg*; F. 34. *tsiər*; 35. *dz'ər*; 27. *tsjan*; 30. *dz'iat*; 64. *tsiəwəd* and *dz'jəwət*; 72. *tsǎn*; 73. *dz'án*; 108. *tswən*; 111. *dz'wət*; 352. *tsǐən* and *dz'ǐən*.

ts ~ tɕ:

B. 93. *tsjəŋg*; 94. *tsjəŋg*; 176. *tsəg*; 179. *tsjəg*; K. 7. *tsəm*; 8. *tsjəm*.

ts ~ ts':

B. 172. *tsuk*; 173. *ts'úk*; 284. *tsóg*; 283. *ts'óg*.

ts ~ s:

B. 238. *tsog*; 241. *sók*; 667. *tsóg*; 668. *sóg*; F. 279. *tswád*; 282. *sjəwəd*; K. 14. *tsam*; 15. *sam*.

ts ~ ɕ:

B. 238. *tsog*; 240. *sók*; 521. *tsók*; 527. *sók*; K. 66. *tsjəm*; 67. *sjəm*.

ts ~ ɕ:

B. 259. *tsók*; 262. *sók*; 285. *tsjog*; 291. *sjog*; 364. *tsók*; 368. *sók*.

ts ~ z:

B. 372. *tsáng*; 371. *zjəŋg*; K. 66. *tsjəm*; 68. *zjəm*.

ts' ~ dz:

B. 495. *ts'ók*; 497. *dzók*.

ts' ~ dz':

B. 221. *ts'áng*; 222. *dz'áng*; 296. *ts'ǐəŋg*; 297. *dz'ǐəŋg*; 319. *ts'jog*; 320. *dz'jog*; F. 65.

ts'wər and *dz'wər*; 131. *ts'wər*: 130. *dz'wər*; 166. *ts'wət* and *dz'iwəd*; 210. *ts'wán*: 211. *dz'wan*; 325. *ts'iar*: 324. *dz'iar*.

ts ~ *tʃ*:

B. 168. *ts'ǐək*: 167. *tʃək*.

ts' ~ *tʃ'*:

F 60. *ts'ǐən*: 61, 62. *tʃ'ǐən*; 223. *ts'ǐwan*: 226. *tʃ'ǐwar*.

ts' ~ *dʒ'*:

B 319. *ts'ǐog*: 321. *dʒ'ǐóg*.

ts' ~ *s*:

B 365. *ts'ǐuk*: 370. *suk*; 383. *ts'ǐog*: 384. *sǐog*; F 26. *ts'ǎn*: 37. *san*; 332. *ts'ǐwən*: 333. *swən*; K. 75. *ts'əm*: 74. *səm*.

ts' ~ *ʃ*:

K 75. *ts'əm*: 76. *ʃǐəm*.

ts' ~ *ś*:

B 168. *ts'ǐək*: 185. *śǐak*.

ts' ~ *z*:

B 383. *ts'ǐog*: 385. *zǐóg*.

dz ~ *tʃ*:

B 181. *dzǐəg*: 180. *tʃǐəg*.

dz ~ *tʃ'*:

B 181. *dzǐəg*: 170. *tʃ'ǐək*.

dz ~ *s*:

F 271. *dzǐwan*: 280. *sǐwan*; 286. *dzǐər*: 287. *sǐər*.

dz ~ *ʃ*:

B 493. *dzǐuk*: 500. *ʃǐuk*; F 125. *dzǐwəd*: 127. *ʃǐwəd*.

dz ~ *ś*:

B 671. *dzǐóg*: 672. *śǐóg*; K 46. *dzǐəp*: 45. *śǐəp*.

dz ~ *z*:

B 450. *dzǐak*: 454. *zǐag*.

dz' ~ *tʃ*:

B 177. *dz'əg*: 179. *tʃǐəg*; K 82. *dz'ǐəp*: 84. *tʃəp*.

dz ~ *tʃ'*:

B 162. *dz'ǐang*: 163, 164. *tʃ'ǐang*; 458. *dz'ǐak*: 459. *tʃ'ǐək*.

dz ~ *dʒ'*:

B 320. *dz'ǐog*: 321. *dʒ'ǐóg*.

dz' ~ *s*:

B 621. *dz'ǐog*: 622. *sǐog*; 673. *dz'ǎk*: 674. *sǐak*.

dz' ~ *ś*:

B 162. *dz'ǐang*: 184. *śǐang*; K 55. *dz'ǐap*: 56. *śǐap*.

tʂ ~ tʂ':

B 167. *tʂɛk*: 166. *tʂ'ɛk*.

tʂ ~ s:

B 666. *tʂɔ̃g*: 668. *sɔ̃g*.

tʂ' ~ s:

B. 316. *tʂ'iang*: 322. *sang*.

tʂ' ~ ʂ:

B 391. *tʂ'iang*: 393. *ʂiang*.

tʂ' ~ ʂ:

B 164. *tʂ'iang*: 184. *ʂiang*.

dʒ' ~ s:

B 80. *dʒ'iong*: 84. *siɔ̃g*; 467. *dʒ'ieg*: 469. *siɔ̃g*.

s ~ ʂ:

B 241. *siɔ̃g*: 240. *ʂiɔ̃g*; 504. *siɔ̃g*: 503. *ʂɔ̃g*; K 74. *səm*: 76. *ʂiəm*.

s ~ ʂ:

B 286. *siang*: 287. *ʂiang*; 293. *siog*: 289. *ʂiok*; 552. *siog*: 553. *ʂiog*; F 48. *siər*: 49. *ʂiər*.

s ~ z:

B 384. *siog*: 385. *ziɔ̃g*; F 144, 145. *siat*: 146. *ziad*.

ʂ ~ ʂ:

B 500. *ʂiuk*: 499. *ʂiuk*.

ʂ ~ z:

K 67. *ʂiəm*: 68. *ziəm*.

p ~ p':

D 91. *pák*: 92. *p'ák*; 104. *pək*: 105. *p'iek*; 138, 139. *piog*: 141. *p'iog*; 186. *pung*: 187. *p'iung*; H 12. *p'iwát*: 11. *p'wát*, 14. *p'iwáđ*; 21. *p'iar*: 22. *p'iar*; 31. *p'wán*: 32, 33. *p'wán*; 152. *pian*: 153. *p'ian*.

p ~ b':

D 30. *pwáng*: 31—35. *b'wáng*; 38. *piəng*: 39. *b'ieŋg*; 46. *piǎng*: 47. *b'ǎng*; 59. *pák*: 57, 58. *b'ák*; 84. *piəng* and *b'ieŋg*; 117. *póg*, 116. *póg*: 118, 119. *b'óg*; 122. *pwəg*: 123, 124. *b'wəg*; 130. *piəg*: 129. *b'əg*; 138, 139. *piog*: 142. *b'iog*; H 20. *p'iar* and *b'iar*, *b'iat*; 38. *p'iwən* and *b'iwən*; 39. *p'iat* and *b'iat*; 56. *p'iwən*: 54. *b'iwən*; 81. *p'iwər*: 82. *b'iwər*; 84. *p'iwət*: 85. *b'iwət*; 115. *piət*: 113. *b'ien*; 120. *p'iwət*, 126. *p'iwəd*: 128. *b'iwər*; 150. *p'iwən*: 151. *b'iwər*.

p' ~ b':

D 97. *p'iung*: 98. *b'iung*; 105. *p'iek*: 106. *b'iek*; 114. *p'iwang*: 115. *b'iwak*; 137. *p'igug*: 135. *b'üuk*; 141. *p'iog*: 142. *b'iog*; 150. *p'óg*: 151. *b'iog*; 155. *p'iók*: 154. *b'iók*; 181. *p'iog*: 182. *b'ióg*; H 29. *p'ian*, 30. *p'ian*: 27. *b'ian*; 77. *p'iwən*: 78. *b'iwən*.

It is easily seen that whereas some of the alternations in these four series are quite rare, others are extremely common and regular and can be said to be some of the principal instruments of the Chinese language in word derivation. As particularly important and normal I wish to point out two:

The alternation of unaspirated surd and aspirated sonant: $t \sim d'$, $k \sim g'$, $ts \sim dz'$, $p \sim b'$;

The alternation of aspirated surd and aspirated sonant: $t' \sim d'$, $k' \sim g'$, $ts' \sim dz'$, $p' \sim b'$.

We have to take up next a very intricate question: whether these four series of initials can be proved to alternate with certain other initials not yet discussed, and whether some of these latter initials have alternations between themselves. There are here six questions which have to be answered. Is there an etymological connection between:

1. The k series and the laryngeal explosive $\cdot$;
2. The k series and χ ;
3. The k series and ng ;
4. The t series and the ts series;
5. The p series and m ;
6. n , $\acute{n}$ and l .

The first four of these questions I think must decidedly be answered in the affirmative; of the last two the good examples are so few as to leave room for doubt:

k etc. $\sim \cdot$:

A 1. $k\check{a}ng$: 19. $\cdot\check{a}ng$; 60. $g'w\check{a}ng$: 69. $\cdot w\check{a}ng$; 186. $k\grave{a}ng$: 201. $\cdot\grave{a}ng$; 245. $kung$: 249. $\cdot ung$; 259. $k\check{u}ng$, 260. $g'\check{u}ng$: 265. $\cdot ung$; 266. $g'iwang$: 273. $\cdot w\check{a}ng$; 328. $g'\check{i}og$: 329. $\cdot\check{i}og$; 353. $k\check{i}og$: 354. $\cdot\check{i}og$; E 1. $k\acute{u}n$ and $\cdot w\acute{a}t$; 10. $k'iw\acute{a}n$: 30. $\cdot w\acute{a}n$; 37. $g'w\acute{e}n$: 49. $\cdot w\acute{e}n$; 114. $k'w\acute{a}n$: 121. $\cdot w\acute{a}n$; 117. $g'iw\check{a}t$: 123. $\cdot iw\check{a}t$; 189. $g'iw\check{a}n$: 192. $\cdot iw\check{a}n$; I 1. $g\grave{a}m$: 11. $\cdot\grave{a}m$; 2. $g'\grave{e}m$: 12. $\cdot\grave{e}m$.

k etc. $\sim \chi$:

A 64. $g'ung$: 66. χung ; 76. $g'w\check{a}ng$: 87. $\chi w\check{a}ng$; 80. $k'iwung$: 88. $\chi iwung$; 140. $k\grave{a}wang$, 141. $g'iwang$: 152. $\chi w\check{a}ng$; 215. $k'\check{u}ng$: 222. $\chi iwung$; 307. $k\check{i}ok$: 308. $\chi\check{i}ok$; 362. $k\check{i}og$: 364. $\chi\check{i}og$; E 32. $k\acute{a}n$: 40. $\chi\acute{a}n$; 73. $k\grave{a}t$: 78. $\chi\grave{a}t$; 105. $g'\acute{a}n$: 109. $\chi\check{a}n$; 129. $k\acute{a}n$, 130. $g'\acute{a}n$: 135. $\chi\acute{a}n$; 167. $g'iw\acute{e}n$: 168. $\chi iw\acute{e}n$; 174. $k'iad$: 175. $\chi\check{a}t$; I 38. $k'\check{a}p$: 39. $\chi\check{a}p$; 77. $g'\acute{e}p$: 80. $\chi\check{a}p$.

k etc. $\sim ng$:

A 250. $k\acute{a}ng$: 257. $ng\check{a}ng$; E 2. $g'w\acute{a}n$: 29. $ngw\acute{a}n$; 79. $k\acute{a}n$, 80. kan : 86. $ngan$; 128. $k'\acute{a}n$, 129. $k\acute{a}n$: 137. $ngen$; 138. $g'iw\acute{e}n$: 142. $ng\check{a}n$; 189. $g'iw\check{a}n$: 190. $ng\check{a}n$; 234. $g'\check{e}t$: 235. $ng\check{a}t$; 307. $kw\check{a}t$: 319. $ngw\check{a}t$; 318. $k\acute{e}r$ and $ng\acute{e}r$; I 40. $g\grave{a}m$: 42. $ng\check{a}m$; 63. $k'\check{e}m$: 66. $ng\check{e}m$.

t etc. ~ *ts* etc.:

B 1. *ḍiāng*: 5. *ṣiāng*; 17. *d'iæg*: 19. *ṣiæg*; 70. *ḍ'iāng*: 83. *ṣiāng*; 69. *təng*: 83. *ṣiāng*; 101. *t'iēng*: 110. *ṣiēng*; 106. *t'iæg*: 111. *ṣiæg*; 151. *ṭiāk*: 165. *ṣṭiāk*; 153. *tək*: 167. *ṣṭək*; 157. *ṭiэг*: 174. *ṣṭiэг*; 203, 204. *təng*: 205. *ṣəng*, 206. *dz'əng*; 230. *tūk*: 231. *dz'ūk*; 237. *t'og*: 238. *ṣog*; 264. *t'āng*: 286. *ṣiāng*; 271. *t'āk*: 288. *ṣiāk*; 278. *ṭióg*: 293. *ṣiog*; 306. *t'iang*, 307. *t'iang*: 316. *ṣ'iāng*; 312. *t'ioг*: 319. *ṣ'iog*, 320. *dz'ioг*; 314. *t'ióg*, 315. *d'ióg*: 321. *dz'ióg*; 330. *t'iāng*: 334. *ṣiēng*; 333. *t'iāk*: 335. *ṣiāk*; 342. *d'ióg*: 343. *ṣióg*, 344. *dz'ióg*; 358. *tieng*, 359. *d'ienг*: 360. *ṣ'ieng*; 397. *ṭiæg*: 400. *ṣiək*, 401. *ṣiæg*; 398. *d'ióg*: 402. *zióg*; 428. *ḍiāk*: 428. *dz'iaг*; 429. *t'iæg*: 431. *dz'iaг*; 446. *d'iek*: 449. *siek*; 465. *d'əg*, 466. *ḍiæg*: 467. *dz'iæg*, 469. *ṣiæg*; 485. *d'ióg*: 495. *ṣ'ióg*; 532. *d'iæg*: 540. *ziæg*; 543. *ṭiung*: 547. *dz'ung*; 557. *t'óg*: 558. *ṣóg*; 560. *tióg*, 559. *d'ioг*: 561. *dz'ióg*; 562. *d'iók*: 565. *sióg*; 600. *ṭióg*: 601. *sóg*; 630. *ḍiэг*: 633. *ṣiэг*; 670. *t'ióg*: 671. *dz'ióg*, 672. *ṣióg*; F 2. *ḍiwan*: 9. *dziwan*; 46. *ṭiən* and *ṣiən*; 50. *t'iən*: 52. *ṣiən*; 55. *ṭiēn*: 56. *ṣiēn*; 63. *ṣwan*: 67. *ṣiwan*; 83. *t'iēn*: 87. *ṣṭiēn*; 112. *tian*: 114. *dz'ian*; 118. *d'iwən*: 122. *ṣiwan*, 123. *dziwan*; 119. *d'iwət*: 127, 128. *ṣiwan*; 129. *t'wər*: 131. *ṣ'wər*, 130. *dz'wər*; 133. *d'wat*: 137. *ṣ'wət*; 192. *d'ien*: 196. *ṣiēn*; 290. *ṭiwan*: 291. *ṣiwan*; 327. *d'iwət*: 328. *dziwəd*; 329. *d'wən*: 333. *swən*; 344. *d'wər*: 345. *dz'wən*; K 3. *ḍiam*: 5, 6. *ṣiam*; 9. *ṭiam*: 7. *ṣəm*, 8. *ṣiam*; 16, 17. *d'iap*: 19. *dziap*; 26. *d'əm*: 27. *dz'iem*; 28. *d'əm*: 31. *ṣiəm*; 40. *d'ām*: 44. *ṣiam*; 41. *d'əm* and *dziem*; 62. *tiam*: 64. *ṣiam*.

p etc. ~ *m*:

D 3. *b'ung*: 12. *mung*; 100. *pāk*: 103. *māk*; H 64. *b'iwən*: 66, 67. *mwən*; 71. *b'iwən*: 72. *mwən*; 75. *piwən*: 76. *miwən*; 84. *piwət*: 90. *miwət*.

n, *ń* ~ *l*:

C 49, 50. *n'óg*, 48. *n'og*: 51. *liog*; G 7. *n'iat*: 14. *liat*; 23. *n'ien*: 17. *liwən*.

The medial (intercalary, subordinate) vowels.

I shall design here the word type without *i*, *i*, *w* (type *kán* etc.) by 0. The theoretically possible alternations are the following:

1. 0 ~ *i*; 2. 0 ~ *i*; 3. 0 ~ *w*; 4. 0 ~ *iw*; 5. 0 ~ *iw*; 6. *i* ~ *i*; 7. *i* ~ *w*; 8. *i* ~ *w*; 9. *i* ~ *iw*.

Most of them are well represented:

0 ~ *i*:

A 4. *g'wáng*: 6. *giwang*, 7. *giwǎng*; 8. *kěng*: 1. *kǐǎng*; 32. *kóg*: 31. *k'íók*; 66. *χung*: 67. *χiung*; 76. *g'wáng*: 77. *k'iwang*; 83. *g'eg*: 84. *g'iæg*; 89. *χǎk*: 90. *χǎk*; 98. *kog*: 100. *g'ioг*; 103. *ngáng*: 105. *ngiang*; 134. *k'áng*: 135. *k'ǐǎng*; 143, 144. *kwэг*: 145. *k'iwэг*; 149. *kog*: 150. *kioг*; 153. *kǎng*: 155. *kǐǎng*; 156. *kwǎng*: 158. *giwэг*; 172. *kəng*: 175. *g'iek*; 176. *kwək*: 177. *giwək*; 184. *káng*, 185. *kǎng*: 186, 187. *k'iang*; 190. *k'ung*: 191. *k'ung*; 196. *kog*: 197. *kioг*; 209. *kwǎng*: 212. *giwэг*; 213. *k'ung*: 216. *k'ung*; 246. *k'óg*: 247. *g'ióg*; 250. *káng*: 252. *k'iang*; 257. *ngǎng*: 258. *ngiəng*; 276. *g'wáng*: 277. *giwang*; 278. *g'wǎk*: 279. *k'iwak*; 288, 289. *ək*: 292. *ǐək*; 302. *kog*: 305. *k'ióg*; 318. *kək*, 320. *kэг*:

316. *k'ǎk*; 332. *g'æg*; 333. *gǐæg*; 336. *g'æg*; 337. *g'ǐæg*; 365. *g'og*; 366. *g'ǐog*; B 29. *d'ák*: 30. *ǐ'ǎk*; 34. *ǐ'ung*; 39. *ǐ'ǐung*; 46. *d'ǎng*; 48. *d'ǐang*; 69. *təng*; 70. *d'ǐeng*; 69. *təng*: 75. *tǐək*; 71. *tung*; 72. *tǐung*; 92. *tsǎng*; 94. *tsǐang*; 150. *ǐ'ák*; 151. *ǐ'ǎk*; 166. *ts'ək*: 168. *ts'ǐək*; 176. *tsəg*; 177. *dz'æg*; 178. *tsǐæg*; 181. *dzǐæg*; 179, 180. *tsǐæg*; 210. *d'ǎng*: 213. *ǐ'ǎg*; 264. *ǐ'ǎng*; 265. *dǐang*; 276, 277. *d'óg*; 280. *dǐog*; 284. *tsóg*; 285. *tsǐog*; 290. *sog*; 292. *sǐog*; 311. *d'og*; 310. *tog*; 313. *dǐog*; 312. *ǐ'ǐog*; 318. *ts'og*; 319. *ts'ǐog*; 322. *sǎng*; 323. *sǐak*; 355. *səng*; 356, 357. *sǐəng*; 370. *suk*; 365. *ts'ǐuk*; 372. *tsǎng*; 371. *zǐang*; 378. *tsǎng*; 379. *dzǐang*; 403. *tǎng*; 405. *d'ǎng*; 407, 408. *ǐ'ǎng*; 417. *d'óg* and *d'ǐog*; 424. *ǐ'óg*; 426. *ǐ'ǐog*; 459. *ts'ək*; 458. *dz'ǎk*; 463. *ts'əg*; 462. *tsǐag*; 465. *d'æg*; 466. *dǐəg*; 470. *d'əng*; 471. *dǐəng*; 480. *d'óg*; 485, 486. *d'ǐog*; 551. *şög*; 553. *şǐog*; 588. *tog*; 589. *ǐ'ǐog*: 601. *sóg*; 600. *ǐ'ǐog*; 614. *tóg*; 612, 613. *ǐ'ǐog*; 673. *dz'ák*; 674. *sǐak*; 675. *tsək*; 676. *tsǐək*; 691. *d'og*; 693. *d'ǐog*; C 1. *lək*; 2. *lǐæg*; 5. *nák*; 6. *nǐak*; 28. *glǎg*; 25. *gliang*; 34. *nǒg*; 35. *nǐog*; 52. *ləng*; 53. *lǐəng*; 70. *log*; 71. *lǐog*; 72. *nəg*; 73. *nǐəg*; D 31—35. *b'wǎng*; 36. *b'ǐwǎng*; 45. *p'ǎng*; 47. *b'ǐǎng*; 53, 54. *b'óg*; 56. *b'ǐog*; 67. *mǎng*; 70. *mǎng*: 68. *mǐwǎng*; 86. *pǔng*; 87. *pǐung*; 126. *pǎng*; 127, 128. *b'ǐǎng*; 129. *b'əg*; 130. *pǐəg*; 148. *pǒg*; 149. *pǐog*; 150. *p'óg*; 151. *b'ǐog*; 152. *b'óg*; 153. *b'ǐog*; 164. *mǒg*; 165. *mǐog*; 186. *pung*; 187. *p'ǐung*; E 1. *kwǎn*, 2. *g'wǎn*, 3—7. *g'wan*: 8. *kǐwan*, 9. *g'ǐwan*, *k'ǐwan*, 12. *gǐwan*; 19. *kwən*; 22. *gǐwən*; 25. *g'wər*; 28. *gǐwər*; 32. *kǎn* and *g'ǐan*; 37. *g'wən*: 39. *gǐwər*; 46, 47. *χwǎr*: 44, 47. *χǐwar*; 53. *kwǎn*: 54. *kǐwǎn*; 74. *ked*; 75. *kǐed*; 95. *ǎn*: 96. *ǐ'an*; 106. *kat*; 108. *kǐǎt*, *g'ǐat*; 120. *kwət*, 119. *k'wət*; 117. *g'ǐwət*; 122. *ǐ'wat*: 123. *ǐ'wat*; 132. *kwǎn*; 133. *kǐwan*; 140. *g'wad*; 139. *gǐwǎt*; 154. *ǐ'wǎn*, 156. *ǐ'wan*: 158. *ǐ'wǎn*, 161. *ǐ'war*; 185. *g'ǎn*; 186. *g'ǐǎn*; 226. *k'ǎn*, 228. *g'ǎn*; 230. *g'ǐǎn*; 231. *k'at*; 232, 233. *g'ǐat*; 300. *k'ǎn*; 302. *g'ǐan*; 312—314. *kwǎd*, 307. *kwǎt*; 315. *kǐwǎd*; 319. *ngwǎt* and *ngǐwǎt*; F 19. *d'wǎn* and *ǐ'wan*; 63. *tswən*; 64. *tsǐwǎd*; 74. *d'ǎn*: 75. *tǐan*; 166. *ts'wət* and *dz'ǐwǎd*; 180. *twər*; 181, 182. *ǐ'wər*, 183, 184. *d'ǐwər*; 185. *ǐ'ǎn*, 186. *d'ǎn*; 188. *ǐ'ǐan*, 187. *ǐ'ǐan*, 189—191. *dǐan*; 214. *tswən* and *tsǐwən*; 214. *tswən*, *ts'wət*; 215. *dzǐwǎd*; 216. *twən*; 217. *ǐ'wən*; 233. *ts'wər*; 234, 235. *sǐwən*, 236. *sǐwǎd*; 259. *d'ǎn*: 262, 263. *d'ǐan*; 279. *tswǎd*; 282. *sǐwad*; 283. *ǐ'ǎn*, 284. *d'ǎn*: 285. *dǐan*; 299. *tǎn*; 301. *tǐan*; 308. *swən*; 309. *sǐwər*; 316. *d'át*; 317. *d'ǐat*; 318. *tǎn*; 319. *ǐ'ǐan*; 333. *swən*; 332. *ts'ǐwən*; 343. *d'wər*; 342. *d'ǐwǎd*; G 1. *nan*: 2, 3. *nǐan*; 25. *lǎt*: 27, 28. *ǐ'at*; H 10. *pwát*, 11. *p'wát*: 12. *pǐwǎt*, 14. *p'ǐwǎd*; 24. *p'wər*; 23. *p'ǐwər*; 26, 27. *b'ǎn*: 27, 28. *b'ǐan*; 66, 67. *mwən*; 69. *mǐwən*; 79. *b'wət*; 81. *pǐwər*; 83. *pwət*; 84. *pǐwət*, 87. *pǐwər*; 125. *pǐəd*; 126. *pǐwǎd*; 155. *mwǎn* and *mǐwǎn*; I 8. *ǎm*: 9, 10. *ǐ'am*; 12. *ǐ'em*; 13, 14. *ǐ'em*; 15—18. *ǎm*: 19, 20. *ǐ'em*; 31. *kap*, 34. *g'ap*; 39. *χǐǎp*; 43. *χǎp*: 44, 45. *χǐǎp*; 50. *ǎp*: 51. *ǐ'ap*; 62. *k'əm*; 63. *k'ǐəm*; 64. *ngam*; 65. *ngǐam*; 77. *g'ǎp*: 80. *χǐǎp*; 83. *k'am*; 84. *k'ǐam*; K 1. *tsám*: 5, 6. *tsǐam*; 7. *tsəm*: 8. *tsǐəm*; 28. *d'əm*: 29. *d'ǐəm*; 36. *tsám*; 38. *dz'ǐap*; 40. *d'ám*; 41. *dǐam*; 42. *d'əm*; 43. *dǐəm*; 57. *d'ám*: 58. *ǐ'am*; 74. *səm*; 76. *sǐəm*; 81. *dz'ǎp*; 82, 83. *dz'ǐǎp*; 87. *nǎp*; 90. *nǐǎp*.

0 ~ i:

A 133. *g'eg*; 132. *g'ǐeng*; 196. *kog*; 198. *kiog*; 303. *g'óg*; 306. *kiog*; 346. *g'og*; 347. *kiog*; B 103. *tók*; 104. *d'íók*; 159. *tog*; 160. *tíog*; 153. *tək*; 152. *tiək*; 235. *d'ók*; 236. *d'íók*; 480. *d'óg*; 484. *d'íog*; 503. *şög*; 504. *síog*; C 12. *log*; 13. *liog*; 34. *nǒg*; 36. *níog*;

D 104. *pək*: 105. *p'iek*; E 84. *kwád*: 83. *kiwat*; 89. *k'wad*: 88. *kiwat*; 106. *kat*: 107. *k'iat*; 128. *k'án*: 129. *kian*; 178. *án*, 179. *an*: 180, 181. *ian*; 203, 204. *k'ər*: 205. *k'iar*; 224. *ken*: 225. *kien*; 305. *kát*; 311. *g'ád*: 309. *k'iat*, 316, 317. *k'iad*; 330. *at*: 331. *ian*; F 307. *ser* and *siər*; 326. *dz'er*: 324. *dz'iar*; 350. *tsed*: 351. *tsiər*; G 11. *glán*: 12, 13. *glian*; I 21. *g'am*: 27. *kiap*; 31. *kap*, 32—34. *g'ap*: 35. *kiap*, 37. *g'iap*; 60. *g'ap*: 59. *k'iap*; 83. *k'am* and *k'iam*; K 32, 33. *d'am*: 34. *d'iam*; 60. *d'áp*: 61. *d'iap*.

0 ~ w:

20. A *g'ǎng*: 21. *g'wǎng*; 50. *g'ǎng*: 49. *g'iwang*; 51. *g'ǎng*: 52. *g'wǎng*; 73. *k'ǎng*: 77. *k'iwang*; 147. *k'ǎg*: 148. *k'iwǎg*; 202. *ǎng*: 203. *ǎng*; 206. *k'ǎng*: 207. *k'wǎng*; 331. *kək*: 330. *k'wák*; 335. *g'ǎg*: 334. *g'iwǎg*; D 24, 25. *mək*: 27. *mwǎg*; 37. *b'ǎng*: 35. *b'wǎng*; 40. *b'ǎng*: 42. *b'wǎg*; 43. *mǎng*: 44. *m'iwang*; 121. *pək*: 122. *pwǎg*; E 45. *ǎg*; 42. *ǎg*: 46, 48. *ǎg*; 52. *k'ian*: 55. *k'iwǎn*; 79. *kán*, 80. *kan*: 81. *g'wǎn*; 99. *k'ən*: 100. *k'wǎn*; 111. *k'ən*: 112. *g'wǎt*; 128. *k'án*: 132. *kwǎn*; 142. *ngǎn*, 143. *ngian*: 139. *g'wǎt*, 138. *g'wǎn*; 153. *ǎn*: 158, 160. *ǎn*; 185. *g'ǎn*: 187. *kwan*; 265. *ǎn*: 266. *ǎn*; 268. *ǎd*: 270. *ǎd*; 291, 292. *g'ian*: 293, 294. *g'iwǎn*; 305. *kát*, 311. *g'ád*: 307. *kwǎt*, 312, 313. *kwǎd*; F 41. *tsian*: 43. *tsiwan*; 112. *tian*: 113. *ǎwan*; 160. *tán*: 161. *d'wǎn*; 259. *d'án*: 265, 266. *d'wǎn*; 261. *tian*: 267. *t'iwán*; 302. *ǎian*: 303. *t'iwán*; G 1. *nan*: 4, 5. *nwǎn*; 7. *ǎiat*: 8. *ǎiwat*; H 21. *pǎg*: 23. *p'ǎg*; 26, 27. *b'ǎn*: 31. *pwǎn*, 32. *p'wǎn*; 68. *mǎn*: 69. *m'iwán*; I 38. *k'ǎp* and *k'iwǎb*; 77. *g'ǎp*: 78. *g'wǎb*; 85. *tǎp*: 86. *twǎb*; K 87. *nǎp*: 89. *nwǎb*.

0 ~ iw:

A 20. *g'ǎng*: 22. *g'iwang*; 109. *ngǎg*: 110. *ng'iwǎg*; 243. *g'ǎng*: 244. *k'iwang*; E 40. *ǎn*: 41. *ǎn*; 128. *k'án*: 133. *k'iwán*; 300. *k'án*: 301. *k'iwǎn*; 305. *kát*, 311. *g'ád*: 308. *g'iwǎt*, 315. *k'iwǎd*; F 163. *tát*: 164. *t'iwat*; 259. *d'án*: 269. *d'iwán*.

0 ~ iw:

A 8. *kǎng*: 9, 10. *kiweng*, 11, 12. *g'iweng*.

i ~ i:

A 58. *g'iwǎng*: 57. *g'iweng*; 129. *k'ǎng*: 130. *kieng*; 192. *k'ǎg*: 193. *g'ieg*; 197. *k'iog*: 198. *k'iog*; 350. *ǎg*: 349. *ǎg*; 352. *ǎg*; 354. *ǎg*: 356. *ǎg*; 363. *g'iog*, 364. *ǎg*: 362. *k'iog*; B 54. *tǎng*: 53. *d'ieug*; 248. *t'iog*, 253. *t'iog*: 245. *tiok*; 302—304. *dz'ǎng*: 305. *dz'iek*; 340. *d'ǎng*: 341. *tieng*; 485. *d'ǎg*: 484. *d'ǎg*; 559. *d'iog*: 560. *tiog*; 579. *d'iog*: 582. *t'iog*; 592. *d'iog*: 593. *d'iog*; 598. *ǎg*: 599. *t'iog*; 605. *d'io*, 607. *t'iog*: 606. *d'iog*; 689. *d'iog*: 688. *d'iog*; C 35. *ǎg*: 36. *niog*; D 38. *pǎng*: 39. *b'ieug*; 84. *pǎng* and *b'ieug*; E 108. *k'iat*, *g'iat*: 107. *k'iat*; E 220. *ǎg*: 221. *ǎg*; F 114. *dz'ǎn*: 117. *ts'iar*; 148. *d'ǎg* and *t'ǎg*; 176. *t'ǎt*: 174. *d'ien*; 248. *ǎd*: 249. *d'iad*; 286. *dz'ǎg*: 287. *siər*; 315. *t'ǎt* and *t'iet*; 314. *tiěn* and *t'ien*; 353. *d'ǎt* and *d'iet*; G 31. *ǎd*: 32. *liər*; H 21. *pǎg*: 22. *p'ǎg*; 30. *p'ǎn*: 29. *p'ǎn*; 153. *p'ǎn*: 152. *pian*; I 29. *g'iam*: 37. *g'iap*; 84. *k'ǎm*: 83. *k'iam*.

i ~ w:

A 1. *k'ǎng*: 3. *kwǎng*; 73. *k'ǎng*: 76. *g'wǎng*; 315. *k'ǎk*: 317. *k'wák*; E 52. *k'ian*:

53. *kwân*; 142. *ngiăn*, 143. *ngian*: 140. *g'wad*; 153. *ĩăn*: 154. *·wân*, 156. *·wan*; 186. *g'ĩăn*: 187. *kwan*; F 105. *twât*: 106. *d'wât*: 104. *ĩ'iat*, 103. *d'iat*; 209. *tsian*: 210. *ts'wân*, 211, 212. *dz'wan*; 261. *tian*: 266. *d'wân*; 311. *điăn*, *đ'ien*: 312. *t'wên*; G 2, 3. *niân*: 4, 5. *nuân*; H 21. *pĩər*: 24. *p'wər*; 27. *b'ian*, 30. *p'ian*: 31. *puân*, 32. *p'wân*; 45. *b'ian*: 48. *puân*; 68. *mien*: 66, 67. *mwên*; 144—146. *b'iad*: 143. *b'wad*.

i ~ w:

A 93. *g'ieŋ*: 94. *g'wěŋ*; E 129. *kian*: 132. *kwân*; 309. *k'iat*, 316, 317. *k'iad*: 307. *kwăt*, 312—314. *kwád*; G 32. *liər*: 33. *lwər*; H 22. *p'ĩər*: 24. *p'wər*; 29. *p'ian*: 32. *p'wân*; 89. *miat*: 88. *mwăt*.

i ~ i̯w:

E 129. *kian*: 133. *kĩwan*; 210. *kiər*: 211. *g'ĩwer*; F 30. *dz'iat*: 31. *dz'ĩwat*; 281. *siən*: 278. *dziwən*; H 22. *p'ĩər*: 23. *p'ĩwər*.

The examples given are sufficient to show that the alternations $0 \sim i$, $0 \sim w$, $i \sim i̯$, and quite particularly the first one (type *kân* ~ *kian*), are among the fundamental means of the Chinese language for creating derivatives from one and the same word stem.

The principal vowels.

By alternations of all the vowels of the Archaic Chinese phonetic system the language has created an *Ablaut* system which is extremely rich and varied, and it is in this respect a true sister language of Tibetan, in which such an *Ablaut* plays a predominant part. I shall exemplify first the alternations of the different varieties of the same principal vowel (*â*: *a*: *ă*; *o*: *ô*: *ô*: *ē*: *e*: *ũ*: *u*), and then the alternations of the different principal vowels.

â ~ a:

For the very numerous cases of type *â*: *ia* (*kân*: *kian* etc.) see alt. $0 \sim i̯$ above. Furthermore:

E 2. *g'wân*: 3—7. *g'wan*; 79. *kân*: 80. *kan*; 121. *wân*: 122. *·wat*; 124. *kwân*: 125. *g'wan*; 154. *·wân*: 156. *·wan*; 178. *·ân*: 179. *·an*; 252. *·ân*: 254. *·at*; 333. *kwân*: 334. *g'wan*; F 210. *ts'wân*: 211, 212. *dz'wan*; G 4, 5. *nuân*: 1. *nan*; H 136. *puât*: 137. *b'wat*; I 71. *glâm*: 69. *klam*; K 1. *tsâm*: 2. *dz'am*.

â ~ ă:

A 76. *g'wâng*: 73. *kĩăŋ*; 134. *k'âng*: 135. *k'ĩăŋ*; 184. *kâng*: 185. *kăŋ*; 207. *k'wâng*: 209. *kwăŋ*, 206. *k'ăŋ*; 250. *kâng*: 257. *ngăŋ*; 323. *·âk*, *·âg*: 324. *·ăg*; D 35. *b'wâng*: 37. *b'ăŋ*; 67. *mâng*: 70. *măŋ*; 91. *pâk*: 92. *p'ăk*; 126. *pâng*: 127, 128. *b'ĩăŋ*; 168. *pâk*: 169. *păk*; 226. *k'ân*, 227. *g'ân*: 228, 229. *g'ăn*, 230. *g'ĩăn*; 305. *kât*, 312—314. *kwád*: 307. *kwăt*, 315. *kĩwăd*; F 73. *dz'ân*: 72. *tsăn*; H 10. *puât*: 12. *pĩwăt*; I 47. *k'âm*: 48. *g'ăm*.

a ~ *ǎ*:

A 6. *giwang*: 7. *giwǎng*; 22. *giwang*: 20. *g'ǎng*; 49. *g'iwang*: 50. *g'ǎng*; 77. *k'iwang*: 73. *k'ǎng*; 168. *k'iang*: 170. *k'ǎng*; 253. *g'iang*: 254. *g'ǎng*; 279. *k'iwak*: 278. *g'wǎk*; B 48. *d'iang*: 46. *d'ǎng*; E 13. *giwan* and *giwǎn*; 56. *g'iwān*: 55. *k'iwǎn*; 108. *g'iat* and *k'iat*; 143. *ngian*: 142. *ngiǎn*; 161. *īwar*: 160. *īwǎn*; 280. *īwan*: 281. *īwǎn*; F 37. *san*: 26. *ts'ǎn*; H 27. *b'ian* and *b'ǎn*.

ō ~ *o*:

A 124. *kōg*: 122, 123. *kog*; B 235. *d'ōg*: 237. *t'og*; 283. *ts'ōg*: 285. *tsiog*; 551. *šōg*: 553. *šiog*; C 34. *nōg*: 35. *nīog*; D 148. *pōg*: 149. *pīog*; 164. *mōg*: 165. *mīog*.

ō ~ *ô*:

A 124. *kōg*: 121. *g'ôk*; 340. *kōg*: 342. *kôg*; 340, 341. *g'ōg*: 343. *g'ôg*, 339. *g'ôk*.

ō ~ *ô*:

A 305. *k'ōg*: 306. *kiog*; 340. *g'ōg*: 345. *k'ôg*; B 235. *d'ôk*: 236. *d'îôk*; 283. *ts'ōg*: 284. *tsôg*; 503. *šōg*: 504. *siôg*; 666. *tsōg*: 667. *tsôg*, 668. *sôg*.

o ~ *ô*:

A 113. *ngiog*: 108. *ngôk*; 122, 123. *kog*: 121. *g'ôk*; 218. *g'og*: 221. *kôg*; D 52. *b'og*, 49. *b'ok*: 53, 54. *b'ôg*; 151. *b'iog*: 150. *p'ôg*; 153. *b'iog*: 152. *b'ôg*.

o ~ *ô*:

A 197. *k'iog*: 199. *k'îog*; 301. *k'og*: 300. *k'ôg*; 302. *kog*: 303. *g'ôg*; 347. *kiog*: 348. *kiôg*; 349. *īog*: 352. *īôg*; 354. *īog*: 356. *īôg*; 363. *giog*: 362. *kiôg*; B 248. *t'iog*, 253. *t'iog*: 255. *t'îog*; 273. *t'îok*: 274. *t'îôk*, 275. *d'îôk*; 280, 281. *d'iog*: 276, 277. *d'ôg*; 285. *tsiog*: 284. *tsôg*; 312. *t'iog*, 313. *d'iog*: 314. *t'îôg*, 315. *d'îôg*; 319. *ts'iog*, 320. *dz'iog*: 321. *dz'îog*; 366. *tsog*: 367. *ts'ôg*; 456. *s'iog*: 452. *s'îôk*; 559. *d'iog*: 560. *tiôg*; 588. *tog*: 589. *t'îog*; 691. *d'og*: 692. *d'ôg*; C 48. *n'iog*: 49, 50. *n'îog*; 70. *log*: 71. *l'îog*; D 181. *p'iog*: 182. *b'îog*.

ô ~ *ô*:

A 121. *g'ôk*: 125. *k'îog*; 344. *k'ôg*: 345. *k'ôg*; D 53, 54. *b'ôg*: 56. *b'îog*; 117. *pôg*: 118, 119. *b'ôg*.

ě ~ *e*:

For the numerous cases of this alternation see alt. *o* ~ *î* and *o* ~ *î* above.

ũ ~ *u*:

A 36. *k'ūk*: 37. *kuk*; 215. *k'ũng*: 214. *k'ung*; 268. *k'ũng*: 270. *k'īuk*; B 42. *d'ũg*: 41. *d'uk*; 123. *tūk*: 125. *t'īuk*; 115. *d'ũng*: 116. *t'īung*; 626. *t'ũng*: 625. *t'ung*; D 86. *pũng*: 87. *p'īung*; 90. *b'ũng*: 94. *p'uk*; 101, 102. *b'ũg*: 98. *b'īung*; 135, 136. *b'īūk*: 137. *p'īug*.

a ~ *e*:

A 1. *k'ǎng*: 8. *kěng*; 6. *giwang*, 7. *giwǎng*: 11, 12. *g'iweng*; 52. *g'wǎng*: 53. *kiweng*; 91. *kāng*: 93. *g'ieug*; 115. *xiang*: 117. *xieng*; 134. *k'ǎng*, 135. *k'ǎng*: 136. *g'ěng*; 184. *kāng*, 185. *kǎng*: 188. *kieng*; 207. *k'wǎng*, 209. *kwǎng*: 212. *giwěng*; 240. *k'iang*: 241.

k'iwěng; 253. *g'iang*, 254. *g'ǎng*; 255. *kjěng*; B 5. *siang*: 6. *sǐg*; 48. *d'iang*: 53. *d'ieng*; 46. *d'ǎng*, 44. *t'ǎng*: 54. *tjěng*; 108. *siang*: 110. *sǐng*; 136. *d'áng*: 138. *d'ieng*; 151. *tjak*: 154. *t'iek*; 265. *djang*, 264. *t'áng*: 266. *tieng*; 338. *ts'áng*: 339. *ts'ieng*; 391. *ts'iang*: 392. *ts'jěng*; 405. *d'áng*, 407, 408. *tjang*: 410. *djěng*, 412. *t'ieng*; 444. *djak*: 445. *djěk*; 458. *dz'jak*: 459, 460. *ts'ek*; 462. *tsiag*: 403. *tsǐg*; 529. *d'ǎk*: 531. *těk*; 660. *djak*: 661. *djěk*; C 58. *liang*: 59. *ljěng*, 60. *lieng*; D 1. *b'ǎng*: 2. *pěng*; 6. *mǎng*: 9. *mieng*; 31. *b'wǎng*: 38. *pjěng*, 39. *b'ieng*; 183. *p'iwang*: 184. *p'jěng*; E 9. *g'iwān*, 12. *giwān*, 13. *giwǎn*: 15, 16. *g'iwēn*; 34. *g'ian*: 36. *g'iwēn*; 52. *k'ian*: 57. *kjēn*, 58. *g'ien*; 63. *kwāt*: 64. *kiet*; 86. *ngan*: 87. *ngjēn*; 117. *g'iwāt*: 118. *g'iwet*; 232, 233. *g'iat*: 234. *g'jēt*; 252. *án*: 253. *ien*; 254. *at*: 255. *jēt*; 268. *iad*: 269. *jēd*; F 30. *dz'iat*: 32. *tsiet*, 33. *ts'iet*; 54. *tán*: 55. *tjēn*; 88. *dz'ian*: 89, 90. *tsjēn*; 175. *t'át*: 176. *t'jēt*; 186. *d'án*: 188. *t'ian*, 189. *djan*: 192. *d'jēn*, 194. *djēn*; 201. *t'át*: 200. *d'iet*; 218. *t'án*: 219, 220. *tjēn*; 271. *dzjwān*: 276, 277. *dzjwēn*; 334. *siang*: 335. *sǐng*; 346. *d'iad*: 347. *djēt*; G 7. *njat*: 9. *njēt*; 50. *liat*: 51. *ljēt*; H 1. *p'iwǎn*: 2. *p'ien*; 29. *p'ian*, 30. *p'ian*: 37. *pien*; 46. *pwán*, 45. *b'ian*: 47. *pien*; 73. *b'iwǎn*: 74. *p'jēn*; 123. *piad*: 124. *pjēd*; 152. *pian*: 154. *pjēn*.

a ~ ə:

A 29. *kǎng*: 30. *kəg*; 73. *kjǎng*: 78. *kjəng*; 92. *g'ǎng*: 114. *xiəng*; 153. *kǎng*, 155. *kjǎng*: 163. *kjək*; 169. *kjǎng*: 172. *kəng*, 175. *g'jək*; 229. *kwǎng*: 231. *g'wəng*; 257. *ngǎng*: 258. *ngjəng*; B 64. *djang*: 70. *d'jəng*; 265. *djang*, 264. *t'áng*: 267. *tjəng*; 335. *giak*: 336. *sǐək*; 406. *tjang*: 411. *təng*; 662. *d'jak*: 663. *djək*; C 24. *liang*: 26. *ljəng*; 56. *nǎng*: 57. *njəng*; D 31. *b'wǎng*: 40. *b'əng*, 42. *b'wəg*; 168. *pák*, 169. *pāk*: 170. *pjək*; E 1. *kwán*, 2. *g'wán*, 3—7. *g'wān*, 12. *giwān*, 9. *g'iwān*, *k'iwān*, 13. *giwǎn*: 19. *kwən*, 17. *k'wən*, 22. *giwən*, 21. *kjwən*; 47. *xiwar* and *xiwər*; 53. *kwán*, 54. *kjwǎn*, 55. *k'iwǎn*: 60. *kwən*, 61. *k'wən*; 63. *kwāt*: 67. *kwət*; 72. *g'iat*: 73. *kjət*; 117. *g'iwāt*, 116. *k'iwāt*: 117. *g'iwət*, 120. *kwət*, 119. *k'wət*; 130, 131. *g'ian*: 134. *g'jən*; 143. *ngjan*, 142. *ngjǎn*: 139. *giwən*; 140. *g'wad*, 139. *giwāt*: 141. *giwəd*; 250. *ɣwán*: 251. *xiən*; 259. *iwǎn*: 260. *jwən*; 263. *giwǎn*: 264. *g'iwən*; 280. *jwān*, 281. *jwǎn*: 282. *jwən*; 300. *k'án*, 302. *g'ian*: 303. *g'ən*, 304. *kjən*; 307. *kwāt*, 308. *giwāt*: 310. *giwət*; 314. *kwād* and *kjəd*; 319. *ngwāt*, *ngjwāt* and *ngwət*; 320. *ngjād*: 321. *ngjəd*; 325. *giwǎn*: 326. *giwən*; 332. *g'jǎn*: 331. *kjən*; F 1. *djwān*: 2. *djwən*; 88. *dz'ian*: 95. *siən*; 114. *dz'ian*: 115. *dz'wən*; 152. *dz'wād*: 153. *dz'iwəd*; 158. *siwad* and *siwət*; 161. *d'wán*: 162. *d'wən*; 210. *ts'wán*, 211. *dz'wān*: 214. *tswən*, 213, 214. *tsjwən*; 228, 229. *tjan*: 230, 231. *tjən*; 271. *dzjwān*: 278. *dzjwən*; 310. *tán*: 311. *djən*, *d'jən*; 337. *ts'iwad*: 338. *swəd*; G 15, 16. *lian*: 17, 18. *ljwən*; 29, 30. *liad*: 49. *ljəd*; H 1. *p'iwǎn*: 3, 4. *pjwən*; 12. *pjwāt*, 14. *p'iwād*: 13. *p'iwət*, 15. *p'iwəd*; 31. *pwán*, 32. *p'wán*, 27. *b'án*, *b'ian*: 38. *pjwən*, *b'jwən*; 40. *b'iwāt*: 41. *b'iwət*; 58, 59. *b'wād* and *b'wət*; 73. *b'iwǎn*: 75. *pjwən*; 88. *mwāt*, 89. *miat*: 90. *mjwət*, 91. *mjwəd*; 92. *man*: 93. *mwən*; 122. *pjād*: 125. *pjəd*; 148. *b'iwǎn*: 149. *b'jwən*; I 4. *g'áp*: 5. *kəp*; 8. *ám*, 9, 10. *jām*: 15—18. *əm*, 19, 20. *jəm*; 21, 22. *g'ám*: 24—26. *g'əm*; 27. *kiap*: 28. *g'əp*; 47. *k'ám*: 49. *k'jəm*; 60. *g'ap*, 59. *k'iap*: 61. *g'jəp*; 64. *ngam*, 65. *ngjəm*: 66. *ngjəm*; 74. *g'iap*: 77. *g'əp*; K 1. *tsám*, 2. *dz'am*, 5. *tsjəm*: 7. *tsəm*, 8. *tsjəm*; 3. *djam*: 4. *tjəm*, 9, 10. *tjəm*; 20. *tjap*: 23. *tjəp*;

36. *tsám*: 37. *ts'iam*; 38. *dz'iap*: 39. *sap*; 40. *d'am*, 41. *diam*: 43. *diam*; 57. *d'am*, 58. *iam*: 59. *d'am*; 64, 65. *tsiam*: 66. *tsiam*; 72. *dz'am*: 73. *dz'iam*; 77. *dz'am*: 78. *ts'iam*.

a ~ ε:

A 184. *káng*: 189. *g'weng*; 229. *kwáng*: 230. *g'weng*; 330. *k'wák*: 331. *kek*; E 9. *k'íwan*, 12. *gíwan*: 20. *k'íwen*; 84. *kwád*, 83. *kiwat*: 82. *g'wet*; 124. *kwán*, 125. *g'wan*: 127. *gíwen*; 128. *k'án*, 129. *kian*: 137. *ngen*; 191. *ñán*: 193. *ien*; 236. *g'wát*: 237. *g'wet*; I 8. *ám*: 12. *em*; 9. *iam*: 13, 14. *iem*; 21, 22. *g'am*: 23. *g'em*; 40. *giam*: 41. *g'em*; 60. *g'ap*, 59. *k'iap*: 61. *g'iep*; 72. *kiam*: 73. *g'em*; 74. *g'iap*: 75. *g'ep*; K 20. *liap*: 22. *liap*; 24. *siam*: 25. *sem*.

a ~ o:

A 106. *ngák*: 108. *ngók*; 120. *g'ák*: 121. *g'ók*; 239. *g'áng*: 242. *kóng*; B 165. *tsiak*: 187. *síok*; 265. *díang*, 264. *t'áng*: 269. *díóng*; 272. *liak*: 273. *liok*; 288. *síak*: 292. *síóg*; 306. *t'iang*, 307. *t'iang*: 308. *t'ióng*; 416. *liag*: 420. *tióg*, 417, 419. *d'ióg*; 450. *dzíak*, 454. *ziag*: 452. *síók*, 456. *síog*; 518, 519. *tsiak*: 521. *tsíók*; 585. *t'íak*: 586. *d'íók*; C 3. *níang*: 4. *nóng*; D 18. *mág*: 21. *móg*; 160. *mág*: 161. *míóg*; 162, 163. *mág*: 164. *móg*, 165. *míog*.

a ~ u:

A 14. *chák*: 15. *chuk*; 20. *g'ǎng*: 25. *g'ǔng*; 51. *g'ǎng*: 54. *kǔng*; 73. *kǐǎng*, 77. *k'íwang*: 79. *kíung*, 80. *k'íung*; 92. *g'ǎng*: 95. *kǔng*; 169. *kǐǎng*: 173. *g'íǔng*; B 46. *d'ǎng*, 44. *t'ǎng*: 56. *tǔng*; 119. *t'ák*: 123. *tǔk*; 120. *d'ák*: 124. *d'ǔk*; 130. *d'áng*: 131. *d'ung*; 141. *tiang*: 143. *tiung*; 151. *liak*: 155. *tǔk*; 198. *táng*: 199. *d'ung*; 228. *d'ǎk*: 230. *tǔk*; 361. *tsáng*: 362. *ts'ung*; 405. *d'áng*, 407, 408. *liang*: 414. *diung*; 406. *tiang*: 413. *d'ǔng*; 498. *sák*: 500. *síuk*, 499. *síuk*; 507. *liak*: 509. *d'íuk*; D 1. *b'ǎng*: 3. *b'ung*; 6. *mǎng*: 12. *mung*; 46. *pǎng*: 48. *p'íung*; 62. *pǎng*: 63. *p'íung*; 79. *mǎng*: 81. *mǔng*; 83. *b'íwang*: 87. *píung*; 89. *b'ǎng*: 90. *b'ǔng*; 92. *p'ák*: 94. *p'uk*; 95. *pǎng*: 97. *p'íung*, 98. *b'íung*; 178. *mág*: 180. *míǔg*.

e ~ ε:

B 101. *t'ǐěng*: 102. *d'eng*; E 64. *kiet*: 69. *gíwet*; 225. *kien*: 224. *ken*.

e ~ ə:

A 145. *kíwǎg*: 148. *kíwǎg*; 288—291. *ek*: 293. *iek*; 296. *kiek*: 298. *kíək*; 334. *g'íwǎg*: 335. *g'íəg*; 336. *g'əg*: 337. *g'íəg*; B 10. *liěng*: 11. *d'íəng*; 20. *liěng*: 21. *d'íək*; 27. *d'íěng*: 28. *t'íəng*; 77. *d'ieg*: 79. *d'əg*; 232. *dz'íəg*: 233. *tsíəg*; 266. *tieng*: 267. *liəng*; 394, 395. *tiəg*: 396, 397. *liəg*; 445. *díək*: 447. *d'ieg*: 448. *d'əg*; 468. *síəg*: 469. *síəg*; 510, 511. *d'ieg*: 513. *liəg*; 630. *díəg*: 631. *liəg*; 633. *síəg*: 632. *tsíəg*; C 64. *lieng*: 65. *liəng*; E 36. *g'íwen*: 37. *g'wən*; 57. *kíěn*, 58. *g'ien*: 60. *kwən*, 62. *k'wən*; 64. *kiet*: 67. *kwət*; 118. *g'íwet*: 120. *kwət*, 119. *k'wət*; 214, 215. *ien*: 216. *ien*; 265. *ien*: 266. *íwən*; 283. *en*: 284. *əd*; F 89. *tsíěn*: 95. *síən*; 121. *dziwěn*: 123. *dziwən*; 232. *d'ien*: 231. *liěn*; 276. *dziwěn*: 281. *síən*; 288. *d'iet*: 289. *t'íəd*; 305. *síěn*: 306. *síən*; 320. *tsíěn*: 321, 322. *ts'íən*; H 74. *p'íěn*: 75. *píwən*; 115. *píět*: 117—119. *píwət*; 116. *píět*: 120, 121. *píwət*; 124. *píəd*: 125. *píəd*, 126. *píwəd*; 141. *píəd*: 142. *píwəd*.

e ~ *o*:

A 96. *k'ǝg*: 99. *k'ǝg*; 109. *ngǝg*: 113. *ngiog*; 143, 144. *kwǝg*, 145. *kǝwǝg*: 149. *kog*, 150. *kǝg*; 193—195. *g'ieg*, 192. *kǝg*: 197. *kǝg*, 198. *kǝg*, 199. *kǝg*; B 58, 59. *ǝǝg*: 61. *d'ǝg*; 85. *dǝng*: 89. *ǝ'ǝng*; 147. *dǝng*: 148. *ǝ'ǝng*; 174. *ts'ǝg*: 182. *tsǝg*; 209. *sieng*: 208. *sog*; 261. *siek*: 262. *sǝg*; 331. *d'ǝng*, 330. *t'ǝng*: 332. *d'ǝng*; 479. *d'ieg*: 486. *d'ǝg*; 578. *dǝg*: 579. *dǝg*; D 129. *b'ǝg*: 131. *b'ǝg*.

e ~ *u*:

A 129. *kǝng*, 130. *kieng*: 131. *g'ǝng*; 210, 211. *k'ien*: 214. *k'ung*, 215. *k'ǝng*; 264. *ǝng*: 265. *ung*; 283. *g'ieg*: 284, 285. *gǝg*; B 8. *ǝng*: 12. *tung*; 54. *tǝng*: 56. *tǝng*; 85. *dǝng*, 88. *dǝng*: 90. *dǝng*; 112. *tieng*: 115. *d'ǝng*; 113. *ǝng*: 116. *ǝng*, 118. *dǝng*; 154. *t'iek*: 155. *tǝk*; 229. *tieng*: 230. *tǝk*; 232. *dz'ǝg*: 231. *dz'ǝk*; 375. *t'ien*: 376. *ts'ung*; 410. *dǝng*: 414. *dǝng*; 442. *tsǝng*: 443. *ts'ung*; 520. *tsǝk*: 523. *tsǝk*; C 31. *lieng*: 32. *lung*; D 2. *pǝng*: 3. *b'ung*; 8, 9. *mieng*: 11, 12. *mung*; 28. *mieng*: 29. *mung*; 173, 174. *mǝg*: 175. *mug*.

ε ~ *ə*:

A 83. *g'eg*: 84. *g'ǝg*; 230. *g'weng*: 231. *g'wǝng*; E 20. *k'ǝen*: 22. *gǝen*; 74. *kǝd*, 75. *kǝd*: 76, 77. *kǝd*; 137. *ngen*: 134. *g'ǝen*; 182. *g'ǝen*: 183. *k'wǝen*; 275. *g'ǝen*, 276. *kǝen*: 277. *kǝen*; F 307. *sǝen* and *sǝen*; 326 *dz'ǝen*: 324. *dz'ǝen*; H 68. *mǝen*: 66, 67. *mǝen*, 69. *mǝen*; I 12. *ǝem*, 13, 14. *ǝem*: 15—18. *ǝem*, 19, 20. *ǝem*; 23. *g'ǝem*: 25. *g'ǝem*; 61. *g'ǝep* and *g'ǝep*; 75, 76. *g'ǝep*: 77. *g'ǝep*; K 22. *ǝiep*: 23. *ǝiep*; 26. *d'ǝem*: 28. *d'ǝem*, 29. *d'ǝem*; 27. *dz'ǝem*: 31. *sǝem*.

e ~ *o*:

A 16, 17. *ǝǝg*: 18. *ǝǝg*; 97. *k'ǝg*: 99. *k'ǝg*, 101. *g'ǝg*; 138. *ǝǝg*: 139. *ǝǝg*; 147. *k'ǝg*: 150. *kǝg*; B 2. *dǝg*: 3. *ǝǝg*; 106. *t'ǝg*: 107. *t'ǝg*; 176. *tsǝg*, 177. *dz'ǝg*, 179. *tsǝg*: 183. *tsǝg*; 247. *ǝ'ǝg*: 253, 254. *ǝǝg*; 267. *ǝǝg*: 269. *dǝǝg*; 300. *dǝǝk*: 301. *dǝǝk*; 382. *dzǝg*: 383. *ts'ǝg*, 384. *sǝg*, 385. *zǝg*; 401. *sǝg*: 402. *zǝg*; 501. *sǝg*: 502. *sog*; 513. *ǝǝg*: 515. *d'ǝg*; 532. *d'ǝg*: 533. *dǝǝg*; 537. *ts'ǝg*: 538. *ts'ǝg*; 570. *d'ǝg*: 571. *tog*; 587. *ǝǝg*: 588. *tog*, 589. *ǝǝg*; 597. *dǝg*: 598. *dǝǝg*; 619. *t'ǝg*: 620. *tǝg*; 623. *t'ǝg*: 624. *t'ǝg*; 644. *tsǝg*: 645. *tsǝg*; 647. *tsǝg*: 650. *dz'ǝg*; D 145. *p'wǝg*: 146. *p'ǝg*.

ə ~ *u*:

A 78. *kǝng*: 79. *kǝng*; 172. *kǝng*: 173. *g'ǝng*; 267. *kwǝng*: 268. *kǝng*; 326. *kǝk*: 327. *gǝk*; 368. *g'ǝwǝg*: 367. *kǝg*; B 411. *tǝng*: 413. *d'ǝng*; 470. *d'ǝng*, 471. *dǝng*: 472. *t'ung*; 505. *dǝng*: 506. *ǝng*; 542. *ǝng*: 543. *ǝng*; 603. *d'ǝng*: 604. *dǝng*; 652. *tsǝk*: 653. *sǝk*; 684. *d'ǝk*: 685. *d'ǝk*; C 61. *ǝng*: 62. *ǝng*; D 10. *mǝng*: 11, 12. *mung*; 122. *pwǝg*, 123, 124. *b'wǝg*: 125. *b'ǝg*; 133, 134. *b'wǝk*: 135, 136. *b'ǝk*; 179. *mwǝg*: 180. *mǝg*; 188. *b'wǝg* and *b'ǝg*.

o ~ *u*:

A 43. *ok*: 44. *ǝk*; 63. *kǝng*: 64. *g'ung*; B 89. *ǝ'ǝng*: 90. *dǝng*; 103. *tǝk*, 104. *d'ǝk*: 105. *ǝk*; 114. *d'ǝng*: 117. *ǝng*; 122. *tǝk*: 123. *tǝk*; 216. *d'ǝng*: 217, 218. *t'ung*; 259. *tsǝk*: 246. *ǝk*; 269. *dǝng*: 270. *dǝng*; 369. *sǝk*: 370. *suk*; 364. *tsǝk*: 365.

ts'ïuk: 389. *tóng*: 390. *tung*; 475. *îïok*: 477. *îïuk*; 508. *d'ïók*: 509. *d'ïuk*; 521. *ts'ïók*, 522. *ts'ïók*: 523. *ts'ïuk*; 594. *d'ïók*: 596. *d'ïuk*; C 66. *nóng*: 67. *n'ung*; D 4. *p'ïók*: 5. *b'ïuk*; 49. *b'ok*: 51. *puk*; 185. *p'ïóng*: 187. *p'ïung*.

Combined alternations.

We now revert to the question posed on p. 90 above. Is it reasonable to combine words which offer, not one alternation but two or more? Let us take the third example given there: E 93. *g'át* »how»: 94. *k'ïär* »how». Semantically the combination is very good; but we have to accept no less than four alternations, concerning every element of the two words. We have attested above the indubitable existence of the four alternations: *g' ~ k'* (very common); *0 ~ ÿ* (very common); *á ~ ə* (very common); *t ~ r* (quite certain and fairly common); are we then authorized to pose the affinity of the two words »how»: *g'át*: *k'ïär*, which supposes those four alternations?

Theoretically, I should say that we are; but I want to emphasize that here, as in all linguistics, the conclusiveness of the argumentation depends upon the question whether we can find many parallel examples, showing the same combination of alternations. If we can, then I think nobody can deny the correctness of our deductions. That is why I wish to give here some fairly extensive examples of double alternations (in order to simplify matters I limit the demonstration to two elements: principal vowel and final consonant):

Alternations *a ~ e* and *ng ~ k ~ g* (*ang ~ ek*, *ang ~ eg*, *ak ~ eng*, *ak ~ eg*):

B 407. *îiang*: 415. *d'ïek*; C 9. *îiang*: 11. *nëk*; A 140. *k'iwang*: 145. *k'iwëg*; 186. *k'iang*: 192. *k'ïeg*; B 5. *şiang*: 6. *s'ïeg*; 48. *d'ïang*: 58. *î'ïeg*; D 127, 128. *b'ïäng*: 129. *b'ëg*, 130. *p'ïeg*; B 333. *î'ïak*: 330. *t'ïëng*; 335. *ş'ïak*: 334. *s'ïëng*.

Alternations *a ~ ə* and *ng ~ k ~ g* (*ang ~ ək*, *ang ~ əg*, *ak ~ əg*, *ag ~ əng*):

A 155. *k'ïäng*: 163. *k'ïək*; 168. *k'iang*, 169, 170. *k'ïäng*: 175. *g'ïək*; B 427. *d'iang*: 428. *d'ïək*; A 73. *k'ïäng*: 84. *g'ïæg*; 92. *g'ï'äng*: 97. *k'ïæg*; 135. *k'ï'äng*: 137. *g'ïæg*; B 1. *d'iang*: 2. *d'ïæg*; 66. *d'iang*: 79. *d'æg*; 108. *s'iang*: 111. *s'ïæg*; 162. *dz'iang*: 177. *dz'æg*; 210. *d'äng*: 214. *d'ïæg*; 243. *î'iang*: 247. *î'ïæg*; 381. *dz'iang*: 382. *dz'ïæg*; 399. *s'iang*: 401. *s'ïæg*; 427. *d'iang*: 429. *î'ïæg*; 535. *ts'iang*: 540. *z'ïæg*; 639. *s'iang*: 640. *s'ïæg*; B 97. *d'ïak*: 99. *d'ïæg*; 119. *t'ák*: 126. *t'ïæg*; 444. *d'ïak*: 448. *d'æg*; 498. *sák*: 501. *s'ïæg*; 507. *î'ïak*: 513. *î'ïæg*; 637. *t'ák*: 638. *d'æg*; B 207. *dz'äg*: 206. *dz'æng*; 416. *î'ïag*: 411. *tæg*; D 18. *mäg*: 10. *mæng*.

Alternations *a ~ o* and *ng ~ k ~ g* (*ang ~ ok*, *ang ~ og*, *ak ~ og*, *ag ~ ok*):

A 118. *k'äng*: 121. *g'ök*; 201. *ïang*: 205. *ïok*; 250. *käng*: 256. *k'ök*; B 265. *d'iang*: 273. *îïok*, 275. *d'ïök*; 286. *s'iang*, 287. *şiang*: 289. *şïok*; C 16. *läng*: 19. *liök*; 24. *liang*: 27. *liök*; D 45. *p'äng*: 49. *b'ök*; A 118. *k'äng*: 122, 123. *kog*; 186, 187. *k'iang*: 197.

kjog, 199. *kjôg*; B 1. *djang*: 3. *îôg*; 48. *d'jang*: 61. *d'ôg*; 242. *djang*: 250—252. *djog*; 265. *djang*: 280. *djog*, 278. *îôg*; 307. *î'jang*: 312. *t'jog*, 314. *t'jôg*; 361. *ts'ang*: 367. *ts'ôg*; 381. *dziang*: 385. *zjôg*; 399. *sjang*: 402. *zjôg*; 407. *îiang*: 418. *d'jôg*; 406. *tjang*: 420. *tjôg*; 440. *d'ang*: 441. *tog*; 554. *t'jang*: 557. *t'ôg*; C 14. *liang*: 15. *liog*; 16. *lâng*: 20, 21. *liog*; 69. *lâng*: 70. *log*, 71. *lôg*; D 6. *măng*: 20. *mog*; 45. *p'ăng*: 53, 54. *b'ôg*; 46. *pjăng*: 55. *pjog*; 62. *pjăng*: 66. *p'jog*; 67. *măng*, 70. *măng*: 71. *mjog*; 79, 80. *măng*: 82. *mjog*; 127, 128. *b'jăng*: 131. *b'ôg*; A 120. *g'ăk*: 122, 123. *kog*, 124. *kôg*; B 119. *t'ăk*: 127. *tôg*; 165. *tsjak*: 183. *tsjog*; 271. *t'ăk*: 279. *tiôg*; 272. *îjak*: 278. *îjôg*; 288. *sjak*: 292. *sjôg*; 323. *sjak*: 319. *ts'jog*; 335. *sjak*: 337. *sjôg*; 450. *dziak*: 456. *sjog*; 581. *dz'ăk*: 583. *tsjog*; B 454. *zjag*: 452. *sjôk*.

Alternations *a* ~ *u* and *ng* ~ *k* ~ *g* (*ang* ~ *uk*, *ak* ~ *ung*, *ag* ~ *ung*):

A 266. *gijwang*: 271. *g'juk*; B 554. *t'jang*: 556. *d'uk*; D 89. *b'ăng*: 94. *p'uk*; A 90. *χjăk*: 88. *χjung*; B 120. *d'ăk*: 116. *îjung*; 119. *t'ăk*: 115. *d'ŭng*; D 91. *păk*: 90. *b'ŭng*; 100. *păk*: 97. *p'jung*, 98. *b'jung*; D 18. *măg*: 11. *mung*.

Alternations *e* ~ *ə* and *ng* ~ *k* ~ *g* (*eng* ~ *ek*, *eng* ~ *eg*, *ek* ~ *eg*):

A 159. *g'ien*: 162. *k'ək*; B 8. *îjeng*: 13. *îjak*; 53. *d'ien*, 54. *tjeng*: 57. *djak*; 68. *tieng*: 74. *təg*; A 136. *g'eng*: 137. *g'jag*; B 101. *t'jeng*: 106. *t'jag*; 110. *sjeng*: 111. *sjag*; 355. *seng*, 356, 357. *sjeng*: 354. *dz'jag*, 353. *tsjag*; 445. *djek*: 448. *d'ag*; 520. *tsjek*: 524. *dzjag*; 531. *tək*: 532. *d'jag*.

Alternations *e* ~ *o* and *ng* ~ *k* ~ *g* (*eng* ~ *ok*, *eng* ~ *og*, *ek* ~ *og*, *eg* ~ *ok*):

A 202. *îjeng*: 205. *îok*; B 101. *t'ien*: 104. *d'îok*; 112. *tieng*: 122. *tîok*; A 8. *keng*: 13. *kog*; 61. *g'ien*: 65. *g'ôg*; 93. *g'ien*: 102. *k'îog*; 210, 211. *k'ien*: 220. *k'iog*; B 53. *d'ien*, 54. *tjeng*: 61. *d'îog*; 101. *t'jeng*: 107. *t'iog*; 112. *tieng*: 127. *tôg*; 340. *d'jeng*: 341. *tieng*: 342. *d'jôg*; 412. *t'ien*: 418. *d'jôg*; C 44. *lieng*: 45. *lôg*; B 154. *t'iek*: 159. *tog*; 239. *siek*: 241. *sîog*; 415. *djek*: 418. *d'jôg*; 642. *d'iek*: 643. *djôg*; 658. *djek*: 659. *djog*; C 39. *liek*: 43. *lîog*; D 15—17. *miek*: 21. *môg*; 159. *miek*: 161. *mîog*; A 109. *ngəg*: 108. *ngôk*; B 510. *d'ieg*: 508. *d'îok*; 563. *d'jag*: 562. *d'îok*; 578. *djag*: 576. *djok*.

Alternations *e* ~ *u* and *ng* ~ *k* ~ *g* (*ek* ~ *ung*, *eg* ~ *ung*, *eg* ~ *uk*):

B 520. *tsjek*: 516. *tsjung*; D 15, 16. *miek*: 11. *mung*; A 24. *kəg*: 25. *g'ŭng*; 193. *g'ieg*: 191. *kjung*; B 58. *îjag*: 56. *tŭng*; D 88. *b'jag*: 87. *pjung*; B 479. *d'ieg*: 477. *îjuk*; 510. *d'ieg*: 509. *d'juk*.

Alternations *ə* ~ *o* and *ng* ~ *k* ~ *g* (*əng* ~ *ok*, *əng* ~ *og*, *ək* ~ *og*, *əg* ~ *ok*):

B 267. *îjang*: 274. *îjôk*; 298, 299. *d'jang*: 301. *djôk*; 470. *d'ang*, 471. *djang*: 475. *îjok*; 603. *d'ang*: 605. *djok*; C 26. *liang*: 27. *liok*; B 411. *tang*: 417. *d'ôg*; 470. *d'ang*, 471. *djang*: 483, 486. *d'jôg*; 528. *djang*: 533. *djôg*; 603. *d'ang*: 606. *d'iog*; D 10. *mang*: 21. *môg*; A 162. *k'ək*: 166. *kog*; 326. *kjak*: 328. *g'jôg*; B 57. *djak*: 61. *d'îog*; 317. *ts'jak*: 319. *ts'jog*, 320. *dz'jog*, 321. *dz'jôg*; 336. *sjak*: 337. *sjôg*; 474. *îjak*: 483. *d'jôg*; 539. *sjak*: 541. *sjôg*; 569. *tək*: 571. *tog*; 572. *tsjak*: 573. *dz'jôg*; C 22. *liak*: 23. *log*; B 106. *t'jag*: 104. *d'îok*; 126. *t'jag*: 122. *tîok*; 179. *tsjag*: 187. *sjok*; 329. *sjag*: 328. *sjôk*; 352. *t'ag*: 350. *djôk*; 597. *djag*: 594. *djôk*; D 172. *b'wag*: 171. *pjôk*.

Alternations *a* ~ *u* and *ng* ~ *k* ~ *g* (*əng* ~ *uk*, *əng* ~ *ug*, *əg* ~ *ung*, *əg* ~ *uk*):

D 40. *b'əng*: 41. *p'ĩk*; 132. *pəng*: 135. *b'ĩk*; A 181. *g'əng*: 182. *kĩg*; D 132. *pəng*: 137. *p'ĩg*; A 84. *g'ĩg*: 80. *k'ĩung*, 79. *kĩung*; 97. *k'ĩg*: 95. *kũng*; B 126. *t'ĩg*: 115. *d'ũng*; 329. *sĩg*: 326. *sĩung*; 524. *dzĩg*: 516. *tsĩung*, 517. *dz'ĩung*; 628. *t'ĩg*, 629. *t'ĩg*: 626. *t'ũng*, 625. *t'ung*, 627. *tũng*; 654. *tsəg*: 655. *şũng*; B 106. *t'ĩg*: 105. *ĩuk*; 126. *t'ĩg*: 123. *tũk*, 125. *t'ĩuk*; 513. *ĩg*: 509. *d'ĩuk*; 597. *dĩg*: 596. *dĩuk*; D 42. *b'wəg*: 41. *p'ĩk*.

Alternations *o* ~ *u* and *ng* ~ *k* ~ *g* (*ok* ~ *ung*, *og* ~ *ung*, *og* ~ *uk*, *ok* ~ *ug*):

A 41. *kók*: 40. *g'ung*; 128. *χok*: 127. *χung*; B 76. *tók*: 72. *tĩung*; 122. *tĩók*: 115. *d'ũng*; 328. *sĩók*: 326. *sĩung*; 605. *dĩók*: 604. *dĩung*; A 65. *g'óg*: 64. *g'ung*; 101. *g'ĩog*: 95. *kũng*; 220. *k'ĩog*: 213. *k'ung*, 216. *k'ĩung*; 221. *kóg*: 214. *k'ung*; 225. *χĩog*: 222. *χĩung*; B 127. *tóg*: 115. *d'ũng*; 225. *d'óg*, 226. *t'óg*: 223, 224. *dĩung*; 417. *d'óg*, 417, 419. *d'ĩóg*: 413. *d'ũng*; 494. *tsĩóg*: 490. *tsung*; 481. *t'óg*: 472. *t'ung*; 606. *d'ĩog*, 607. *t'ĩog*: 604. *dĩung*; D 21. *móg*: 11. *mung*; 55. *pĩog*: 48. *p'ĩung*; 66. *p'ĩog*: 63. *p'ĩung*; 74. *mog*: 73. *mũng*; 82. *mĩog*: 81. *mũng*; A 272. *kĩóg*: 270. *k'ĩuk*; 328. *g'ĩóg*: 327. *giuk*; B 107. *t'ĩog*: 105. *ĩuk*; 127. *tóg*: 123. *tũk*, 125. *t'ĩuk*; 159. *tog*, 160. *tĩóg*: 155, 156. *tũk*; 253, 254. *ĩog*: 246. *ĩuk*; 367. *ts'óg*: 365. *ts'ĩuk*; 486. *d'ĩóg*: 477. *ĩuk*; 515. *d'óg*: 509. *d'ĩuk*; 538. *ts'og*: 536. *tsũk*; 550. *dz'óg*: 549. *dz'uk*; 557. *t'óg*: 556. *d'uk*; C 42, 43. *lĩóg*: 40. *luk*; A 280, 281. *kĩók*: 284, 285. *giũg*.

Alternations *a* ~ *e* and *n* ~ *t* ~ *d* (*an* ~ *et*, *ad* ~ *et*, *at* ~ *en*):

E 230. *g'ĩăn*: 234. *g'ĩět*; 252. *·án*: 255. *·iět*; F 346. *d'iad*: 347. *ĩiět*; 297. *tsat*: 296. *dz'ĩen*.

Alternations *a* ~ *ə* and *n* ~ *t* ~ *d* ~ *r* (*an* ~ *ət*, *an* ~ *əd*, *an* ~ *ər*, *at* ~ *ən*, *at* ~ *əd*, *at* ~ *ər*, *ad* ~ *ən*, *ad* ~ *ət*, *ad* ~ *ər*):

E 53. *kwán*: 67. *kwət*; 259. *·iwăn*: 261. *·iwət*; F 107. *ts'wan*: 111. *dz'wət*; 132. *dĩwan*: 133. *d'wət*; H 111. *p'ĩwăn*: 120, 121. *pĩwət*; E 302. *g'ian*: 314. *kĩəd*; F 210. *ts'wán*: 215. *dzĩwəd*; H 112. *pĩwăn*: 126. *pĩwəd*; E 1. *kwán*, 2. *g'wán*, 3. *g'wan*: 25, 26. *g'wər*; 40. *χán*: 45. *χĩər*; 90. *giwăn*: 92. *giwər*; 135. *χian*: 136. *χĩər*; 149. *kwán*, 150. *g'wán*: 152. *giwər*; 157. *·wan*: 162. *·wər*; 199. *χán*: 200. *χĩər*; F 27, 28. *tsian*: 34. *tsiər*, 35. *dz'ĩər*; 172. *twán*: 180. *twər*; 304. *tsian*: 306, 307. *siər*; G 1. *nan*: 10. *nĩər*; H 45. *b'ian*: 52. *pĩər*; E 63. *kwát*: 60. *kwən*, 61, 62. *k'wən*; 139. *giwăt*: 138. *giwən*; 184. *kĩwăt*: 183. *k'wən*; 305. *kát*: 303. *g'ən*; F 109. *tswăt*: 108. *tswən*; 164. *tĩwat*: 162. *d'wən*; H 39. *pĩat*, *b'iat*: 38. *pĩwən*, *b'iwən*; 136. *pwăt*: 135. *pwən*; E 72. *g'iat*: 76. *kĩəd*; 139. *giwăt*: 141. *giwəd*; 244. *kĩat*: 245. *kəd*; 305. *kát*: 314. *kĩəd*; 319. *ngi'wăt*: 321. *ngi'əd*; G 26. *lăt*, 27. *lĩat*: 31. *lĩəd*; H 11. *p'wăt*, 12. *pĩwăt*: 15. *p'iwəd*; 114. *piat*: 125. *pĩəd*; E 91. *giwăt*: 92. *giwər*; 93. *g'át*: 94. *k'ĩər*; 305. *kát*: 318. *kər*; F 4. *tĩwat*: 6. *dĩwər*; 30. *dz'iat*: 34. *tsiər*, 35. *dz'ĩər*; 257. *sĩat*: 258. *siər*; E 24. *giwad*: 22. *giwən*; 140. *g'wad*: 138. *giwən*; 311. *g'ád*: 303. *g'ən*; F 279. *tswád*: 278. *dziwən*; E 103. *kád*: 102. *k'ĩət*; 315. *kĩwəd*: 310. *giwət*; F 134. *t'wád*: 133. *d'wət*; 158. *şiwad* and *şiwət*; H 14. *p'iwəd*: 13. *p'iwət*; H 58, 59. *b'wád* and *b'wət*; 122. *pĩad*: 117, 118. *pĩwət*; E 24. *giwad*: 28. *giwər*; 103. *kád*: 104. *g'ĩər*; 247. *kād*: 248. *k'ər*; F 4. *tĩwad*: 6. *dĩwər*; 22, 23. *ĩiad*: 24, 25. *t'ĩər*; 39. *dĩwad*: 40. *ĩiwər*; 93. *tsi'ad*: 94. *tsiər*.

Alternations $e \sim \text{ə}$ and $n \sim t \sim d \sim r$ ($en \sim \text{əd}$, $en \sim \text{ər}$, $et \sim \text{ən}$, $et \sim \text{əd}$, $et \sim \text{ər}$):

E 283. ʔn : 284. ʔəd ; H 97. $m\dot{i}\ddot{e}n$: 108. $m\dot{i}\ddot{e}d$; 113. $b'\dot{i}\ddot{e}n$: 125. $p\dot{i}\ddot{e}d$; E 214. $\text{.}\dot{i}\ddot{e}n$: 217. $\text{.}\dot{i}\ddot{e}r$; F 89. $ts\dot{i}\ddot{e}n$: 94. $ts\dot{i}\ddot{e}r$; 98. $ts\dot{i}\ddot{e}n$: 99. $ts\dot{i}\ddot{e}r$; 196, 197. $\acute{s}\dot{i}\ddot{e}n$: 198. $\acute{s}\dot{i}\ddot{e}r$; 305. $\acute{s}\dot{i}\ddot{e}n$: 306, 307. $\acute{s}\dot{i}\ddot{e}r$; G 37, 38. $l\dot{i}\ddot{e}n$: 39. $l\dot{i}\ddot{e}r$; H 18. $b'\dot{i}\ddot{e}n$: 21. $p\dot{i}\ddot{e}r$, 22. $p'\dot{i}\ddot{e}r$; 47. $p\dot{i}\ddot{e}n$: 52. $p\dot{i}\ddot{e}r$; 95. $m\dot{i}\ddot{e}n$: 110. $m\dot{i}\ddot{e}r$; F 53. $d'\dot{i}\ddot{e}t$: 50. $t'\dot{i}\ddot{e}n$; 288. $d'\dot{i}\ddot{e}t$: 289. $t'\dot{i}\ddot{e}d$; E 273. $k'\dot{i}\ddot{e}t$: 274. $k\dot{i}\ddot{e}r$; F 21. $t\dot{i}\ddot{e}t$: 25. $t'\dot{i}\ddot{e}r$; 32. $ts\dot{i}\ddot{e}t$, 33. $ts'\dot{i}\ddot{e}t$: 34. $ts\dot{i}\ddot{e}r$, 35, 36. $dz'\dot{i}\ddot{e}r$; 169. $d'\dot{i}\ddot{e}t$: 170. $t\dot{i}\ddot{e}r$; 176. $t'\dot{i}\ddot{e}t$: 178. $t\dot{i}\ddot{e}r$; 200. $d'\dot{i}\ddot{e}t$: 202, 203. $t\dot{i}\ddot{e}r$; 237. $d'\dot{i}\ddot{e}t$: 238. $d'\dot{i}\ddot{e}r$; 349. $dz'\dot{i}\ddot{e}t$: 351. $ts\dot{i}\ddot{e}r$; H 19. $p'\dot{i}\ddot{e}t$: 22. $p'\dot{i}\ddot{e}r$.

I have quoted my materials so extensively in order to prove, by these ample series of parallel cases, that we are perfectly justified in operating with combined alternations. Hence the examples on p. 90 above: $k'ung$: χiog ; $\acute{d}iang$: $t\acute{o}k$: $d\acute{e}g$; $g'\acute{a}t$: $k'\dot{i}\ddot{e}r$; $ng\dot{i}\ddot{a}n$: $g'wad$; $s\acute{a}n$: $dz'w\acute{e}r$ are far from being so impossible as they look at first sight. They fall within series of well-established alternations, and there is in principle nothing against considering them as being polar extremes within large and richly varied word families.

FINAL REMARKS.

There are two more questions which should be touched upon before finishing this preliminary investigation of Chinese word families.

One of them is this: are all the materials adduced here *homogeneous*, i. e. do they belong to one and the same language, one Archaic dialect? If not, if there are certain words which come, so to speak, from a side-track, which have been incorporated in literature by picking up dialectal words outside the main branch of the language and in their dialectal phonetic garb, then this would necessarily disturb our circles and endanger our results. I must frankly point out that in a few cases there is a certain risk of such a source of error. Indeed, I have inserted in my tables half a dozen words which the ancient sources directly indicate as being dialect words (such are B 496, B 530, F. 243, H 9). But on the whole this risk is very small, since I have operated mainly with the most common words of the language (practically all of them are such as are to be found in Soothill's little Pocket Dictionary).

The second question concerns the grammatical nature of the alternations. We have seen thousands of examples in which the language by their aid has formed parallel words for the same notion or phonetically more or less differentiated words for kindred notions. But do the alternations not sometimes serve as *express-*

sions for purely grammatical functions in a narrower sense? They certainly do, but this is an extremely complicated topic to which I hope to revert in another work. I shall point out here, merely as suggestive examples, a series of cases, in which our alternations studied above are expressions for different parts of speech or similar grammatical distinctions.

- B 517. *tsiung* a follower (noun): 517. *dz'jung* to follow (verb);
A 340. *kōg* a school (noun): 340. *g'ōg* to study (verb);
D 122. *pwæg* back (noun): 123, 124. *b'wæg* to turn the back (verb);
B 345. *tieng* («a fixer») anchor (noun): 347. *d'ien* to fix (verb);
B 353. *tsiæg* child, son (noun): 354. *dz'iaeg* to breed, beget (verb);
E 271. *kán* shield (noun): 272. *g'án* to ward off (verb);
H 38. *piwən* to divide (verb): 38. *b'iwən* a part (noun);
B 134. *tìōng* middle, centre (noun): 135. *d'ìōng* the middle one (adj.);
B 191. *tàng* to become long (verb): 191. *d'iang* long (adj.);
B 248. *tìog* dawn (noun): *d'ìog* («the matutinal one», adj. =) morning ceremony, audience;
B 649. *tsōg* early morning (noun): 650. *dz'ōg* black (adj.);
E 32. *kán* dry (adj.): 32. *g'ian* (heat and light:). Heaven (noun);
E 129. *kian* to see (active): 129, 130. *g'ian* to be seen, appear (passive);
F 325. *ts'iar* (equal:) mate, wife (noun): 324. *dz'iar* equal (adj.);
H 152. *pian* side (noun): 153. *p'ian* oblique (adj.);
K 45. *siəp* moist, wet (adj.): 46. *dz'iap* marsh (noun);
F 85. *t'ied* to go to (verb): 86. *t'ied* to cause to go to (causative verb to the preceding);
F 96. *t'iwət* to come out, go out (verb): 97. *t'iwət* to eliminate, degrade (causative verb to the preceding);
B 445. *d'iek* to change (verb): 445. *d'ieg* (changeable:) easy (adj.);
A 323. *ák* bad (adj.): 323. *ág* to hate (verb);
F 127, 128. *siwət* to lead (verb): *siwəd* leader (noun);
F 341. *siwat* to speak (vb. intrans.): *siwad* to speak to, to address (vb. trans.);
B 29. *d'ák* to measure (verb): 29. *d'ág* a measure (noun);
B 428. *d'iak* to eat (verb): 428, 431. *dz'iaeg* food (noun);
D 154. *b'ók* to return (verb): 154. *b'óg* again (adv.);
A 229. *kwáng* wide (adj.): 232. *k'wák* to widen (verb);
E 139. *giwăt* to say, said (vb. pres., past.): 138. *giwən* to have said (perfect);
B 552. *siog* small (adj.): 551. *šōg* small quantity (noun);

- A 129. *kǐəng* neck (noun): 130. *kieng* to behead (verb);
 E 142. *ngǐǎn* to talk (verb): 143. *ngǐan* a saying (noun);
 H 138. *b'wát* base, foot, root (noun): 137. *b'wat* to uproot (verb);
 B 355. *sǝng* to bear (verb): 356. *sǝng* (maternity:) clan (noun), 357. *sǝng* innate qualities (noun);
 F 216. *twən* solid, firm: 217. *ǝwən* (to make solid:) inculcate (verb);
 G 7. *nǐat* hot (adj.): 8. *nǐwat* to burn, to heat (verb);
 H 83. *pwat* not (general): 84. *pǐwat* not willing, not able (modal sense): 87. *pǐwər* is not (with a noun for predicate): 90. *mǐwət* don't (imperative): 91. *mǐwəd* not yet (perfective).
 K 90. *nǐəp* to enter (vb. trans. and intrans.): 87, 88. *nəp* to introduce (to cause to enter, causative): 89. *nwəb* the interior (noun).